Education of Humanity

On the Convergence of Nations

Krishna Pagadala

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In memory of Richard A. Easterlin (1926–2024), who asked the right question.

In memory of Jane Goodall (1934–2025), who saw what the juvenile dependency period does.

In memory of Mahbub ul Haq (1934–1998), who knew that people are the real wealth of a nation — and left one step for the rest of us.

Abstract

Wallace's theory explained how species differentiate under isolation. This paper documents the inverse: one species, post-1960, converging across every political system and continent onto a shared developmental phenotype — the paired thresholds of TFR below 3.65 and life expectancy above 69.8 (the 1960 United States values). By 2022, 154 countries have crossed. The outcomes tracked under "human development" — fertility, longevity, offspring survival — are the classical fitness components of life-history theory; convergence onto the phenotype is the event this paper predicts and

explains. Pre-transition local fertility patterns and post-crossing trajectory dynamics are outside its scope; it addresses the crossing event in the post-decolonization window and its generational predictor.

Humans have the longest juvenile dependency of any species — nearly two decades of plastic brain development scaffolded by dedicated adult teachers. This is the evolved substrate for high-bandwidth vertical cultural transmission. Cultural transmission is not uniquely human; elephants, orcas, and apes all transmit learned behaviour. What is uniquely human is cultural transmission applied to tools that rebuild the environment, and to the specialisation of individuals across them. Fire was the first; agriculture, metallurgy, literacy, and medicine followed — each a culturally-transmitted tool that manipulates the environment at a scale no individual could invent alone. Formal schooling is the latest and highest-bandwidth payload ever delivered through this evolved channel.

I argue that education is necessary and sufficient for the developmental phenotype at population scale. **Necessity:** of the 154 countries that have crossed, every one through mass schooling. **Sufficiency:** every country that reaches the educational threshold crosses within one human generation of doing so, under the post-decolonization substrate. Because the transmission operates through the *home niche* — the near-older humans embedded in the child's daily life across the juvenile dependency window — the predicted lag between input and outcome is ~25 years: the biological turnover time, not a policy-response interval. An educated home niche carries norms, health behaviours, and the expectation of school completion forward to children who exercise these as household decisions a generation later. Across a 185-country panel (1950–2015) spanning four generational depths — one century — this is what the data show. A great-great-grandparent's school completion predicts their great-great-grandchild's child-survival outcomes a century later. Seven empirical signatures are deduced from the biology in advance — generational timing, multi-generational persistence, asymmetric disruption, income independence, universality, amplification at low baselines, famine resistance — and tested against the panel; all seven hold.

Easterlin (1981) identified mass schooling as the cause of uneven development and asked why the whole world wasn't yet developed; his answer was largely set aside while the field pursued institutions, geography, and factor accumulation. Tested on its own terms with the field's own methods, the alternative collapses: income's predictive power decays within a sin-

gle generation, and once education's contribution is residualized out, income predicts nothing. Provision without educated populations produces no durable outcomes. Institutions are built by educated populations, not the reverse.

Keywords: life-history theory; extended juvenile dependency; vertical cultural transmission; developmental niche construction; phenotypic convergence; fitness components; mass schooling; home niche; cultural-transmission niche; human development; demographic transition

1. The Convergence

Between 1960 and 2022, 154 of 185 countries crossed the two thresholds that define human development: total fertility below 3.65 children per woman and life expectancy above 69.8 years — the United States values of 1960. Figure 1 traces the accumulation. The ascent is not gradual. The share of the world's population living in a country that has crossed climbs slowly from 13% in 1961 to roughly 20% by 1993, jumps when China crosses in 1994, passes 50% in 2001, and reaches 80% by the late 2010s. One species, across every political system and every continent, is converging onto a single developmental phenotype.

This convergence is the paper's empirical anchor. What varies between countries is the speed of arrival. Korea compressed the transition into thirty-five years; Spain took four-hundred-and-fifty. The mechanism is the same. What varies is how fast the state acts to extend it.

I argue that the mechanism is biological. Humans alone have a juvenile dependency period long enough — nearly two decades of plastic brain embedded with dedicated adult teachers — for cultural transmission to load the population with knowledge deep enough to produce the convergence (Konner 2010; Hrdy 2009). Formal schooling is the latest and highest-bandwidth payload ever delivered through that channel. Once an educated population is in place, development is no longer a policy target pursued through a list of interventions; it is the population's own expression of what the juvenile dependency window permits. Chapters 2 through 6 establish the biology. Chapter 7 states the predictions the biology entails. Chapter 8 tests each one against a 185-country panel spanning 1950–2015. Chapters 10 and 11 confirm the mechanism through natural experiments and country trajectories. Chapter 14 takes up the frameworks the biological claim must displace. Chapter 15 states what remains: the decision that no policy calculation dispenses from.

By "education" I mean the educated population, not the school system. The school system is the delivery mechanism; the educated adult is the operative unit. A country can rebuild schools within a decade; it cannot rebuild an educated population in less than a generation (Cambodia, Section 10.2). The transmission mechanism is the *home niche*: the near-older humans — parents,

grandparents, older siblings, near-kin — embedded in the child's daily life across the ~18-year juvenile dependency window. An educated home niche carries knowledge, norms, health behaviours, and the decision to educate the next child forward automatically. Mass schooling adds the *school niche* alongside it; children in a country under active expansion absorb from both at once. Population-level development requires the state to extend the school niche to households whose home niche still carries illiterate cultural transmission — not to make education work, but to reach the generation that the home niche alone cannot lift.

1.1 Defining Development

I define a country as developed when it has simultaneously achieved a **total fertility rate (TFR) below 3.65** and **life expectancy above 69.8 years** — the 1960 United States values (World Bank WDI).

These two measures are chosen because they are intrinsically valuable — not as proxies for something else, but as the things themselves. The UN's Human Development Index (UNDP 1990) already treats health and education as intrinsically valuable alongside income. This definition takes that logic to its conclusion: it retains the two outcomes that are ends in themselves and drops the one that is not.

Life expectancy is the most fundamental measure of human welfare. Every other outcome — income, freedom, knowledge, dignity — presupposes being alive. A country where people die at 45 has failed on the most basic term, regardless of its GDP or institutional quality. Life expectancy is not one metric among many; it is the precondition for all others.

The number of children a woman bears defines the shape of her entire adult life in ways that no other variable does. Seven children means two decades of continuous pregnancy and nursing, followed by a further decade of child-rearing before the last child is self-sufficient (Section 6.7). The difference between seven children and two is not incremental; it is the difference between a life fully absorbed by reproduction and a life that permits education, work, civic participation, or rest.

These are also the two most direct population-level expressions of whether

the cognitive transformation produced by education has occurred. Life expectancy tracks whether people have the planning horizon, health knowledge, and behavioural capacity to sustain their own survival. TFR tracks whether women have moved from biological fate to conscious reproductive choice — precisely the cognitive autonomy that underlies every other dimension of human flourishing.

Every competing metric of development is either a component of life expectancy and fertility — child mortality is a component of LE, nutrition is a cause of LE variation, sanitation prevents child death that shows up in LE — or a consequence of the cognitive transformation that moves them (institutional quality, income, democracy). No input can outrank the summary of itself. No consequence can outrank its cause. LE and TFR are not the best items on a list of development metrics — they are what the list is measuring.

No country considered developed — by any definition, under any framework — has life expectancy below 69.8 or TFR above 3.65. Even Sen's capabilities — health, bodily integrity, political participation, practical reason — all presuppose being alive and having cognitive autonomy over one's reproductive life. Every capability on Sen's list, examined for its generative source, has educated cognition as its enabling precondition. On his own framework, LE and TFR are not two capabilities among many; they are the enabling conditions for the capability to have capabilities.

GDP is excluded: measuring development through income would assume the answer. For readers who include income in their definition: education predicts future GDP per capita as strongly as it predicts LE and TFR (Table 7). Crossing dates under three threshold specifications shift by 10–30 years but the ordering holds (Table A8). Countries with high expansion rates cross any threshold within 14–16 years. The definition chosen here is the conservative one — it strengthens the alternative's position by removing from the outcome the one variable where income might have independent standing.

Why the 1960 USA? It is the natural baseline: 1960 marks the start of mass decolonization, and the USA was the unambiguous global hegemon. A country that crosses both thresholds has surpassed the richest nation on earth on the metrics that matter for human welfare. The USA in 1960 was mid-transition — its TFR of 3.65 was the baby boom peak, and it did not complete its own

fertility transition until 1972. A TFR below 3.65 is itself an achievement: in 1960, only the developed world — Europe, Japan, and a few settler states, roughly a quarter of humanity — had fertility that low. The remaining three-quarters of humanity averaged a TFR above 6 — the human baseline.

The countries that had already crossed both thresholds by 1960 were those with the longest prior histories of mass education: Scandinavia, the Netherlands, France, Switzerland, the United Kingdom, Australia, and New Zealand. Japan — despite a TFR of 2.0 — had life expectancy of 67.7, just below the threshold; it crossed in 1964. By 2022, 154 countries have crossed both thresholds, representing 80% of the world's population. Figure 1 traces this accumulation over time.

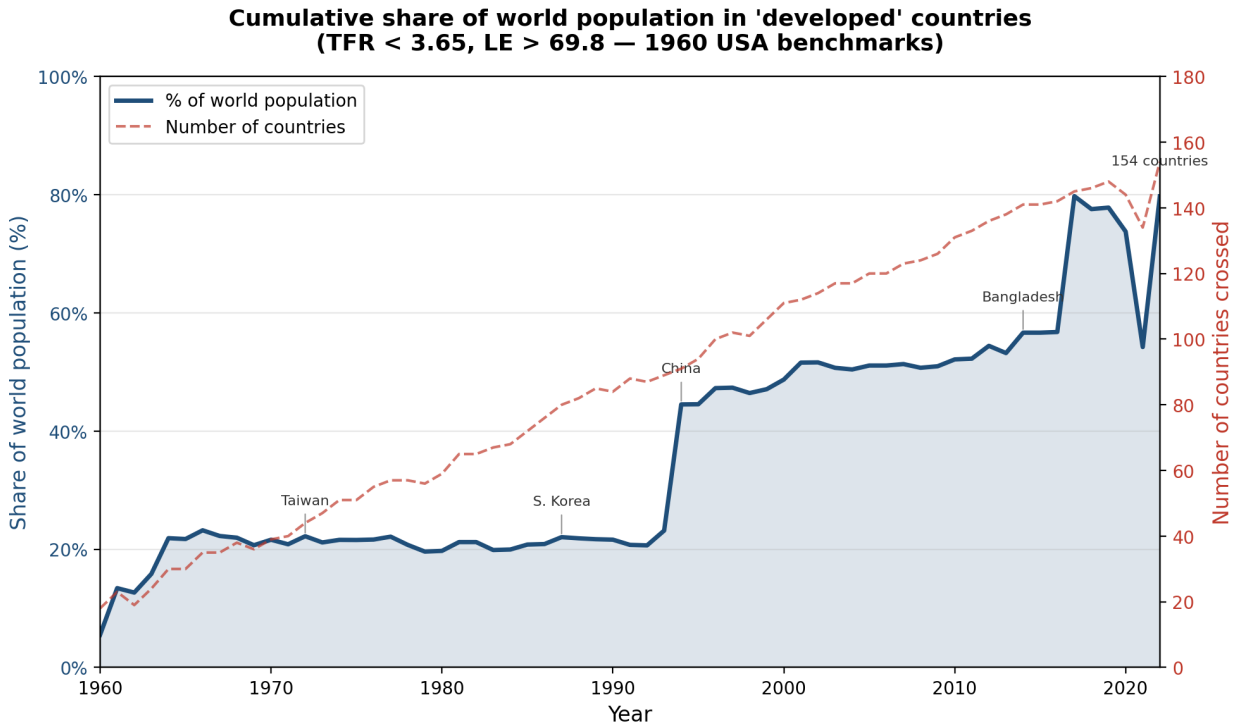

Figure 1: Cumulative share of world population crossing both development thresholds (TFR < 3.65, LE > 69.8), 1960–2022. Slow climb from 13% (1961) to ~20% (1993); vertical jump in 1994 when China crosses; 50% by 2001; 80% by 2017–2022. *(Source: World Bank WDI, WCDE v3. See* `scripts/figures/cumulative_developed.py`*)*

1.2 What the Convergence Requires

If one species is converging onto one phenotype through every political system, every ideology, and every income level, the cause cannot be any specific

political system, ideology, or income level. The mechanism must be singular because the convergence is singular. Two intellectual traditions — one asking what development is, the other asking what humans are — point to the same mechanism.

In development economics, Haq and Sen broke the identification of development with income when they built the Human Development Index (UNDP 1990). Sen's capabilities framework (1999) made explicit that what matters is what people can do and be, not what they earn — and he pointed to Kerala, Sri Lanka, and China as the right anomalies, welfare outcomes far above what their incomes predicted. Easterlin (1981) identified mass schooling as the binding constraint and asked why it had not reached the whole world. Lutz and Kebede (2018) showed that the classic Preston curve — the apparent relationship between national income and life expectancy — tightens when education replaces income on the horizontal axis. The income–mortality link was education proxying through income all along.

In evolutionary biology, Konner (2010) documented that the human juvenile dependency is the longest of any species — roughly eighteen years of plastic-brain exposure to dedicated adult transmitters, most of it observational rather than formal; schooling is a late, high-bandwidth intensification of the same channel. Hrdy (2009) showed that a dependency this long was only viable because humans evolved as cooperative breeders: mothers, fathers, grandparents, older siblings, and alloparents share the burden of provisioning children that no pair of parents could carry alone. The household I measure is the one Hrdy identifies. Boyd and Richerson (1985) formalised parent-to-child cultural transmission as the mechanism by which human populations accumulate knowledge across generations — at higher bandwidth than the genetic channel, because transmission is continuous across the dependency window rather than a single reproductive event. Hawkes (2003) closed the loop: post-reproductive human longevity, unique among primates, evolved to support grandchild survival, which is why longevity and fertility move together as one developmental signature.

These findings form a chain. Hamilton (1964) formalised inclusive fitness: behaviour that raises the reproductive success of kin is selected even when it lowers the actor's own. Boyd and Richerson extended the logic from genes to

culture: knowledge, norms, and the decision to educate are themselves heritable and subject to the same selective pressures, with the adults embedded in the child's daily life the highest-fidelity route. The home niche is inclusive fitness operating through cultural rather than genetic inheritance — kin-directed investment in descendants, carried by the extended juvenile dependency period and supported by the post-reproductive longevity that evolved alongside it.

The two traditions converge in this paper's central claim: that education is the necessary and sufficient cause of human development at the population scale, that income and provision are downstream of the same biological channel, and that the channel operates through the cultural-transmission niche (home and school together) across generations. The rest of the paper states the mechanism and tests it.

1.3 Why the Mechanism Has Been Missed

Every major approach to human development — income-first, institutions-first, provision-first, governance-first — was built by people standing on educational foundations they could not see. Adam Smith (1776) observed the pin factory and built a theory of wealth from the division of labour. The specialisation he described required workers already loaded with the tools of literacy, numeracy, and trade-specific knowledge through the cultural-transmission channel; he never asked where that loading came from. Scotland's Education Act of 1696 had been producing educated Scots for eighty years by the time Smith wrote the *Wealth of Nations*; he took educated labour as given — the same structural blindness that makes education invisible to educated populations today (Sections 6 and 8.5).

Japan, with negligible fossil fuels, industrialised within a generation of the Meiji education reforms; Nigeria, sitting on one of the world's largest oil reserves, did not. Every major development framework has made the same structural move: mistaking the products of education for its causes. The policy consequence is dispersion: resources spread across interventions, with education funded as one line item rather than recognised as the mechanism that makes the others work.

Smith's blindspot had a twin. The same framework that took educated labour

as given also took abundant coal as given (Wrigley 2010) — but both were outputs of the same channel: centuries of culturally-transmitted tools (schooling, metallurgy, mining, steam engineering), each rendered invisible by the prosperity it produced. Energy capture is not a separate precondition under cultural transmission; it is cultural transmission's oldest output, from fire forward. The empirical work in this paper is about the operative layer because that is where cross-country variance lives. The channel — from fire to formal schooling — runs under every case the paper analyses and every prediction it makes (Section 2).

2. The Longest Juvenile Dependency

2.1 The Dependency Window

The human juvenile dependency period is the longest of any species. A chimpanzee reaches competence and full subsistence at around seven years. A human takes roughly eighteen — more than twice as long, spent with the body still growing, the brain still being built, and the learner embedded in a household of adults whose daily behaviour is the curriculum (Konner 2010). Across mammals, dependency length tracks brain growth and the complexity of the behavioural repertoire that must be acquired before independence; no other species has stretched the window this far.

The length is not a byproduct of slow biology. It is the channel itself. Eighteen years of plastic brain, embedded with an adult caregiver, is what allows a cultural inheritance of thousands of years of accumulated technique, norm, and knowledge to be installed deeply enough that the learner will transmit it in turn. A shorter window — five years, ten — could transmit a skill. Only eighteen can transmit a cognitive architecture.

The window was built on a substrate of cooperative breeding. Human infants are the most helpless of any primate, and human mothers could not have borne the energetic load alone; fathers, grandmothers, siblings, and non-kin caregivers share care across the window, and the allomothering pattern is itself part of the species adaptation (Hrdy 2009). The grandmother contribution is evolutionarily load-bearing: human females live decades past menopause because post-reproductive women raised grandchildren's survival and trans-

mission odds enough to select for the long post-reproductive lifespan (Hawkes 2003). Dependency, cooperative breeding, and the long post-reproductive adult life are three parts of a single architecture.

2.2 How the Window Was Built

A brain that learns for eighteen years is metabolically extraordinary. The human brain represents roughly 2% of body mass and consumes about 20% of resting energy — an energetic share unmatched among mammals. Herculano-Houzel's (2016) comparative work on neuron counts across species shows that the human brain is not anomalous in structure — it is a primate brain at the predicted size. What is anomalous is the energetic budget that sustains it. No other primate can afford a brain this large because no other primate can extract the calories.

Wrangham (2009) identifies the substrate of the substrate: cooking. Cooking raised the net caloric yield of food enough to fund the expensive brain and, critically, to shorten gut transit time so that the energy budget could be redirected from digestion to cognition. Cooking is a culturally-transmitted technique; it requires the channel it later expands. Once the threshold was crossed, the architecture became self-reinforcing: cooking funded the brain, the brain extended the dependency window, the window allowed cooking technique (and everything that followed) to be transmitted with higher fidelity and deeper absorption than any pre-cooking ancestor could manage.

Every organism is an energy-extraction system — the substrate of life is shared across every species (Smil 2017). What makes the human architecture different is not the energy extraction but the capacity to accumulate and transmit techniques for extraction across generations (Wrigley 2010; Smil 2017). That capacity requires the eighteen-year window. Without it, the architecture of cumulative cultural transmission does not begin.

2.3 What the Window Is For

The window is long enough for cumulative cultural knowledge to be absorbed so deeply that it becomes the learner's own baseline — not content recalled but cognition restructured. Other species transmit behaviour; only humans

transmit an installed operating system. What the learner gains by eighteen is not a body of facts (most facts will be forgotten within a decade) but a cognitive architecture shaped by sustained exposure to the accumulated tradition: planning horizon, categorical literacy (the brain reorganisation formal schooling produces; see Section 4.2), symbolic reasoning, and the dispositional expectation that their own children will pass through the same channel.

This is not a description of deficiency. Eighteen years of dependency is not an unfortunate delay before the human becomes useful; it is the channel that makes the adult useful at all. Every capacity the developed adult will later exercise — household planning, health behaviour, reproductive choice, civic participation, transmission to the next child — arrives through what was installed in the window.

2.4 The Window Supports a Continuous Dose

Educational dose within the dependency window is continuous, not thresholded. The learner who enters at five and leaves at eighteen receives thirteen years of the cognitive architecture; the learner who leaves at nine receives four. The biological substrate does not switch on at nine years of schooling and off at twelve; the substrate is the window itself, and schooling loads into the window at whatever depth the population supports.

Evidence from populations given the full option confirms this. In Singapore, a country that has run the dose to the end of the dependency window for decades, lower secondary completion in the cohort aged 20–24 is effectively universal (~99%), upper secondary completion is ~96%, and tertiary completion is 73% — roughly three-quarters of the cohort continuing into formal education past the age at which agency transfers from parent to child.[1] When the option is given, most of the cohort takes it. This is what a biological architecture for continuous cultural loading looks like at the population level: deeper loading is chosen wherever it is available.

The choice to measure development at lower secondary completion in this paper (Section 5.4) is a methodological choice driven by data availability and empirical sufficiency, not a substrate claim. The substrate permits more. The

[1] WCDE v3, both sexes, age 20–24, 2020 vintage. See `scripts/singapore_continuation.py`.

nine-year measurement is the floor at which the mechanism reliably crosses the development threshold (Section 5.4); it is not the ceiling of what biology supports.

2.5 Why No Other Species Develops

Cultural transmission is not uniquely human. Elephants transmit migration routes and water-hole knowledge through matriarchs; orca pods have culturally distinct hunting techniques and vocal dialects; chimpanzees have regionally varied tool traditions, the most famous of which is termite fishing (Goodall 1986; de Waal 2013). What these species transmit is behaviour — a technique refined within an ecological niche and passed to the next generation by observation. What they do not transmit is a cognitive architecture for restructuring the environment at scale.

The constraint is the dependency window. A chimpanzee juvenile has around seven years of plastic learning embedded with an adult before independence. Seven years is long enough to absorb termite fishing, social hierarchy, and the foraging repertoire of the group. It is not long enough to absorb writing, arithmetic, the accumulated technical tradition of metallurgy or medicine, and the cognitive disposition to add to that tradition. The human window — eighteen years, with cooperative caregiving and post-reproductive grandmother contribution stabilising it — is the minimum architecture that cumulative culture requires. No other species has the window; no other species develops.

Chapter 3 turns from the window itself to what has been loaded through it: the tool sequence that has rebuilt the human environment at every scale.

3. Cultural Transmission and the Tools That Rebuild Environments

The window of Chapter 2 is what humans load; this chapter is what gets loaded. The content of the channel is a tool sequence that has rebuilt every environment humans occupy. No other species has a tool sequence of this kind, because no other species has the window that makes a tool sequence possible.

3.1 Cultural Transmission Across Species

Cultural transmission — learned behaviour passed across generations — is not uniquely human. Elephant matriarchs transmit migration routes and waterhole knowledge across decades of drought cycles that no single elephant would experience. Orca pods have culturally distinct hunting techniques and vocal dialects that persist across generations, differentiated from neighbouring pods by transmission, not genetics. Chimpanzees have regionally varied tool traditions; Goodall's (1986) four-decade record at Gombe documents termite fishing, nut cracking, and leaf medicine varying by community, each learned by juveniles watching adults. De Waal's (2013) primate work extends this to social cognition, coalitional behaviour, and proto-normative enforcement — sophisticated transmission, carried within an ecological niche.

The cooperative dimension of human cognition is continuous with this. Hare and Woods (2020) document how self-domestication produced the cognitive substrate for cooperative learning that humans require: reduced reactive aggression, greater tolerance of strangers, sustained joint attention. Domesticated foxes show the same syndrome when selected over generations for tameness. The cognitive architecture that lets a human child sit in a classroom for thirteen years learning from a non-kin adult is built on a primate sociality that was reshaped over evolutionary time toward cooperative attention.

What every non-human species transmits is behaviour — a technique refined within an ecological niche and passed by observation. What none of them transmits is a cognitive architecture that rebuilds the niche itself.

3.2 The Human Ratchet

The distinctive feature of human cultural transmission is that each generation builds on the last without losing prior gains. Tomasello (2014) calls this the ratchet: cumulative cultural evolution that advances and does not slip back. Chimpanzees learn termite fishing; their descendants do not invent termite-fishing 2.0. Humans learn writing; their descendants invent printing, then typewriters, then computers, then the internet, with each layer stacked on the last. The ratchet requires both directions of fidelity — a child who absorbs the

tradition deeply enough to transmit it, and a cohort in which enough children absorb deeply enough that the tradition survives any individual's death or failure to transmit.

The ratchet runs through the eighteen-year window of Chapter 2. A five-year window cannot support a ratchet because five years is not long enough to absorb the accumulated tradition before the learner is asked to transmit it. Each tick of the ratchet is a generation; each tick requires the window; without the window there is no ratchet; without the ratchet there is no cumulative culture.

3.3 The Tool Sequence

Fire is the first. Cooking raised the net caloric yield of food enough to fund the metabolically expensive human brain (Wrangham 2009), which funded more transmission, which funded more tools. Every subsequent enlargement of the human population rode the same channel: agriculture displaced foraging, animal traction and wind displaced human muscle, fossil fuels displaced all of it (Wrigley 2010; Smil 2017). Each step is a culturally-transmitted tool that manipulates the environment at a scale no individual could invent or reinvent alone, and no individual masters all of them. Cultural transmission distributes tools across a specialising population; the population-level tool-stock grows because each child acquires a different toolkit through the same channel.

Writing compresses time. A technique refined over three generations can be written down and read by someone in a different continent in a different century. Arithmetic compresses space: a single symbolic system allows trade, taxation, navigation, and surveying to coordinate across populations that never meet. Metallurgy, chemistry, medicine, and engineering each load a body of accumulated technique through the same eighteen-year window into a specialised fraction of the cohort. The tool-stock is no longer held in any individual brain; it is held in the population-level transmission system, and each generation's contribution is the sum of every specialisation in it.

Formal schooling is the latest and highest-bandwidth layer. What took centuries to diffuse through observational transmission, and what parents alone could never transmit beyond their own trade, schooling loads into differentiated cohorts within twelve years. The species-level adaptation — cumula-

tive, specialised tool-making through extended juvenile dependency — is the same; only the loading rate and the breadth of specialisation have changed. Mass schooling became materially possible because the fossil-fuel economy produced the surplus to sustain schools, teachers, and the extraction of children from productive labour for the duration of their juvenile dependency. Scotland in 1696 could afford Knox's parish schools. Most of the world in 1696 could not. Chapter 4 turns to why schooling specifically — among all the things that have loaded through the window — is the highest-bandwidth payload ever delivered.

3.4 Specialisation Requires Loaded Labour

Smith (1776) opened his treatise with the pin factory: ten workers specialising produced thousands of pins per day where one unaided worker produced only twenty. The division of labour was the engine of the wealth of nations. Smith observed specialisation and built a wealth theory around it. He did not ask where the literate, numerate, trade-aware workers came from.

Scotland's 1696 Education Act had been producing them for eighty years before Smith was born. Knox's network of parish schools, extended through the eighteenth century into the Scottish system of free primary education, produced a population that could read contracts, measure quantities, apprentice into trades, and coordinate across the non-kin relationships that specialisation requires. The pin factory worked because the population had already been loaded through the cultural-transmission channel with categorical literacy, numeracy, and the cognitive disposition to hold a specialised role within a cooperative structure of non-kin. What Smith saw as the cause of wealth — specialisation — was a downstream consequence of the substrate that made specialisation possible.

Every industrial revolution since has followed the same pattern. Populations without prior cultural-transmission depth do not specialise into productive factories; populations with it do. The Meiji reform put schools before factories. Korea's post-war expansion put universal primary education in place before the first heavy industrial plant was built. Where the substrate is absent, the factory produces nothing but frustration; where it is present, the factory runs. Chapter 8 will show what this looks like at the 185-country scale.

3.5 Geography as Substrate, Not Mechanism

Diamond (1997) argues that geography — the latitudinal orientation of continents, the distribution of domesticable plants and animals, the east-west diffusion corridor of Eurasia — explains why some populations developed earlier than others. At the deep-time scale, this is a claim about what substrate was available for the tool sequence to begin at all. Populations in geographies that supported agriculture, cereals, and large domesticable mammals could cross the threshold into the culturally-transmitted tool sequence; populations in geographies that did not had nothing to load.

But geography is the substrate for which environments could be rebuilt, not the mechanism that builds the capacity to rebuild them. Every geography that humans occupy has seen educational expansion produce the developmental phenotype in the twentieth and twenty-first centuries. The variance in outcomes is no longer geographic; it is educational. Chapter 14 returns to Diamond at greater length. For the present chapter the point is structural: geography set the starting conditions for the cultural-transmission channel, but the channel itself is what did and continues to do the work.

4. Education as the Highest-Bandwidth Payload

Chapter 3 traced the tool sequence that has loaded through the eighteen-year window. Formal schooling is the latest, and by a considerable margin the highest-bandwidth, layer ever delivered. This chapter asks why — what schooling does in the window that other transmission channels cannot.

4.1 Why Schooling Is the Highest-Bandwidth Layer

Twelve years of structured cognitive immersion delivers what centuries of observational diffusion could not. Before mass schooling, literacy spread through apprentice-trader-scribe chains, monastery scriptoria, and the slow diffusion of devotional texts through communities that built religious reading aloud; the doubling time for literacy rates in pre-schooling Europe was on the order of centuries. Within a generation of universal schooling, populations went from near-zero functional literacy to near-universal, and in the

same window acquired numeracy, the capacity to read a contract, the capacity to follow a written health protocol, the capacity to hold a symbolic role in a bureaucracy, and the disposition to expect their own children to do the same.

The bandwidth advantage is structural. Observational transmission loads one technique at a time into one learner at a time, and the learner acquires only what is within sight of an adult already fluent. Schooling loads a cognitive architecture into an entire cohort in parallel, using specialised teachers whose job is transmission itself. Knowledge that previously required direct access to a practitioner — arithmetic, mechanics, natural history, the operation of a written legal system — becomes reliably accessible to every child who passes through. The channel did not become faster; the architecture of the channel changed. Schooling is the first transmission technology in human history that scales transmission itself.

4.2 Categorical Brain Reorganisation

What schooling delivers is not content but a reorganisation of the cognitive architecture that processes content. Herculano-Houzel's (2016) comparative-neuroanatomy work establishes the substrate: the human brain carries the largest neuron count of any primate, and the energetic budget that this neuronal density requires is what the long dependency window and cooperative breeding evolved to sustain. The substrate is evolved; what schooling does is put it to use in ways that never develop without schooling.

Dehaene's (2009) neuroimaging work on literacy shows that learning to read physically restructures the brain's visual processing pathways. A specialised region in the left fusiform cortex — the visual word form area — becomes tuned over years of reading to respond preferentially to letter strings over faces, objects, or other visual stimuli. Adults who never learned to read do not have this specialisation; their visual cortex processes text no differently from any other pattern. The reading brain is a physically different brain from the non-reading brain, produced by sustained exposure to a specific cognitive task during the dependency window. The 2010 extension of this work (Dehaene et al. 2010) traced the timing: the reorganisation begins early and continues to deepen through the schooling years.

Dehaene's (1997) parallel work on numeracy shows the same pattern for quan-

titative reasoning. Specific parietal-cortex regions activate in numerate adults during symbolic numerical tasks that remain dormant in populations without formal arithmetic training (see also Gordon 2004; Frank et al. 2008 for corroboration from the Pirahã natural experiment: an adult population with no numerical vocabulary cannot reliably distinguish sets of 5 from sets of 7). The capacity to manipulate quantities symbolically — which underlies every aspect of modern economic, scientific, and administrative life — is a culturally-installed cognitive technology that the brain does not develop without schooling.

This is what *categorical literacy* names in this paper: the categorical reorganisation of the brain through formal education, through which literacy, numeracy, and symbolic reasoning become capacities the adult exercises automatically across every domain of life. The reorganisation is not a metaphor. It is a physical restructuring of cortical pathways that schooling produces and whose absence leaves the brain in its unschooled state.

4.3 Duration Over Fidelity

What schooling installs is a cognitive reorganisation, not content. The content is the vehicle; the reorganisation is the payload. Heyes (2018) calls literacy and numeracy *cognitive gadgets* — culturally installed capacities that require duration of exposure to develop in any given individual, not fidelity of instruction. The install is slow and deep, not fast and precise.

The vehicle is disposable. A software engineer does not remember tenth-grade biology; a doctor does not remember trigonometry; a lawyer does not remember the periodic table. If the mechanism operated through content, education would depreciate as content was forgotten. It does not: the 1950s-schooled cohort, trained with slide rules and pre-computer physics, made the same health, fertility, and transmission decisions in the 1980s that a 1980s-schooled cohort makes now. The reorganisation was installed by the duration of schooling, not the specific techniques it used. The bandwidth is in the years, not the curriculum: a country that teaches twelve years of almost anything worth structured study will produce categorically literate adults; a country that teaches two years of the optimally-designed curriculum will not.

The Hanushek-Woessmann “knowledge capital” line argues the opposite —

that test scores, not years, capture the human-capital content that drives growth (Hanushek & Woessmann 2008, 2012); within that framework, populations that achieve many years of schooling with low test quality have gained nothing. The panel disagrees: years predict, tests do not. Section 8.5 develops the argument at the operative level (completion, not just duration) and tests it directly against the test-score frame.

4.4 Why Poorly-Nourished Populations Learned to Read Anyway

Every successful educational expansion in the historical record was made by populations that were, by contemporary standards, poorly nourished. Scotland's 1696 parish school population, Meiji Japan's 1872 cohort, Korea's postwar 1950s learners, Cuba's 1961 literacy brigades — none of them had anything like modern nutrition. They learned to read, did the arithmetic, crossed the categorical-literacy threshold, and transmitted to their children.

The reason is structural. The brain energetics that sustain the long dependency window were made possible by cooking, which Wrangham (2009) dates to roughly two million years ago. Once cooking crossed the threshold that made the energetically expensive human brain metabolically possible, the dependency window did the rest. The neural substrate for categorical literacy is what evolved over that time; contemporary nutrition affects the ceiling of performance and the speed of acquisition but not the substrate itself (Grantham-McGregor et al. 2007). A poorly-nourished child in 1696 Scotland learned to read because the substrate was already in place; the reorganisation was installed, even if contemporary nutrition would now deepen attainment and accelerate acquisition.

The practical consequence is that hunger, insecurity, and poor nutrition are part of the human baseline, not preconditions that must be resolved before education can begin. Every historical case of successful educational expansion ran through poor nutrition, not after resolving it. The substrate evolved in populations whose nutrition was highly variable and often poor by modern standards; it was not built for abundance. The decision to educate does not wait on the resolution of any of these conditions; resolving these conditions is what educated populations then do.

5. Generational Transmission and the State

5.1 The Generational Transmission Mechanism

A child absorbs culture from whoever is embedded in their daily life across the ~18-year juvenile dependency window. The set of near-older humans the child absorbs from — parents centrally, but also grandparents, older siblings, near-kin, and surrounding adults in the cooperative-breeding unit Hrdy (2009) describes — is the **CT niche**. Channel and content are inseparable: the near-older humans are the transmission, their practices are the content, and the child's brain is shaped by the specific niche it spends its window with. What the child becomes is what the niche carries, sustained for eighteen years.

Two CT niches matter at population scale. The **home niche** is the inherited cultural-transmission regime carried by the household's near-older humans. The **school niche** is the regime the state constructs through schools, teachers, and curriculum. In pre-state populations only the home niche exists, and the child grows up inside whatever regime the household carries. After mass schooling begins, children live in two niches simultaneously — they absorb from both. Their cognitive trajectory depends on how much of each they absorb.

Two regimes of CT exist and they are categorically different. Illiterate CT is oral, local, embodied, tied to the speaker's presence. Literate CT is written, translocal, externalised, decoupled from any single speaker. The cognitive reorganisation literacy produces (Section 4.2) is not a degree of literacy more or less: it is a different cognitive regime. The home niche and the school niche each carry one regime or the other; what matters for a child's outcomes is which regime their total absorbed exposure tracks.

The home niche does more than carry its own regime forward. It modulates the child's capacity to absorb the school niche. A child whose home niche is fully illiterate enters school without language scaffolding, without books, without literate role models, without a bridge between home and school: they absorb less of the school niche per year of enrollment. A child whose home niche carries partial literacy — an older sibling who finished school, a mother who reads, a grandparent who keeps ledgers — enters school prepared, and

absorbs more. A child whose home niche is fully literate has home and school reinforcing each other; absorption is near-complete. This is what the literature describes as the schooling–learning gap (Pritchett 2013; Hanushek & Woessmann 2015): enrollment is exposure to the school niche; literacy is absorbed CT; absorption depends on both niches, not one.

This architecture produces an intrinsic ratchet. Each generation's young adults who pass through school become the next cohort's near-adults in the home niche. The next cohort's home niche is therefore slightly more literate than the previous one, their absorption of the school niche is greater, and the generation they produce is more literate still. The ratchet is self-amplifying, and absent catastrophic state failure it is one-way: literate near-adults cannot become illiterate, and they gate forward what they carry. The state's role is to set the pace of clicking. No state contribution: the ratchet stalls. Modest state effort under competing priorities: roughly one notch per generation, crossing in ~60 years. Singular priority: multiple notches per generation, crossing in ~25 years. Section 5.3 develops the regimes; the cases (Chapter 11) trace them.

The ~25-year lag that runs through this paper is the biological turnover time: the interval for one full generation of children to pass through the school niche and become the near-adults of the next cohort's home niche. It is not the interval between policy and response. Policy is the click of the ratchet; the lag is how long the cohort absorbed under the click takes to become parents themselves. A woman completing lower secondary at age 15–18 has children reaching school age 20–30 years later. Taiwan's development crossing arrived ~20 years after educational expansion began: one generation. Kerala's arrived at ~65 years: three generations of slow ratchet-clicking before enough of the home niches carried literate CT for household decisions to produce population-level development outcomes. Across longer windows the same signature holds: the grandchild's attainment reflects the grandparent's through two sequential cycles (~50 years), the great-grandchild's through three (~75 years).

How does the literacy of the adults produced by the ratchet translate into fertility, life expectancy, and child survival? Through the decisions made in the household they head, jointly between both parents, with mothers carrying

heavier weight on proximate care. Fathers co-determine schooling enrollment (especially for daughters), resource allocation, and fertility negotiations — without them, educated mothers cannot always act. Mothers dominate proximate care: feeding, sanitation, early-symptom recognition, health-seeking, and the spacing of births. Caldwell's classic result — that maternal education is the single strongest predictor of child survival, independent of income (Caldwell 1979; Cleland & Van Ginneken 1988) — reflects this routing. The composition-by-level result (Section 8.9) operationalises it by outcome: primary completion moves TFR because basic literacy makes family size a choosable parameter; lower and upper secondary move LE and U5MR because the deeper cognitive reorganisation lets a mother evaluate symptoms, weigh distant risks, and act on them.

The architecture explains three properties of the transmission. Non-depreciation: what the child absorbs persists for life because what survives is not content but the cognitive reorganisation that content exposure produced (Section 8.5); literate near-adults cannot revert to illiterate CT. Amplification: at low baselines, where home niches carry illiterate CT, the state's school niche moves children who have no home-niche literacy at all, and each year of schooling clicks the ratchet by a large increment — the $\beta_g > 1$ result, where β_g is the ratio of the child generation's education gain to the parental baseline (Section 8.7). At high baselines, home and school niches are both literate; additional state effort produces smaller increments. Catastrophic reset: because literate CT lives in the near-adults of the home niche and the teachers of the school niche, destroying the educated population leaves the next generation's plastic brains with a home niche stripped of literate CT and a school niche with no-one to teach. Cambodia (Section 10.2) is this test: schools rebuilt within a decade, the home niche gutted of literate adults, progress stalled for a generation. Knox did not invent generational transmission; he built a singular-priority school niche alongside the home niche, accelerating a ratchet as old as the species.

The human baseline — the state before formal education reaches a population — is high fertility, high child mortality, low life expectancy, hunger, and disease. This is not a description of poverty. It is a description of the species when only the home niche exists and that niche carries the illiterate CT regime. Hunger and insecurity are part of the baseline, not preconditions

that must be resolved before education can begin. Every successful educational expansion in the historical record was made through poor nutrition and insecurity, not after resolving them (Chapter 11).

The pathway runs: absorption from the school niche restructures cognition; restructured cognition restructures intent (family size becomes a choice, child survival becomes an expectation); the fertility transition follows; fewer children releases resources per child; those resources raise the literacy of the next generation's home niche, which lifts absorption of the next generation's school niche. Each pass clicks the ratchet.

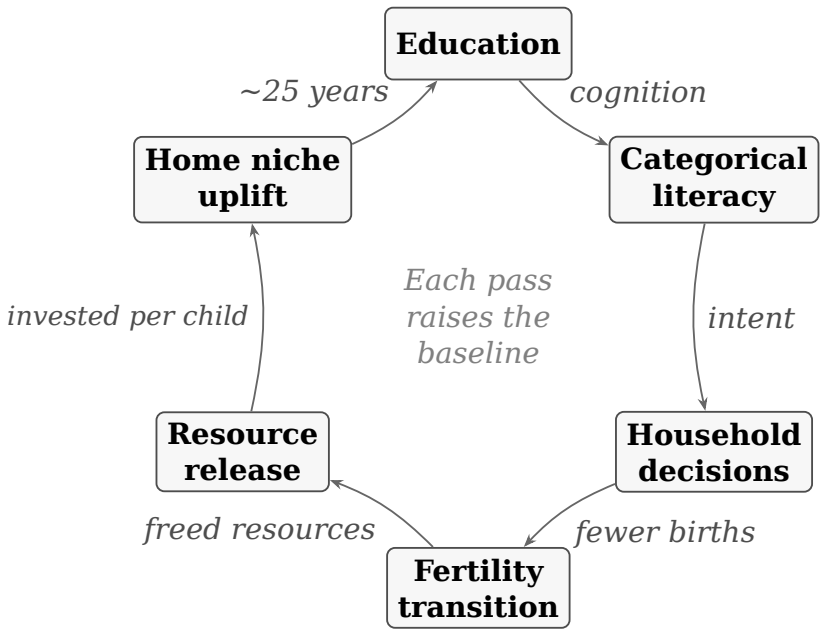

Figure 2: The generational transmission loop. Absorption from the school niche installs categorical literacy; categorical literacy shapes household decisions; household decisions drive the fertility transition; fewer children per household releases resources; those resources raise the literacy of the next generation's home niche, lifting absorption of the school niche. Each pass through the ~25-year cycle clicks the ratchet one notch.

5.2 From Action to Talk: How Education Reaches Beyond the Household

Education's effects start in the household but do not stay there. The four radii shown in Figure 3 extend education's effects outward from the Household decisions node of the generational loop (Figure 2), with decreasing durability as the radius widens.

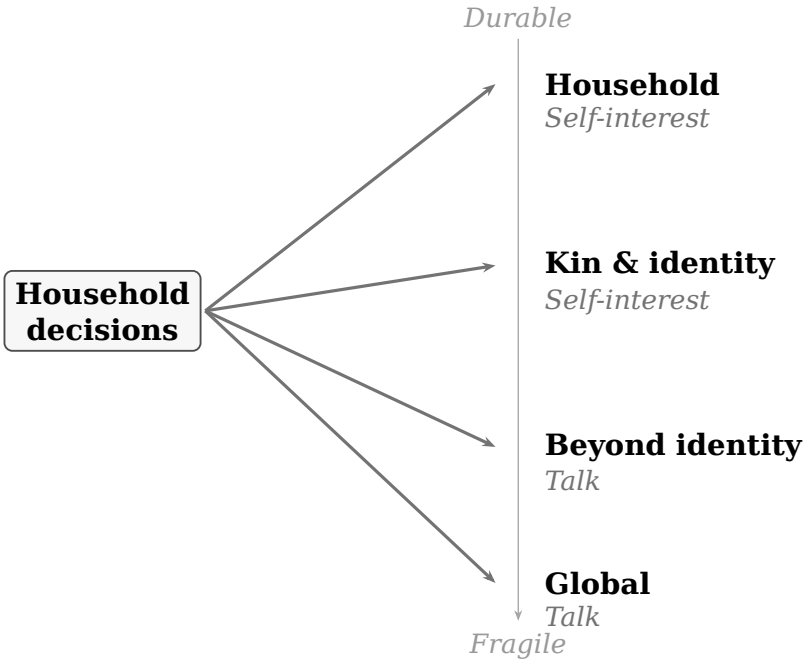

Figure 3: Four radii of educational effect. Education's effects extend outward from the Household decisions node of the generational loop (Figure 2), with durability decreasing as the radius widens: self-interest compounds; talk requires continuous external commitment.

The home niche operates through action — the household decisions of educated adults. But education's effects extend beyond the household, and the nature of that extension changes as the radius widens. Two things widen it simultaneously: education expands the boundary of who counts as "my people" — coalitional psychology (Tooby & Cosmides 2010) allows flexible group boundaries that education stretches from kin to nation — and it releases surplus, as educated parents have fewer children and nearly all survive, freeing time, money, and attention. The arc widens (who you invest in) and the constraint loosens (what you can invest).

The result is four radii of effect, each weaker than the last:

1. **Household.** The home niche is action: the parents raise the child in front of them, and the 18-year biological channel guarantees delivery. Runs on self-interest.

2. **Kin and identity.** Investment is still concrete and compounding. Runs on self-interest extended by expanded identity.

3. **Beyond identity.** Education becomes talk — advocacy, policy campaigns, diluted across every good cause simultaneously. This is where the competing-priority pathway (Section 5.1) originates.

4. **Global.** External provision — vaccines, aid, health delivery — funded by the surplus of already-developed populations. Talk, diluted further across borders and competing claims.

The axis that predicts durability is motivational: the home niche compounds because it runs on self-interest. Every non-education intervention requires someone to keep serving past the point where self-interest would redirect them. Revolutionary commitment, donor altruism, state fiscal discipline: all are fragile because all run against the grain of normal human motivation. Education runs with it. Non-education intervention buys time; educational investment changes trajectory.

5.3 The State: Reach, Not Mechanism

The state is not a substitute for the home niche. It constructs the second niche — the school niche — that, combined with the home niche, produces the transition state in which children absorb from both simultaneously. Its reach is the number of households whose children absorb the school niche. Its pace is how thickly it builds that niche: how many children enrolled, for how many years, at what quality.

Lutz and Kebede (2018) document the empirical signature of this at the global scale: state-led educational expansion shifts the fertility curve down and the life-expectancy curve up at a speed determined by how quickly the state reaches households whose home niche still carries the illiterate CT regime. Their redrawing of the income–health relationship turns income-axis diagrams into education-axis diagrams, and the fit improves. The state matters because it sets the pace of the ratchet at the population scale; it does not determine whether, once children have absorbed the school niche, the adults they become will carry literate CT into the next generation's home niche. That is guaranteed by the biology of the dependency window.

The operative question, in any country that has not crossed the development threshold, is not whether the state will invest in education — all states claim

to — but whether it invests with *singular priority*. Three regimes recur:

- **No priority.** The state does not build schools, does not train teachers, does not ensure access. Only the home niche exists, and it gates forward whatever CT regime it inherited. Pre-industrial populations lived in this regime for millennia.
- **Competing priority.** The state builds schools and trains teachers, but education competes with health, nutrition, security, governance, and other demands for budget and political attention. The school niche exists but is thin: modest enrollment, few years, variable quality. Absorption per child is limited, the ratchet clicks one notch per generation, and the development threshold is crossed at a multi-generational pace. Chapter 11 traces this through Sri Lanka's earlier phases, India, and much of sub-Saharan Africa.
- **Singular priority.** The state makes education the unconditional focus and builds the school niche ahead of need — schools before children exist to fill them, teachers before students exist to teach, standardisation before local pressure demands it. Absorption per child is high because the school niche is thick. The ratchet clicks several notches per generation, and the development threshold is crossed within one. Knox's Scotland, Meiji Japan, post-war Korea, Taiwan, and revolutionary Cuba all ran this regime.

The household operates within whichever school niche the state provides. Bangladesh from the 1990s illustrates the home-niche/school-niche interaction: the state-led push for girls' secondary education reached households whose home niche carried illiterate CT, and the girls who absorbed the school niche then carried literate CT into the next generation's home niche, making the household decisions — fertility, health, investment — that produced the next generation's gains (Mamun & Bongaarts 2022; Section 22). Mullainathan and Shafir (2013) describe the bandwidth tax of the illiterate home niche: evaluating delayed, abstract returns requires the cognitive reorganisation that only absorption from a literate niche produces. The state's reach constructs the school niche; the home niche, once uplifted by a first-generation school-niche absorber, compounds the gain across generations.

5.4 Why I Measure at Lower Secondary Completion

The biological substrate (Chapter 2) supports continuous educational dose up to roughly the age of agency transfer. I measure development at lower secondary completion — nine years of schooling — for two reasons.

First, data. WCDE v3 supplies long-run lower-secondary completion estimates back to 1875 across the panel; higher-secondary and tertiary completion series cover narrower windows and are available for fewer countries. A panel of 185 countries across 1950–2015 requires a measure that every country has a series for. Lower secondary completion is the deepest cross-country-comparable measure the data supports.

Second, sufficiency. Primary completion alone is empirically insufficient for the full developmental phenotype. Fertility falls hardest at the primary-to-secondary transition (Lutz & Kebede 2018; Section 8.9 documents the composition-by-level result in the panel), but life-expectancy gains and the next-generation completion signal accumulate with deeper exposure. Nine years is the empirical floor at which the mechanism reliably crosses the development threshold (Chapter 8).

The nine-year measurement is a methodological choice, not a substrate claim. The substrate permits more — Singapore's roughly three-quarters tertiary completion rate in the 2020 cohort (Section 2.4) is evidence that, when the option is given, most of the cohort chooses to continue. The biological window supports continuous dose; the panel data supports measurement at the nine-year floor. The two are consistent: the floor is what the panel can observe, not what the biology requires.

6. Why Education Is Invisible

The mechanism described in Chapters 2–5 is invisible to the populations it produces. This is not an accident of policy discourse; it is a structural consequence of the mechanism itself. The capacity to rank education as fundamental requires a cognitive frame that only education produces; and for households without it, the return is unimaginable. This chapter makes the invisibility precise and then extends the argument to the frameworks through

which this paper itself will be evaluated.

6.1 Education Leaves No Physical Signature

An educated person is physically indistinguishable from an uneducated one. Same body, same biological drives, same appearance. Wealth builds visible infrastructure; disease marks the body; education rewires cognition and leaves no trace on the surface. The only observable signal is behavioural change — and the disciplines misread it. They see smaller families and call it wealth flows, or contraceptive access, or patriarchy relaxing, or telenovelas on television. They see longer lives and call it health systems. They see rising productivity and call it economic growth. Every discipline theorises the downstream behaviour without tracing it back to the cognitive reorganisation that produced it. Education has no physical signature, so the signature it does leave — changed decisions, across every domain, for the rest of the person's life — is assigned to whatever domain notices it first.

6.2 Invisible From Inside

Educated people rank education as important — one of many important things, alongside health, nutrition, governance, and security. They cannot categorise it as *fundamental* because that would require seeing their own cognition from outside. The act of ranking priorities is itself an educated act; the cognitive architecture that assembles competing inputs into a weighted comparison is the product of the substrate the paper describes. The medium cannot be seen by those thinking through it.

Pinker's (2018) *Enlightenment Now* is a representative case. Pinker attributes the post-1800 rise in life expectancy, literacy, wealth, and peace to "reason, science, humanism, and progress." The rise itself was education. Literacy is education by definition; life expectancy, wealth, and peace are what educated populations produce — longer lives because educated adults care differently for themselves and their children, material gain because educated labour can be loaded with specialised skill, reduced violence because educated populations have fewer children, releasing the demographic and Malthusian pressure that drives conflict; and reason, science, and humanism are what schooling installs in the mind, the cognitive residue of the same mechanism. Pinker

has described education twice — once as outcome, once as mechanism — without recognising that formal schooling is the common cause of both sides. The book is a compendium of the downstream outcomes of the mechanism in Chapters 2–4 without a chapter tracing them to schooling. A reader inside the educated frame reads this as persuasive because the frame is what produces the reading. From outside the frame, the chain is one step, not two: mass schooling produced successive cohorts of educated adults, and those cohorts produced the measurable gains Pinker documents.

Subjective wellbeing confirms the blindspot: across 147 countries, self-reported happiness tracks income, not education — people feel richer, not more educated, so they attribute progress to wealth.[2] The signal educated populations receive from their own lives is material improvement, not cognitive transformation; the transformation is the medium through which they perceive the improvement.

6.3 Unimaginable From Outside

For uneducated households, the cost of schooling is concrete and immediate: the child not working, school fees, uniforms, distance to the school, the opportunity cost in daily labour and childcare. The return is abstract, delayed by a generation, and has no precedent in the household's experience. The frame required to evaluate the return — to imagine the cognitive reorganisation of one's future grandchild, and the household decisions that reorganisation will enable, and the population-level outcomes that will cascade — is itself the product of the education the household lacks.

Mullainathan and Shafir's (2013) work on the cognitive bandwidth tax of poverty is load-bearing here. Scarcity consumes the bandwidth required to evaluate abstract, delayed returns. An uneducated household under daily scarcity cannot process a multi-decade educational investment with the same cognitive apparatus an educated, resource-secure household brings; the apparatus itself is partially produced by the return they are being asked to imagine. The transformation is not just invisible — it is unimaginable. This is why state reach matters (Chapter 5 §5.3): the state has to deliver the first generation of schooling into households that cannot voluntarily choose it because

[2] See `scripts/happiness_education.py` and `scripts/figures/fig_happiness_edu_gdp.py`.

the choice itself requires capacities they do not yet have.

The people who could identify education as fundamental cannot, because they are inside it; the people who would benefit most cannot, because they are outside it. This is the structural form of the invisibility.

6.4 The Dilution Mechanism

Education simultaneously widens who the educated person invests in (literacy connects her to populations she has never met) and creates surplus to invest (fewer children, healthier children, longer productive life). The combination disperses investment across every cause simultaneously: health, poverty, gender equality, climate, governance, animal welfare, the arts. This is not a failure of educated people; it is the structure of educated behaviour. The educated adult, by the time she can evaluate priorities, has absorbed the frame in which priorities are plural by default.

The result is a global development discipline that treats education as one input among many. This is what the competing-priority regime (Chapter 5 §5.3) looks like at the intellectual level: the educated frame produces dispersion as its native output, and the institutions built by educated people inherit the dispersion. Education sits at number four in the SDG framework not because the framers mis-specified, but because the frame through which they specified is itself the output of education operating at its widest radius.

The clearest contemporary demonstration of the dilution mechanism is the United Nations Sustainable Development Goals (UN 2015). The SDG framework was designed by the most credentialed development institutions on earth, with full access to the evidence I cite — including Haq and Sen's own Human Development Index, Easterlin's identification of mass schooling as the binding constraint, and the Preston Curve literature showing education proxying through income. The result: SDG 4, Quality Education, listed fourth among seventeen parallel goals — alongside SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 3 (Good Health), SDG 5 (Gender Equality), SDG 8 (Decent Work), and SDG 10 (Reduced Inequalities). Every one of these is a downstream product of education. Income is what educated populations generate — stripped of education's contribution it predicts nothing (Section 8.9). Hunger is what educated households escape — not through provision but

through the fertility transition that releases resources per child, the planning horizon that converts subsistence into surplus, and the household-level cognitive and behavioural capacity that prevents food shocks from becoming famines: of 21 famines since 1950, 19 occurred where lower secondary completion was below 50%; of 17 comparable shocks in high-education populations, none produced famine (Section 10.1). Gender equality is what the fertility transition produces first — the shift from biological fate to conscious reproductive choice that occurs at primary education, before labour force participation, before political voice, before any of the dimensions SDG 5 measures independently.

The SDG framework does not reflect ignorance of the evidence. It reflects the mechanism: the educated people who designed it experienced income, food security, health literacy, and institutional capacity as the normal baseline from which you design frameworks — not as products of education that the uneducated world does not yet have. Education at number four, competing for budget allocations against hunger programs that feed children today, poverty interventions with visible short-term outcomes, and health delivery systems that save lives this year, is not an oversight. It is what the competing-priority regime looks like when institutionalised at the global scale by people who cannot see they are inside it.

6.5 Historical Exceptions Confirm the Rule

Inside the development frame, education competes with health, nutrition, governance, and security in a cost-benefit comparison — and it always loses, because its returns are generational while the others' are immediate and visible. A child dying of diarrhoea today is an emergency; a child not sent to school today is no one's emergency — until her child arrives as one in 2050.

Every historical breakthrough bypassed this comparison. Knox, Meiji Japan, Korea, and Cuba made education non-negotiable for reasons that preceded any cost-benefit calculation — salvation, existential military threat, Cold War survival, revolutionary ideology. Each of these bypasses was produced by external compulsion that overrode the educated default of treating education as one priority among many. Knox wrote under the conviction that every Christian must read the Bible; Meiji reformers under the threat of colonisation;

Park Chung-hee under Cold War military necessity; Castro under revolutionary ideology. In every case, the singular-priority decision was made *against* the default educated frame of plural competing priorities, not from within it.

This pattern confirms the invisibility claim. Without external compulsion, the educated frame produces dispersion (Section 6.4), and dispersion produces the competing-priority regime (Chapter 5 §5.3). With external compulsion strong enough to override the educated default, the singular-priority regime becomes possible — and where it has been attempted, the development threshold has been crossed within one generation (Chapter 11). The policy consequence — what it takes to replicate a bypass from evidence rather than contingency — is Chapter 15.

6.6 This Applies to This Paper

The argument of this paper applies to its own production, and to the production of every argument, discovery, and specification in the history of ideas. I received categorical literacy through formal schooling. The biological components of the mechanism — Konner on juvenile dependency, Boyd and Richerson on cultural transmission, Hawkes on post-reproductive lifespan, Dehaene on literacy's neural reorganisation — and the development components — Haq, Sen, Easterlin, Lutz and Kebede — arrived through the same transmission channel the paper describes. The synthesis is that channel operating on itself: an instance of what educated minds do with accumulated payload, no different in kind from any other output of the mechanism.

Every foundational specification is produced this way. Smith wrote from within an educated Scotland whose parish schools had been building literate labour for a century before him; he took the literate workforce as given and theorised the division of labour it enabled. Wallace wrote from the Malay Archipelago, having left formal schooling at fourteen; he carried the accumulated tradition of Linnaean taxonomy, Lyellian geology, and Malthusian population dynamics — each diffused to him through books and lower-tier British schooling rather than the Edinburgh–Cambridge elite circuit (Wallace 1870). Darwin reached the same synthesis independently because the tradition was ready. Shannon wrote from within the mathematical-engineering tradition of Nyquist, Hartley, Wiener, and Turing. Mendeleev assembled Döbereiner,

Newlands, and Meyer into the periodic table; Meyer reached a parallel arrangement the same year. In each case the specifier is a node in a transmission chain that has reached the threshold where the chain's own mechanism can be articulated from within.

Traditions specify their own mechanisms at readiness-moments. The specific author is contingent; the readiness is not. Parallel synthesis — Leibniz and Newton, Wallace and Darwin, Meyer and Mendeleev — is the usual signature of a tradition arriving at its own self-description. What matters is that the accumulated components reach the point where synthesis becomes possible, not who performs the synthesis.

This closes the invisibility argument of the previous section. The capacity to see the mechanism from inside is produced by the mechanism itself, operating across enough generations of accumulated knowledge that the channel becomes describable in its own terms. Education is invisible at shallow depths of accumulated understanding and specifiable at sufficient ones. The present synthesis is the tradition Haq and Sen opened in 1990 reaching the depth at which its own operative channel can be named. The paper is the channel speaking about itself — late, contingent, and no more exterior to the phenomenon than any other educated output ever has been. What separates this synthesis from competing ones is empirical survival (Section 8), not its provenance.

6.7 Demographic Structure and the Fertility Transition

Educated women have fewer children. The decline is continuous, steepest at the bottom, and requires no threshold to trigger.

Fertility declines continuously as female secondary completion rises. The evolutionary mechanism is a shift in reproductive strategy, not an override of the reproductive drive: in high-mortality environments, the optimal strategy is quantity — have many children because some will die. Education changes both inputs to this calculus. Child survival rises (fewer births needed to achieve the same number of surviving children), and the planning horizon extends — a literate parent can see that investing more in fewer children produces better outcomes than hedging with numbers. The drive to maximise offspring success is unchanged; what changes is the strategy that serves it.

The steepest fertility decline occurs at primary education (Section 8.9), where women are barely literate. What primary education gives is capacity over preferences that already exist — the capacity to read a contraceptive instruction, understand a health protocol, act on a desire not to be pregnant continuously. Yet the demographic headwinds are also steepest here: where total fertility rates are typically above 4, population growth is rapid, and the gains from each educated cohort are diluted across a larger and faster-growing next generation. As completion rises, fertility falls, child survival improves, and household resources per child increase. The home niche is a household mechanism. States can create conditions for these decisions; they cannot make them.

Human biology enforces the scale of these decisions. A woman who begins bearing children at fifteen and has seven will still be pregnant or nursing at thirty-five — and the last child, born then, still needs daily care when she is in her mid-forties. High fertility is not merely correlated with constrained lives; it is the constraint.

Even in populations where life expectancy at birth is low, women who survive to childbearing age typically live into their fifties or beyond; the average is dragged down by infant and child death, not by adults dying at forty. But survival does not mean freedom. The grandmother hypothesis in evolutionary biology holds that human females live decades past menopause precisely because post-reproductive women raised grandchildren's survival odds. The long post-reproductive lifespan is itself part of the transmission architecture. The species evolved long post-reproductive lives because it evolved extended juvenile dependency because it evolved cultural transmission. Together they are the biological signature of the human developmental niche.

Fertility decline is not a proxy for women's empowerment — it is the first expression of it. Education gives capacity over preferences that already exist: contraceptive availability alone does not produce the shift; education does. The further dimensions of empowerment that the literature measures — labour force participation, household bargaining power, political voice, economic independence — follow after fertility falls, as smaller families free time, resources, and attention. Education produces the first move; the rest cascades from it.

7. The Prediction

The claim this chapter tests is that education is the necessary and sufficient population-scale input for the developmental phenotype.

A causal model, if real, entails empirical patterns that the data must show (Pearl 2009). These predictions follow from the biology set out in Chapters 2–5, not from the data; they are deduced from the mechanism, not fitted to outcomes. Chapter 8 is the formal check. If the mechanism is real — if extended juvenile dependency creates a biological channel for cultural transmission, and formal education is the most powerful payload ever delivered through that channel — then seven empirical patterns must hold. The outcomes across which those patterns must appear — fertility, longevity, offspring survival — are the classical fitness components of life-history theory (Stearns 1992).

1. **Generational timing.** Because the juvenile dependency window closes around age 18 (Section 2.1) and educated children reach household agency — first marriage, first pregnancy, first independent health decision — in the early-to-mid twenties, education at time T should register in development outcomes at $T + \sim 25$, not at 10 or 15 years, the intervals that direct income transfer or institutional change would operate through.

2. **Multi-generational persistence.** Because Parental Transmission (Chapter 5) runs through the dependency window in every generation, a parent's educational depth installs cognition that the child installs in turn, so education's predictive power should decay slowly across generational depths (50, 75, 100 years). Income, which has no transmission channel equivalent in bandwidth or duration to the dependency window, should collapse within one generation.

3. **Asymmetric disruption.** Because the payload lives in the educated adult — the loading is through the dependency window, not through the institution that delivered it (Sections 5.1 and 5.3) — destroying the educated population should reset generational progress, while destroying institutions, income, or infrastructure while leaving the educated population intact should not.

4. **Income independence.** Because the causal path runs education → cognitive capacity (Chapter 4) → household decisions (Chapter 5) → outcomes, income stripped of education's contribution should have no independent predictive power for development outcomes: it sits downstream of the same mechanism that produced it. Operationalised as residualization against education (Frisch-Waugh-Lovell), and tested in Section 8.9.

5. **Universality.** Because extended juvenile dependency and cumulative cultural transmission are species-level traits (Chapters 2–3), not institutional products, the pattern should hold across political systems, cultures, geographies, and colonial histories.

6. **Amplification at low baselines.** Because state reach brings first-generation learners into the school niche on top of what the home niche already carries (Section 5.3), each generation's attainment should exceed the parental baseline by the most where that baseline is lowest; at high baselines the gap should close as ceiling compression sets in. Formally: the generational amplification ratio β_g (defined Section 5.1) exceeds 1 at low baselines and approaches 0 at high ones.

7. **Famine resistance.** Because the household-level cognitive and behavioural capacity that drives the fertility and mortality transitions (Section 5.1, Chapter 6) also drives collective response to supply disruption — substitution, rationing, migration, political pressure for relief — educated populations should not starve even under severe food shocks. Uneducated populations facing equivalent shocks should produce famines. Exceptions should require external force physically preventing educated populations from accessing available food.

Chapter 8 tests all seven in the panel; Chapter 10 tests them in natural experiments where the design is cleaner than the cross-country regression can deliver.

8. The Panel

Education predicts every development outcome one generation forward; income, once education's contribution is removed, predicts none of them. These results are not found by searching for a pattern — they are the seven predic-

tions of the biological mechanism (Chapter 7), tested against 185 countries over 65 years. All seven hold.

Each subsection below shows where one of the seven biological predictions (Chapter 7) appears in the 185-country panel. The predictions were deduced from the mechanism chain in Sections 2–5 before the data were examined; their empirical signatures appear in the ordering, decay structure, and subsample invariance the mechanism specifies, not in any identification strategy imposed on the regression. The statistical tests below formalise what the cases in Section 11 demonstrate historically.

8.1 What This Section Adds

Three literatures converge on the substrate this section tests. The *evolutionary-anthropology* substrate is the foundation: Konner (2010) on extended juvenile dependency; Wrangham (2009) on the caloric substrate for the expensive brain; Hrdy (2009) and Hawkes (2003) on cooperative breeding and the post-reproductive life; Herculano-Houzel (2016) on the energy budget that sustains the primate brain at human scale; Tomasello (2014) on the cumulative-cultural ratchet; Dehaene et al. (2010) on categorical literacy as physical neural reorganisation. These authors specified the mechanism. They did not test whether it scales into the population-level developmental phenotype that concerns policy. This section is that test.

The *human-development* tradition specified the phenotype. Mahbub ul Haq (HDR 1990) operationalised development as a population-level outcome beyond GDP; Sen (1999) argued capabilities, not commodities, are the object of interest. This paper agrees with Sen on what counts as development and disagrees on mechanism: Sen's growth-mediated / support-led dichotomy is two fiscal routes to the same educated-population substrate, not two separate causal channels.

The *development-economics* tradition asked whether the phenotype follows from income. Easterlin (1981) answered no — mass schooling was the binding constraint — and asked why it had not reached the whole world. Lutz and Kebede (2018) showed that replacing income with education on the x-axis of the income–health relationship largely absorbs the over-time upward shift: the income–mortality link was education proxying through income all

along. This section extends both on a 185-country 1950–2015 panel with a 25-year forward lag and a Frisch–Waugh residualisation that strips education's contribution from GDP before the comparison. What's new here: seven pre-registered predictions from the biological mechanism, tested simultaneously; income, once education's contribution is removed, predicts nothing at every cutoff and every lag tested.

8.2 Temporal Structure and Mechanism Testing

Education predicts income one generation forward (Table 7). Because education produces income, GDP sits on the causal path (education → GDP → life expectancy, fertility). Controlling for GDP strips out part of education's effect and biases the result against the mechanism (Pearl 2009; Angrist & Pischke 2009, ch. 3). Regressing outcomes on income compares education's product against the education that produced it; the proximity of income to the outcome gives it an unearned advantage in the comparison.

The temporal structure of the regressions provides the ordering. Parental cohort precedes child cohort by biological necessity, ruling out reverse causality at the individual level. The lag-decay shape (Figure 5; Section 8.8) is the signature of the mechanism: education and income produce structurally different decay profiles that autocorrelation cannot explain. The natural experiments that test Prediction 3 (Chapter 10) — Cambodia (population destruction → generational shadow); Sri Lanka (civil war, home niche unbroken, development on schedule); Myanmar (73 years of military rule, outcomes tracked education); China (barefoot doctors dismantled, LE continued rising); Korea vs Philippines (identical colonial base, divergent state commitment); and the Asian Financial Crisis (income wiped, education untouched) — produce outcomes that pure autocorrelation cannot predict.

8.3 Data

The analysis covers 185 countries from 1950 to 2015, with education measured as the share of 20–24-year-olds who finished lower secondary school.

Education data are from WCDE v3 (Lutz et al. 2021): lower secondary completion rates for the 20–24 age cohort, both sexes, which reflects completed ed-

ucation rather than enrolment. All education percentages in this paper refer to this measure unless otherwise noted. Coverage: 185 countries, 1950–2015 at five-year intervals. The core generational panel uses 1975–2015 (1,665 country-years); the residualization analysis (Section 8.9) extends to T=1960–1990 with education interpolated to annual values for precise threshold identification. Long-run analysis uses a 28-country subsample extending back to 1900, selected for data availability; the generational coefficient varies systematically with baseline education level (Section 8.7). GDP data: World Bank, constant 2017 USD, per capita, log-transformed. Life expectancy, TFR, and under-5 mortality: World Bank WDI. Parental education: each country's lower secondary completion lagged 25 years (T−25).

The 15 USSR republics inherit Goskomstat's 1960–1990 reporting inflation on lower-secondary completion; Barro-Lee v3.0 partially corrects the level but leaves residual bias on life expectancy and child mortality. Chapter 9 documents the four phenotype signatures of the anomaly and confirms that every headline result in this paper survives — and strengthens under — exclusion of those 15 republics.

8.4 Descriptive Statistics

The analysis panel is 185 countries observed at 5-year intervals 1975–2015 (1,665 country-years; 1,466 observations after merging the World Bank WDI GDP series, which covers 178 of 185 countries). Parental lower secondary completion is observed at $T-25$; child completion, log GDP per capita, life expectancy, TFR, and under-five mortality at T.

Table 1. Period means, six core variables. Five-year observation bands. Each variable shifts monotonically across the sample window as the global educational expansion propagates into the developmental phenotype. Full summary statistics (n, mean, sd, min, max) and the complete 185-country list are in Appendix A.2.8. Produced by `scripts/tables/summary_stats.py`.

Variable	1975–1989	1990–2004	2005–2015
Parental lower-sec completion (%, $T-25$)	27.0	42.7	56.2
Child lower-sec completion (%, T)	52.3	62.2	71.1
Log GDP per capita (const. 2017 USD)	8.07	8.19	8.52

Variable	1975–1989	1990–2004	2005–2015
Life expectancy at birth (years)	61.4	65.5	70.1
Total fertility rate (births/woman)	4.53	3.63	2.95
Under-5 mortality (per 1,000 live births)	98.8	66.8	39.2

Parental completion rises 29 pp over the sample; child completion rises 19 pp; total fertility halves (4.53 → 2.95); life expectancy rises 8.6 years; under-five mortality falls by 60%. The parental column lags child by one generation (25 years) by construction: the 1975–1989 parental mean of 27% corresponds to the 1950–1964 cohorts of those same countries.

8.5 Completion as the Operative Variable

I measure whether people finished school, not how well they scored on tests. Completion is the right measure for a generational mechanism.

Formal education — even at the lowest quality — represents a categorical discontinuity from oral cultural transmission. The neuroscience is unambiguous: literacy physically restructures the brain's visual processing pathways (Dehaene et al. 2010), and numeracy recruits parietal circuits unavailable to innumerate populations (Gordon 2004; Frank et al. 2008). These are not marginal improvements on existing capacities — they are new cognitive technologies — *categorical literacy* — that do not develop without schooling, regardless of intelligence or environment. An educated person knows bacteria cause disease; an uneducated person does not — there is no halfway position between germ theory and miasma.

Why exposure, not learning, is the operative variable. Section 4.3 established the payload: cognitive reorganisation, not content retention. The panel uses completion — not years alone, not test scores — because completion is where the mechanism registers at the generational scale. What formal schooling provides is not a specific curriculum but sustained exposure to literate culture: structured time, abstract categorisation, written language, numeracy, adult authority organised around knowledge transmission. The longer the child remains inside this environment, the deeper the reorganisation. Primary gives capacity over reproduction — the ability to read a contra-

ceptive instruction, understand a health protocol, act on desires that already exist. Secondary gives capacity over health navigation, economic participation, civic engagement. Tertiary gives capacity over complex systems. Each level is more time inside the categorical change.

A child who completed nine years of poor-quality schooling and scored badly on every assessment has still spent nine years inside an environment that reorganised their cognition categorically relative to a parent who never entered the building. The test score measures how well the payload was delivered. Completion measures whether the child was inside the channel long enough for the reorganisation to consolidate. The life-course data confirm it: no generation in any country has completed school at a lower rate than its parents, and the life-course returns to education show no decay with age — exactly what a reorganisational mechanism predicts and a content-retention mechanism does not.

What the curriculum's breadth is for. Years of sustained exposure to biology, chemistry, history, mathematics, and literature produce adults who spent years inside five different formal knowledge systems — each with its own mode of abstraction, its own vocabulary, its own relationship between evidence and conclusion. The breadth of the curriculum is not an attempt to produce polymaths. It is sustained duration of exposure across multiple modes of formal reasoning, each contributing to the same cumulative reorganisation. Tests measure whether the student retained the content of each subject; the mechanism operates through the years the student spent inside the reorganising environment.

The generational gate. The distinction sharpens at the generational gate. The educated parent's role in the home niche is not to teach their child what they learned in school. It is to hold the expectation that their child belongs in school — to see completion as normal, to monitor attendance, to assume that the educated world is where their child will live. This expectation is the gate. An uneducated parent cannot open it, because they have no experience of the world on the other side. An educated parent opens it automatically — through self-interest, not ideology — because school is where they came from and where their child belongs. The gate does not require the parent to be a

good teacher. It requires the parent to have been a student.

This is also why the grandparent effect is strongest at low baselines. Where education is scarce, the grandparent is a second educated person in the household — and, in a village where education is rare, a visible exemplar to the wider community. At high baselines, the entire society holds this expectation and the grandparent's independent contribution disappears — exactly the pattern the data show (grandparent β significant below 50% parental completion, zero above it).

From reorganisation to specialisation. The reorganised individual naturally specialises. Categorical literacy — the capacity for systematic learning, abstract reasoning, structured problem-solving — is not an end state. It is the precondition for further skill acquisition. The educated individual can absorb trade skills through apprenticeship, on-the-job training through employment, formal knowledge through college, deep expertise through a PhD. The specific pathway does not matter. What matters is that each pathway requires the cognitive infrastructure that formal schooling built. The uneducated individual is not excluded from specialisation by lack of opportunity alone. They are excluded because the capacity to learn a structured skill systematically — to read a manual, follow a protocol, internalise an abstract procedure — was never constructed.

The empirical confirmation. A direct test partitions the outcomes by the cognitive depth each requires. On a 104-country cross-country panel with USSR republics excluded, using Angrist et al. (2021) HLO test scores and WCDE completion: for TFR, quantity (primary completion) dominates at $\beta_z = -0.64$ ($t = -7.4$); adding test-score quality contributes $\beta_z = -0.25$ ($t = -2.9$), taking R^2 from 0.70 to 0.72. Primary completion alone drives the fertility transition. For child mortality (log U5MR) the partition reverses: quantity goes insignificant ($\beta_z = -0.16$, $p = 0.09$); quality dominates ($\beta_z = -0.70$, $t = -7.6$, R^2 = 0.78). For life expectancy: quantity insignificant ($t = -1.1$); quality dominates ($\beta_z = +0.83$, $t = +7.7$, R^2 = 0.62). The three outcomes separate by cognitive demand. *Fertility is a threshold flip that runs on completion even where schooling is shallow; child survival and adult longevity require cognitive depth and do not respond where credentials are hollow.*

Read alone, the U5MR and LE results vindicate Hanushek — but the vindication is the mechanism reasserting itself, not an alternative to it. A continuous lag sweep (Chapter 9, §9.6, and Figure 10) shows that HLO secondary today is best predicted not by current schooling but by the *parental* generation's completion 10–25 years earlier; R^2 at the great-grandparent lag (60 years) remains near 0.49. Hanushek's test scores are not a rival framework. They are the paper's quantity measure integrated across three generations of parental transmission. The horse race partitions the way the biology predicts, and the partition is what Chapter 9 develops: hollow education is not education, but exposure still matters for TFR.[3]

8.6 Empirical Strategy

Each country is compared only to its own past. The question: does parental education predict what happens to their children one generation later?

Primary specification — country fixed effects regression:

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \varepsilon_{it}$$

In plain language: a country's education level today is predicted by its own education level 25 years ago (the parents' generation). The α_i term gives each country its own baseline — so the test is not "which countries are more educated?" but "within the same country, does the previous generation's education predict the next generation's?" This absorbs everything permanent about a country: institutions, geography, culture, colonial history. What remains is change over time. The 25-year lag is not a researcher choice — it is the time-to-agency interval derived from the biological mechanism (Prediction 1).

Comparison specifications replace parental education with income, or include both:

$$E_{it} = \alpha_i + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

[3] See `scripts/hanushek_horse_race_comprehensive.py`, `scripts/hlo_is_parental_education.py`, and `scripts/hlo_vs_education_lag_sweep.py`; earlier within-country robustness in `scripts/robustness/completion_vs_years_vs_tests.py`.

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

For development outcomes (life expectancy, fertility, child mortality), I measure outcomes 25 years after the education measurement: education at time T predicts outcomes at T+25. Temporal ordering rules out reverse causation; time-varying confounding is handled by the natural experiments in Section 8.2:

$$O_{i,t+25} = \alpha_i + \gamma_1 E_{it} + \gamma_2 O_{it} + \varepsilon_{it}$$

Including the initial outcome level (O_{it}) is deliberately conservative: it absorbs the country's prior trajectory, so the test asks whether education predicts improvement beyond whatever trend was already underway.[4]

Residualization strategy. If education causes income, part of income's apparent predictive power is really education's effect in disguise. To test whether income contributes anything on its own, the design strips out the portion of income that education produced and uses only the remainder. Mechanics and results are in Section 8.9 (Table 10).[5]

On year fixed effects. Adding country-and-year fixed effects on the full 185-country panel collapses the parental coefficient from 0.483 to 0.083 — the mechanical signature of the staggered-adoption pathology, not evidence against the mechanism. Three-quarters of the 2WFE weight sits on comparisons between newly-expanding countries and already-high-completion controls (Goodman-Bacon decomposition, `scripts/robustness/goodman_bacon_decomposition.py`). The Callaway-Sant'Anna (2021) estimator, limited to clean treatment-timing comparisons, recovers the effect: +7.9 pp at the generational lag and +21.4 pp by year 35, with pre-treatment coefficients indistinguishable from zero (Figure 6). Full decomposition weights, event-study pre-trend tests, and the continuous-treatment

[4] Both-sexes completion is used throughout; female-only specifications produce stronger effects (Section 8.9).

[5] Applied across three education levels, four outcomes, and multiple sample definitions including an entry-cohort design (each country enters when it crosses a given threshold; sample closes at a ceiling). See `scripts/residualization/` (`education_vs_gdp.py`, `education_vs_tfr.py`, `child_education.py`, `by_entry_ceiling.py`).

discretisation caveat are in Appendix A.1. I keep the one-way country-FE specification as the headline because it corresponds to the "how much do educated countries transmit to their children" test the paper sets up.

8.7 Education vs. GDP as Predictors of Attainment (Prediction 2 and Prediction 6)

Where it matters most — countries still building their education systems — education predicts the next generation's attainment 3× better than income does. This is the test for Prediction 2 (education persists across generations, income does not) and Prediction 6 ($\beta_g > 1$ at low baselines).

Among countries where the expansion is still active — parental completion below 30%, where most of the remaining 20% of humanity sits — a one-percentage-point rise in parental lower secondary completion raises child completion one generation later by **1.38 pp** (country-FE, clustered SE 0.08, $t = 16.4$; Table 3, column 1). Adding contemporaneous log GDP per capita to the same regression leaves the parental coefficient essentially unchanged (1.27, SE 0.06) and attaches only a weakly significant 5.02 pp per log-unit of income (Table 3, column 2) — on a percent-for-log-point metric, education's effect is estimated nearly five times more precisely than GDP's on the same sample. Allowing curvature (column 3) reveals diminishing returns and raises the linear coefficient to 2.04: the linear specification in column 1 understates the effect at low baselines. The two-way fixed-effects column (column 4) shows the coefficient attenuates to 1.57 but remains significant at $p < 0.01$, unlike the full-panel collapse discussed below. Within-country explained variance reinforces the point: education $R^2 = 0.70$ vs GDP $R^2 = 0.21$ at the same cutoff (single-regressor fits; Table 5). Education's coefficient leads GDP's at every cutoff from 10% to 90%, with R^2 ratios between 2.0× and 3.3×.[6]

Table 3. Country fixed-effects regressions of child lower secondary completion on parental education with increasing controls. Dependent variable in every column: child lower secondary completion (percent, 20–24 cohort, both sexes, year t). Sample restricted to country-years with parental completion below 30% at year $t - 25$ (active expansion); identical

[6] See `scripts/residualization/by_gdp_cutoff.py` for the full coefficient, SE, and t-statistic table at every cutoff.

629 country-years / 105 countries across all four columns. Produced by `scripts/tables/table_1_stepwise.py`.

	(1)	(2)	(3)	(4)
Parent edu ($t-25$)	1.376***	1.270***	2.039***	1.570***
	(0.083)	(0.059)	(0.273)	(0.409)
Parent edu^2	—	—	-0.026***	-0.019*
			(0.008)	(0.010)
Log GDP (t)	—	5.02*	4.57*	3.99
		(3.04)	(2.75)	(3.00)
Country FE	Yes	Yes	Yes	Yes
Year FE	—	—	—	Yes
R^2 (within)	0.699	0.724	0.746	0.717
Observations	629	629	629	629
Countries	105	105	105	105

Country-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. *Column (1) is the headline one-way FE. Column (2) adds contemporaneous log GDP to test the bad-control concern; the parental coefficient barely moves (1.376 → 1.270). Column (3) adds parent^2: the negative coefficient implies diminishing returns (β_g compression near the ceiling). Column (4) adds year FE; the parental coefficient attenuates to 1.570 but stays significant at $p < 0.01$ — the full-panel 2WFE collapse in Table A1 does not occur here because restricting to the active-expansion subsample minimises the staggered-adoption pathology that drives it. GDP is excluded from the main regression as a bad control; Table 10 residualises it before the comparison.*

Table 5. Education versus GDP as predictors of child lower secondary completion, across three representative cutoffs. Each row is a single-regressor within-country fit. The education/GDP R^2 ratio holds throughout the 10%–90% range (full 10-row sweep in Table A3). Produced by `scripts/residualization/by_gdp_cutoff.py`.

Cutoff	Edu β	Edu R^2	GDP β	GDP R^2	Edu/GDP R^2
<10%	2.022	0.590	15.535	0.296	2.0×

Cutoff	Edu β	Edu R^2	GDP β	GDP R^2	Edu/GDP R^2
<50%	1.053	0.697	17.311	0.247	2.8×
Full	0.483	0.533	15.787	0.245	2.2×

Country fixed effects; clustered standard errors. "Cutoff" restricts the sample to country-years with parental lower secondary completion below the threshold. The education/GDP R^2 ratio never reverses across the full 10%–90% sweep; the ordering confirms Prediction 6.

Not driven by one region or one era. Fifteen subsample re-estimations of Column 1 (six regions, two eras, three within-sample GDP terciles) all yield positive coefficients significant at $p < 0.01$ (Section 8.10, Table 12); the headline is not an artefact of East Asia, the post-Cold-War period, or any income band.

The decisive test is the shape across lags (Figure 5; Section 8.8).

Each generation gains more than it inherits (Prediction 6). The generational amplification coefficient β_g measures how much more educated the children's generation is relative to the parents'.[7] If children merely inherited their parents' education level, β_g would be 1 or less. In fact it exceeds 1: at the <20% cutoff, β_g=1.032 (t=5.8); at <10%, β_g=1.019 (Table A1). The state adds reach on top of what households transmit.

The amplification is strongest where baselines are lowest: among countries below 20% completion, the next generation's gain is 2.86× the parents' level (education explains 70% of the variation); below 50%, 1.83× (explains 79%); below 90%, 1.24× (explains 77%).[8] The all-country panel (1900–2015) produces β_g=1.04 — stronger, not weaker, than the post-1975 panel. The long-run number pulls slightly higher because it reaches back into the low-baseline era, where state reach adds more on top of what the home niche alone carries; the post-1975 panel is weighted toward countries already partway up the S-curve, where ceiling compression is starting to bite.

[7] Three related quantities appear below: the pooled coefficient, subsample gain-ratios at low baselines, and within-country sliding-window estimates (Figure 4). All exceed 1 at low baselines and approach zero near the ceiling.

[8] See `scripts/robustness/beta_by_ceiling_cutoff.py`.

The S-curve within countries. Within individual countries, β_g varies systematically with baseline education level (Figure 2). The United States shows β_g=1.9 when its baseline was 13% (early 20th century), declining to β_g=0.08 when its baseline reached 92%. Korea shows β_g=6.5 at 1% baseline, declining through 3.6, 1.8, and 0.2 as its baseline rose to 59%. Taiwan follows a nearly identical trajectory (β_g=5.1 at 1.2%). The Philippines (Section 15) shows a flatter curve: β_g=4.4 at 1.5% declining more slowly to 0.4 at 49%. The S-curve structure is robust across all countries with sufficient historical depth. Bangladesh shows β_g≈2 throughout its observed trajectory (1–18% baseline), consistent with a country still in the steep part of the S-curve. India follows a similar pattern at β_g≈1.6 through 28% baseline.

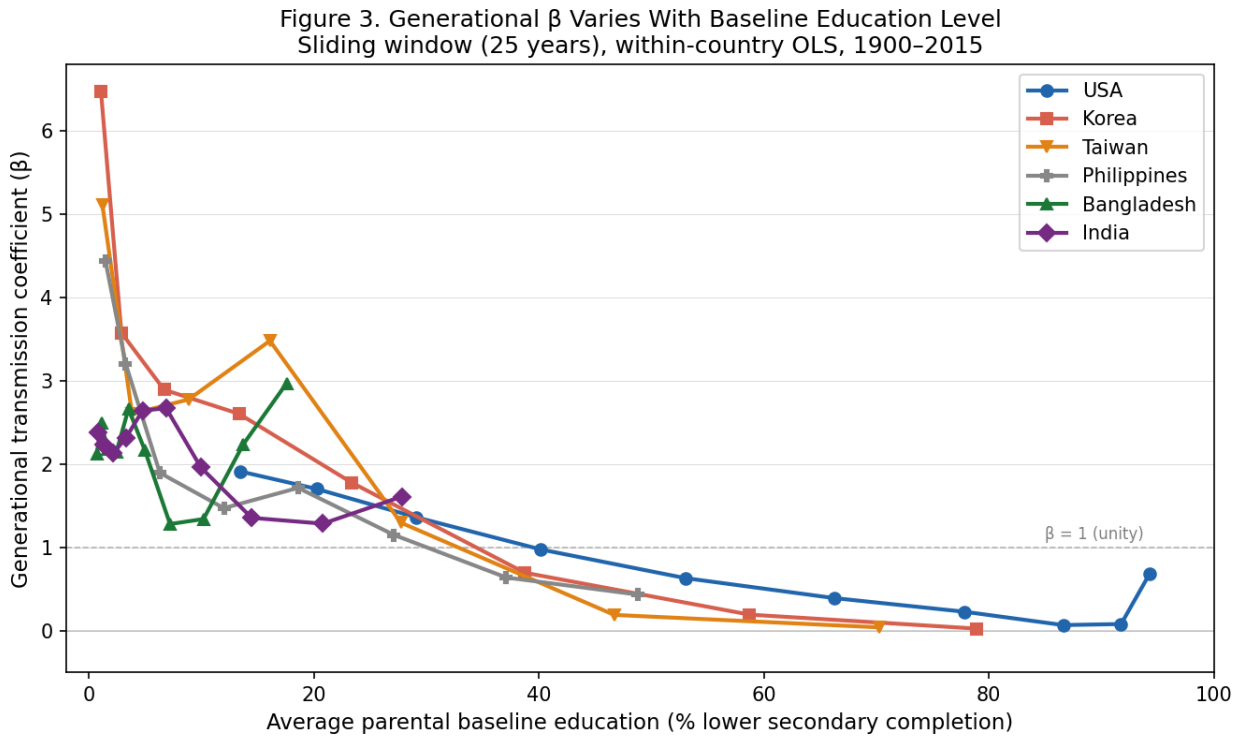

Figure 4: Generational β_g varies systematically with baseline education level. Each point: 25-year sliding window (6 child cohorts) within one country, against the window's parental baseline. $\beta_g > 1$ at low baselines reflects state reach; $\beta_g \to 0$ at high baselines reflects ceiling compression. *(WCDE v3 long-run cohort reconstruction 1875–2015, broader than the 28-country regression panel in Table A7. See* `scripts/figures/beta_vs_baseline.py`*.)*

β_g approaching zero at high baselines is ceiling compression, not diminishing returns (distinct from Lutz & Kebede's cross-level finding that education's effect on life expectancy shows no diminishing returns from primary

through tertiary). Non-depreciation holds universally — the Cambodia test (Section 10.2).

8.8 Education Predicts Development Outcomes 25 Years Forward (Prediction 1)

Prediction 1 says outcomes at $T+25$ follow education at T with a one-generation lag. Knowing how educated a country's parents were predicts how long their children will live, how many children they will have, and how much they will earn one generation later — the panel shows the signature the mechanism specifies.

Table 7. Fixed effects regressions: development outcomes one generation forward on education, controlling for initial outcomes. Panel A reports education predicting development outcomes forward; Panel B reports the symmetric reverse regression (GDP predicting education forward) to demonstrate mutual endogeneity.

Panel A: Education(T) → development outcomes(T+25). n=927 (179 countries) for GDP row; n=1,295 (185 countries) for LE and TFR rows.

Outcome (one generation forward)	Education β	Initial outcome β	R^2 (within)
Log GDP per capita	+0.011***	+0.182***	0.387
Life expectancy (years)	+0.109***	+0.301***	0.382
Total fertility rate	−0.032***	+0.039*	0.362

The reverse regression tests whether GDP predicts future education with comparable strength — it does not once parental education is controlled.

Panel B: Log GDP(T) → education(T+25) — reverse regression

Outcome (one generation forward)	Log GDP β	Initial edu β	R^2 (within)
Lower secondary completion	+14.42***	—	0.263
Lower secondary completion	+2.865***	+0.485***	0.495

Country fixed effects; country-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. *Panel A: n=927 (179 countries) for GDP row; n=1,295 (185 countries) for LE and TFR rows (GDP coverage is sparser). Panel B: n=927 for GDP-education matched obs.*

Income alone explains 27% of child education variation across the full panel (Panel B row 1). But add the parents' education to the same model and income's apparent contribution collapses — the model's explanatory power jumps from 27% to 50%, almost entirely from education, not income.

Each 1-percentage-point gain in lower secondary completion predicts, one generation forward: 1.1% higher GDP per capita, 0.109 more years of life, and 0.032 fewer children per woman — all from within-country change over time, controlling for each country's starting point.

The education-vs-GDP predictive gap holds for child education at every cutoff tested.[9]

For life expectancy the gap is even wider: among countries below 30% completion, education explains 33% of the variation vs 2% for GDP — a 14× gap. Among countries below 10%, education explains 51% while GDP explains 1.6%. Even that small GDP signal vanishes once education's contribution is stripped out (Section 8.9). The same pattern holds for the other outcomes: at completion below 50%, fertility shows a 32× edu-over-GDP ratio (edu R^2=0.359, GDP R^2=0.011); at completion below 10%, log under-5 mortality shows a 39× ratio (edu R^2=0.570, GDP R^2=0.015).[10]

[9] See `scripts/residualization/by_gdp_cutoff.py`.
[10] See `scripts/residualization/cutoff_all_outcomes.py`.

Lag-decay shape (Prediction 2)

Education's signal survives four generations across every outcome tested. Income's collapses within one.

Each lag corresponds to a generational relationship, and the relationship names what the number actually measures. The same pattern holds for all four development outcomes the paper cares about:

Lag	Gens	Relationship	LE	TFR	U-5 mort.	Child edu
0	—	contemporary	0.743	0.817	0.814	—
25	1	parent → child	0.597	0.693	0.820	0.722
50	2	grandparent → grandchild	0.411	0.452	0.664	0.388
75	3	great-grandparent → great-grandchild	0.284	0.258	0.502	0.227
100	4	g-great-gp → g-great-grandchild	0.223	0.134	0.395	0.152

Standardized within-country $|\beta|$: outcome on lower-secondary completion at T−lag. Each series demeaned by country mean and standardized by pooled within-country SD. Signs: + for LE and child education, − for TFR and U-5. Under-5 mortality is logged; child education at lag 0 is identity and not shown. 142 countries, outcome panel 1960–2015. All coefficients significant at $p < 0.001$; minimum $|t|$ at lag 100 is 10.9 (TFR).

Every outcome's coefficient survives four generations. Under-5 mortality carries the deepest signature: $|\beta| = 0.395$ at century depth with $|t| = 34.8$, roughly twice the strength of any other outcome and the only one where $|\beta|$ exceeds 0.3 after four intervening households. Life expectancy, fertility, and child education show the same smooth monotonic decay, all still significant at 100 years ($|t|$ of 18.4, 10.9, and 12.4 respectively). Roughly a century of generational transmission running through three intermediate households leaves a statistically detectable trace in every dimension of human development the paper defines. (R^2 values show the same monotonic decay pattern but are not directly comparable across outcomes because they conflate effect size with predictor variance and outcome noise; see `checkin/outcomes_r2_by_lag.json` for the R^2 decay curves.)

No other development input has comparable reach. Income's R^2 is already fading by lag 20–25, and it cannot be tested beyond that — not because the data is missing but by construction. Pre-literate subsistence societies had uni-

formly low income (~$400–600 per capita, constant 2017 USD), so there is no cross-country income variation to test before literate populations existed. The uniformity is not offered as evidence for the mechanism; it is a reminder of what the mechanism had to produce before income could become a variable at all. Institutions, quality scores, and provision fail the same generational-depth test: none survives at the depth education reaches. This is not a separate finding for each alternative — it is a single methodological principle derived from the mechanism itself. Backfilling pre-1960 income at $500 does not rescue income's predictive power for any of the four outcomes: the backfilled-GDP coefficient crosses zero and loses significance by lag 55–60 for life expectancy, fertility, log under-5 mortality, and child education (within-R^2 drops below 0.01 at the same points), while education's coefficient remains significantly signed through lag 100.[11]

The compounding-generations signature (Prediction 2)

The mechanism in Section 5 predicts that where a country first crosses the 10% lower secondary completion threshold, the effect on child completion should grow with the generational window, not plateau at $T + 25$. Parents who crossed raise children more likely to cross, who raise children still more likely to cross. The Callaway-Sant'Anna event study confirms this shape directly: +1.3 percentage points immediately, +6.6 at ten years, +14.9 at twenty-five years, +21.4 at thirty-five years. Pre-treatment coefficients are indistinguishable from zero — countries that will expand and countries that will not follow identical trajectories before the crossing. The signature is the monotonic compounding through generations after the crossing, not the cross-sectional level at any single lag.

The grandparent channel

The lag-decay results show that education predicts across multiple generations. A sharper test: does the grandparent's education predict outcomes *independently*, after controlling for the parent's own education?

For fertility in the full panel, the answer is no — grandparent education adds nothing once the parent's is controlled (β_{gp}=+0.001, p=0.80). The parent ab-

[11] See `scripts/robustness/backfill_all_outcomes.py`.

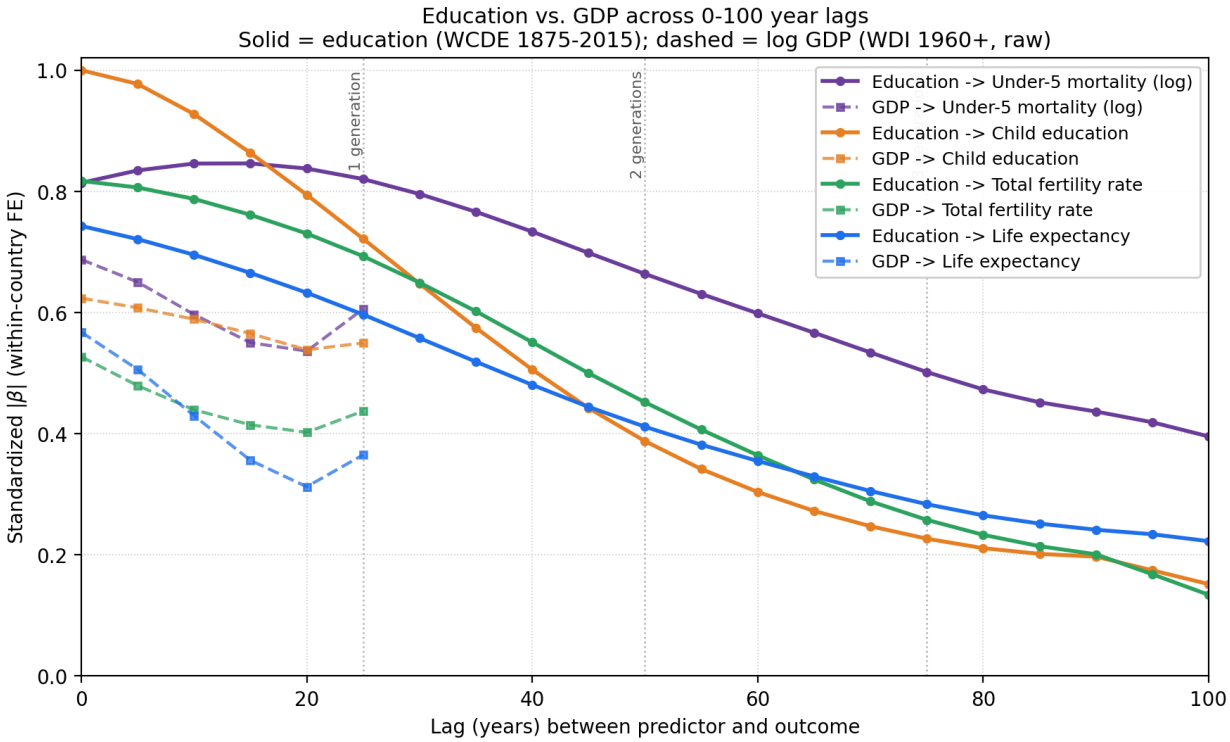

Figure 5: Education vs. GDP effect size across lag lengths (within-country FE). Solid: standardized $|\beta|$ for lower secondary completion predicting four development outcomes at lags 0–100 years. Dashed: log GDP predicting the same outcomes. Education beats GDP at every outcome and every lag. GDP curves terminate at lag 25 (raw WDI coverage; backfilled GDP loses significance by lag 55–60, see §A.2). Generational horizons (1–3) marked. *(WB WDI and WCDE v3; see* `scripts/robustness/lag_coefficients.py`*.)*

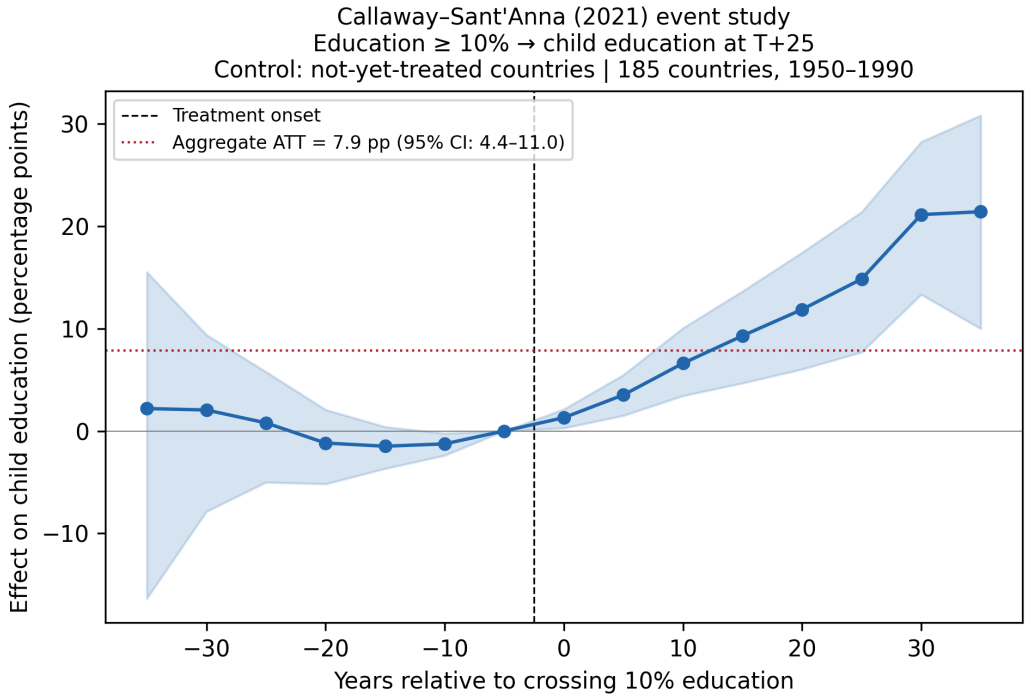

Figure 6: Generational compounding after the 10% threshold crossing. Treatment: country crosses 10% lower secondary completion. Outcome: child education at $T + 25$. Control group: countries not yet treated at each comparison point. Shaded band: 95% confidence interval (500-replication clustered bootstrap). Pre-treatment coefficients near zero; post-treatment effects grow monotonically from $+1.3$ to $+21.4$ percentage points over 35 years — the compounding-generations signature the mechanism in Section 5 predicts. Callaway-Sant'Anna (2021) estimator. Full decomposition and staggered-adoption diagnostics in Appendix A.1. *(See `scripts/robustness/callaway_santanna.py`.)*

sorbs the grandparent's entire contribution to fertility. But restrict to countries where education is still scarce (parent completion below 50%) and the grandparent effect emerges: β_{gp}=−0.059 (p=0.0003), nearly twice the parent's own coefficient (β_p=−0.033). R^2 jumps from 0.39 to 0.46. Where education is scarce, having an educated grandparent independently reduces fertility beyond what the parent's own education predicts. The effect is not gendered: grandfather and grandmother education produce comparable coefficients in the low-education subsample (β=−0.054 and −0.050 respectively), consistent with oblique transmission — in a village where education is rare, any educated elder is a visible exemplar.

For life expectancy, the grandparent effect is significant across all education levels (β_{gp}=+0.070, p<0.0001), adding 3.6 percentage points of R^2 beyond the parent alone. Roughly half the grandparent's total effect on LE runs through the parent; half is direct. For grandchild education, the grandparent coefficient is β_{gp}=+0.271 (p<0.0001), with an R^2 gain of 5.2 points.

The largest grandparent effect is on child survival. Regressing log under-5 mortality on both generations' lower secondary completion in the full panel, the grandparent coefficient is larger in absolute terms than the parent's (β_{gp}=−0.018, β_p=−0.016; $|\beta_{gp}/\beta_p|$=1.16, both p<0.0001). Within-R^2 rises from 0.377 to 0.561 once grandparent education is added — a gain of 18.4 percentage points, roughly five times the life-expectancy gain and the largest grandparent contribution of any development outcome the paper tracks. The grandmother's cohort carries an independent signal for child survival that, in magnitude, matches the mother's own education — the signature expected if the mechanism is kin transmission of child-care practice (sanitation, feeding, vaccination) within the household and kin radius.[12]

The pattern is what the generational gate predicts (Section 8.5): at low baselines each educated person in the household adds independent weight; at high baselines the parent has internalised everything the grandparent could offer. The grandparent effect on fertility is real, but it operates precisely where it matters most — the steepest part of the transition.[13]

This generates a falsifiable prediction distinguishing the kin/community-

[12] See `scripts/robustness/grandparent_effect_all_outcomes.py`.
[13] See `scripts/robustness/grandparent_effect.py`.

radius account from a pure household account. Any educated elder inside the child's daily kin/community radius — co-resident grandparent, village elder, aunt, older cousin — should raise grandchild outcomes beyond what the parent's education alone predicts. Equivalent elders present only on paper (deceased, non-resident, absent from daily life) should not. The channel runs through physical presence and daily interaction at Radii 1–2 (household and kin/identity; Section 5.2), not through inherited status. That the grandfather and grandmother coefficients come out nearly equal in the low-education subsample (β=−0.054 and −0.050) is consistent with the same reading: any educated elder works, not specifically the grandparent of the same sex. DHS co-residence data can test the presence-versus-absence prediction directly.

8.9 GDP Has No Independent Effect (Prediction 4)

Prediction 4 says income carries no independent predictive power once education's contribution is removed. Take each country's income and residualise out the part that education produced; use only what remains to predict development outcomes. The residual predicts nothing — with one bounded exception in post-2000 child mortality, examined below.

Table 10. Residualized GDP analysis: education vs. raw GDP vs. GDP after removing education's contribution. Entry-cohort design (entry ≥ 10%, ceiling ≤ 90%), country fixed effects, lower secondary completion, T=1960–1990, lag=25 years.

Outcome (T+25)	Edu R^2	Raw income R^2	Income R^2 (edu removed)	p-value
Life expectancy	0.472	0.179	**0.003**	0.56
Total fertility rate	0.478	0.175	**0.000**	0.87
Child education	0.524	0.303	**0.008**	0.31
Under-5 mortality	0.284	0.046	**0.023**	0.11

"Income R^2 (edu removed)" is the predictive power of income after stripping out education's contribution. Country-clustered SEs. n=822 (152 countries) for LE and TFR; n=856 (157 countries) for child education; n=787 (147 countries) for under-5 mortality. See `scripts/residualization/education_vs_gdp.py`,

`scripts/residualization/education_vs_tfr.py`, `scripts/residualization/child_education.py`.

Reading the table directly: a one-percentage-point rise in lower secondary completion at T predicts a one-generation-later gain of $+0.11$ years of life, -0.032 births per woman, and a -2.7 percentage-point drop in log under-five mortality; the same-magnitude response to residualised income (income with education's contribution stripped out) is statistically indistinguishable from zero in every row. The R^2 columns summarise the same thing at the country-panel level: education explains a non-trivial share of within-country variation across all four outcomes (0.284 to 0.524), while the leftover-income R^2 is at most 0.023 in any row, and the coefficient carrying it fails conventional significance ($p \geq 0.11$ in all four). Education is doing essentially all the predictive work.

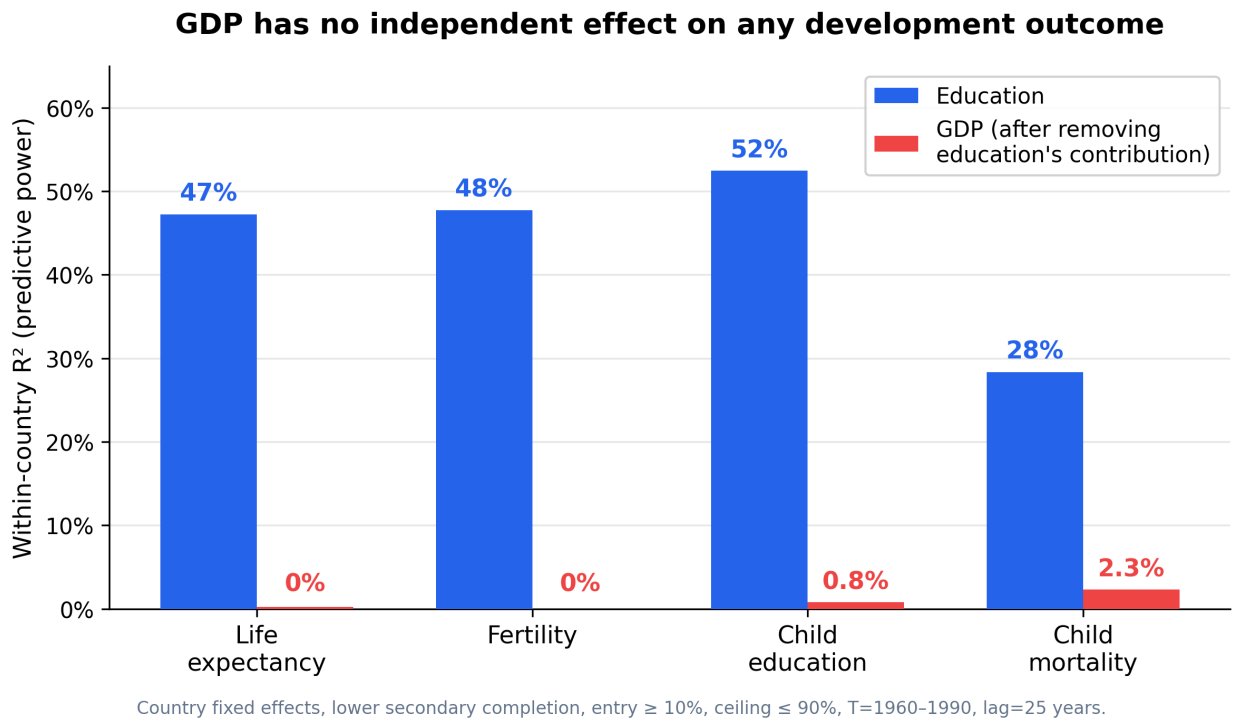

Figure 7: How much does education vs. income predict development one generation forward? Blue bars: education's predictive power. Red bars: income's predictive power *after stripping out what education produced*. Income predicts nothing on its own. Entry-cohort design (entry $\geq$ 10%, ceiling $\leq$ 90%), country fixed effects, lower secondary completion, T=1960–1990, lag=25 years. *(See `scripts/figures/edu_vs_resid_gdp_bars.py`)*

Child mortality: one bounded exception. Child mortality is the only out-

come where residualized income shows a marginal independent effect. Before 2000, leftover income explains 0.9% (not statistically significant); after 2000, 2.7% (p=0.03). The effect appears only at intermediate education levels and vanishes below 20% completion in any period. The paper's claim does not require every residual to be education: income, through external provision (GAVI founded 2000, vaccine delivery through global supply chains), can produce a small independent effect on child mortality in the post-2000 window where such provision scaled. Whether the surviving signal is genuinely income or is global education funded by donor surplus — vaccines developed, delivered, and paid for by already-developed populations — the channel is fragile by construction: it runs on donor altruism, not self-interest (Section 5.2, outer radius), and development assistance for health is already declining.[14]

Composition by education level

Primary education drives fertility decline; deeper education drives life expectancy gains.

The *composition* of the effect varies by level. Primary education is the strongest predictor of fertility decline: its coefficient on TFR at $T+25$ is larger than lower or upper secondary's, and the resulting within-country fit is tightest at this level (R^2=0.65 vs 0.48 for lower secondary, 0.32 for upper secondary). The first educated generation — the categorical jump from no schooling to literacy — is where fertility falls hardest, universally, across every country in the dataset regardless of culture, religion, or economic structure. Caldwell (1976) predicts fertility decline when children shift from economic assets to costs; if correct, the steepest decline should track economic modernisation, not primary education. It tracks primary education, in agrarian and industrial contexts alike. The mechanism is not economic calculation; it is cognitive capacity. Life expectancy gains, by contrast, accumulate with deeper education: the upper-secondary coefficient dominates, and the within-country fit is correspondingly higher than at primary (R^2=0.48 vs 0.41). This is the structure that Lutz & Kebede (2018) identified when redrawing the income–mortality curve with education on the x-axis: the fertility transition is triggered by the first level, while the education–health relationship tightens at higher levels.

[14] See `scripts/robustness/u5mr_by_edu_level.py`.

Residualized GDP R^2 never exceeds 0.03 for life expectancy and is zero for fertility, at every level.

The education–life expectancy relationship shows no diminishing returns — linear from primary through tertiary (Lutz & Kebede 2018). Among 74 countries with lower-secondary completion above 85% in 2010, college completion carries a within-group correlation of r=0.45 with life expectancy — a 5.7-year gradient between lowest and highest college-completion quartiles among countries that have already solved the lower-secondary problem.[15]

Robustness and additional tests

Residualized GDP R^2 stays below 0.003 for life expectancy and fertility across all lag lengths tested; under-5 mortality reaches 0.019 (Table A5). The confidence intervals do not overlap: education R^2 [0.33, 0.59] vs residualized GDP R^2 [0.00, 0.03] (500-replication bootstrap). An alternative estimation method designed specifically to correct for bias in short panels (Anderson-Hsiao instrumental variables) produces results within 3–10% of the main estimates.[16] Results are unchanged when time-varying institutional quality scores (Polity5 polity2) are added (Appendix 14.4). Female education produces stronger results than the aggregate on every outcome.

The parental income test asks a simple question: does family income predict child education independently, or only because richer parents tend to be more educated? Parental income alone predicts child education moderately (β=14.4, R^2=0.263). But add parental education to the same regression and income's coefficient collapses by 80% to β=2.9 (p=0.16), adding almost nothing (R^2=0.006) beyond education alone. Education's coefficient barely moves (β=0.489 from 0.540) while income's collapses. Income transmits through education, not vice versa.

The Asian Financial Crisis (1997–98) tests Prediction 3: GDP collapsed across five countries (Indonesia −14.5%, Thailand −8.8%, Malaysia −9.6%, Philippines −3.0%, Korea −6%). Lower secondary completion continued uninterrupted in all five; Indonesia gained 5.4pp while losing 14.5% of GDP. Income was destroyed; the educated population was not; development continued (see

[15] See `scripts/cases/college_le_gradient.py`.
[16] See `scripts/robustness/robustness_tests.py`.

`scripts/robustness/asian_financial_crisis.py`).

Once education is included, initial fertility barely predicts subsequent fertility one generation forward (Table 7, Panel A, row 3: initial outcome β=0.039). Demographic momentum does not drive the transition — educated women's decisions do.

8.10 Universality Across Subsamples (Prediction 5)

Prediction 5 says the mechanism is universal: it appears in every region, every era, and every income band, not only in the East Asian cases on which much of the growth literature has concentrated. Two tests, panel and residual.

Panel, fifteen subsamples. Re-estimating Column 1 of Table 3 on six regions, two eras, and three within-sample GDP terciles produces fifteen positive coefficients, every one significant at $p < 0.01$ (Table 12). Every β exceeds unity; Sub-Saharan Africa, where most of the remaining 20% of humanity sits, tracks the full-sample headline. The era and tercile splits rule out post-Cold-War and income-band artefacts.

Table 12. Subgroup robustness of the headline specification. Column 1 of Table 3 (child lower-sec $\sim$ parental, country FE, clustered SE) re-estimated on regional, temporal, and income-tier subsamples of the 629-obs / 105-country active-expansion sample. Produced by `scripts/tables/table_1_subgroups.py`.

Subsample	β	SE	R^2	N	Countries
Full (headline)	**1.376*****	0.083	0.699	629	105
By region					
Sub-Saharan Africa	1.220***	0.109	0.702	301	40
Middle East & N. Africa	1.145***	0.176	0.745	61	12
South Asia	2.444***	0.515	0.720	57	8
East Asia & Pacific	1.491***	0.097	0.823	84	17
Latin America & Caribbean	1.272***	0.091	0.829	112	23
Europe & North America	2.057***	0.156	0.762	14	5
By child-cohort period					
Pre-1990 (1975–1985)	1.364***	0.172	0.575	279	99

Subsample	β	SE	R^2	N	Countries
1990–2015	1.332***	0.116	0.621	338	72
By GDP tercile (within sample)					
Low	1.295***	0.153	0.663	208	33
Middle	1.276***	0.091	0.761	205	44
High	1.393***	0.137	0.667	205	53

Country-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. *All fifteen subsamples reject the null at* $p < 0.01$*; every* β *exceeds unity. Headline row repeats Table 3 Column 1 for reference.*

Residuals, eight over-performers. Some countries are educating children far beyond what their parents' generation would predict. The 2015 country-FE residuals from the headline specification identify eight: they are uniformly poor.

Table 14. Countries delivering education above generational baseline, 2015 FE residuals. The residual measures how far a country exceeded its own historical trend.

Country	FE residual above baseline	GDP per capita (2015)
Maldives	+34.9 pp	$9,645
Cape Verde	+26.3 pp	$3,415
Bhutan	+26.1 pp	$2,954
Tunisia	+25.5 pp	$4,015
Nepal	+17.8 pp	$876
Viet Nam	+16.0 pp	$2,578
Bangladesh	+15.8 pp	$1,224
India	+14.1 pp	$1,584

Nepal at $876 (2015), Bangladesh at $1,224, Vietnam at $2,578. Income is not the prerequisite for educational over-performance.

A full robustness battery — alternative cutoffs (full 10-row sweep), alternative lag lengths (15–30 years), specification robustness (PPML, log outcomes,

period-length aggregation, balanced subpanel, within-year cross-cohort, Wooldridge strict-exogeneity), data-source robustness (Barro-Lee replication, post-1970 subset), and development-threshold robustness — appears in Appendix A.2. None reverses the ordering, and no check reduces the education/GDP R^2 ratio below 2.0×.

9. Data Credibility: The Goskomstat Anomaly

The paper's empirical engine is lower-secondary completion in the WCDE v3 reconstruction (Lutz et al. 2018), chosen because it reaches back to 1875 on a consistent definition and passes the phenotype tests laid out in Chapter 8. On the 185-country panel, WCDE disagrees meaningfully with Barro-Lee v3.0 (Lee & Lee 2016) for exactly fifteen countries: the Soviet republics. WCDE reports Kazakhstan at 94% lower-secondary completion in 1970; Barro-Lee reports 49% reached-secondary for the age 25–34 cohort. WCDE reports Latvia at 99%; Barro-Lee reports 57%. WCDE reports Russia at 95%; Barro-Lee reports 73%. Fifteen republics, a mean gap near 40 percentage points, all in the same direction.

This chapter asks what the mechanism says about that disagreement. The answer is unusual and worth stating plainly. *The reported Soviet education numbers fail the phenotype test that the rest of the panel passes.* A country at 95% lower-secondary completion in 1970 should, by the mechanism, produce a fertility trajectory over the following generation indistinguishable from Spain's and a child- mortality trajectory indistinguishable from South Korea's. The Central Asian republics did neither. Their TFR trajectories sat on top of Iran's and Turkey's for six decades. Their under-5 mortality trajectories were overtaken by Iran's between 1990 and 2010 — Iran, whose reported lower-secondary completion was 22% in 1970 against Kazakhstan's 94%. If the reported gap had been real, this reversal would be impossible.

The chapter treats the Soviet reporting inflation not as a footnote but as a diagnostic: a natural experiment run in reverse, where the input side is corrupted and the phenotype reveals where. The diagnostic yields three payoffs. First, a credibility case for the panel: all headline results survive exclusion of the fifteen republics, and in fact strengthen. Second, a reconciliation with the Hanushek–Woessmann "knowledge capital" line (2008, 2012): the HLO test-

score measure, which the field has read as an alternative to completion-based education, turns out to be a multi-generation integrated output of completion itself. Third, a sharpened thesis on what schooling transmits. **Hollow education is not education — but exposure still matters for TFR.** Fertility is a norm decision that flips at the completion threshold even where learning is shallow; child survival and adult longevity require cognitive depth and do not flip where credentials are hollow.

9.1 The Anomaly

Of the 154 countries that have crossed the developmental threshold by 2022, 28 were post-socialist — 15 USSR republics plus 13 Warsaw Pact and Yugoslav successors. The post-socialist crossers follow a pattern nothing else in the panel matches. Among market-economy crossers, the median country reached 90% lower-secondary completion 18 years *before* crossing the developmental threshold. Among post-socialist crossers, the median country reached 90% at the crossing year. Among the USSR core, Turkmenistan crossed 52 years after the reported date at which it reached 90%. Either these republics had reached 90% decades before their fertility and longevity trajectories responded — a 50-year mechanism gap with no precedent — or the 90% was not what it claimed to be.

The rest of the chapter tests the second hypothesis. Under the paper's mechanism, reported education at year T should produce phenotypic response by $T+25$. Where it does not, either the mechanism is wrong or the reported education is wrong. The mechanism passes for 171 of 185 countries in the panel; the 14 where it fails (Uzbekistan drops out of the panel on data coverage) share a common origin: one statistical office.

9.2 What WCDE Reports vs What Was Observable

Three ground-truth checks against contemporaneous non-Soviet neighbors — Iran, Turkey, Afghanistan, Pakistan — with comparable pre-1917 educational baselines:

Gender parity in 1970. Central Asia reports a mean female-minus-male lower-secondary completion gap of -2.7 pp (near parity). Iran, Turkey,

Afghanistan, and Pakistan report a mean gap of -15.6 pp, with several countries at -11 to -18 pp. Near-parity in 1970 rural Central Asia is not consistent with the family structure and labor patterns documented in ethnographic sources for the same decade. The Baltics' reported near-parity is plausible (pre-1940 European educational infrastructure); the Central Asian figure is not.

Primary minus lower-secondary. By construction, primary completion must exceed lower-secondary completion: one cannot complete the latter without the former. In 1980, non-Soviet neighbors show a mean primary-minus-lsec gap of $+21.0$ pp — a fifth of the cohort completes primary without continuing to lower-secondary. Central Asia shows $+1.6$ pp: virtually every child who completes primary is reported as completing lower-secondary as well. This holds across individual Central Asian republics. Near-complete continuation is not what normal school systems produce; it is what uniform reporting produces.

Raw 1970 levels. Kazakhstan 94%, Turkmenistan 95%, against Iran 22%, Turkey 22%, Pakistan 16%, Afghanistan 6%. A 70-percentage-point gap from comparable pre-1917 baselines, built in half a century, is not what rural agricultural populations produced in this period anywhere else on the planet. Later cognitive evidence is consistent: Kyrgyzstan's 2009 PISA score of 350 sits at the low end of the global distribution; Turkmenistan, Uzbekistan, and Tajikistan do not participate in international assessments; Baltic PISA scores at 500–535 are consistent with their reported education and with separate pre-1940 educational infrastructure.

9.3 The Phenotype Test

The mechanism predicts that educational input at T produces phenotypic output at $T+25$. Where the predicted output does not appear, the input was not real. Figure 8 shows the test for under-5 mortality.

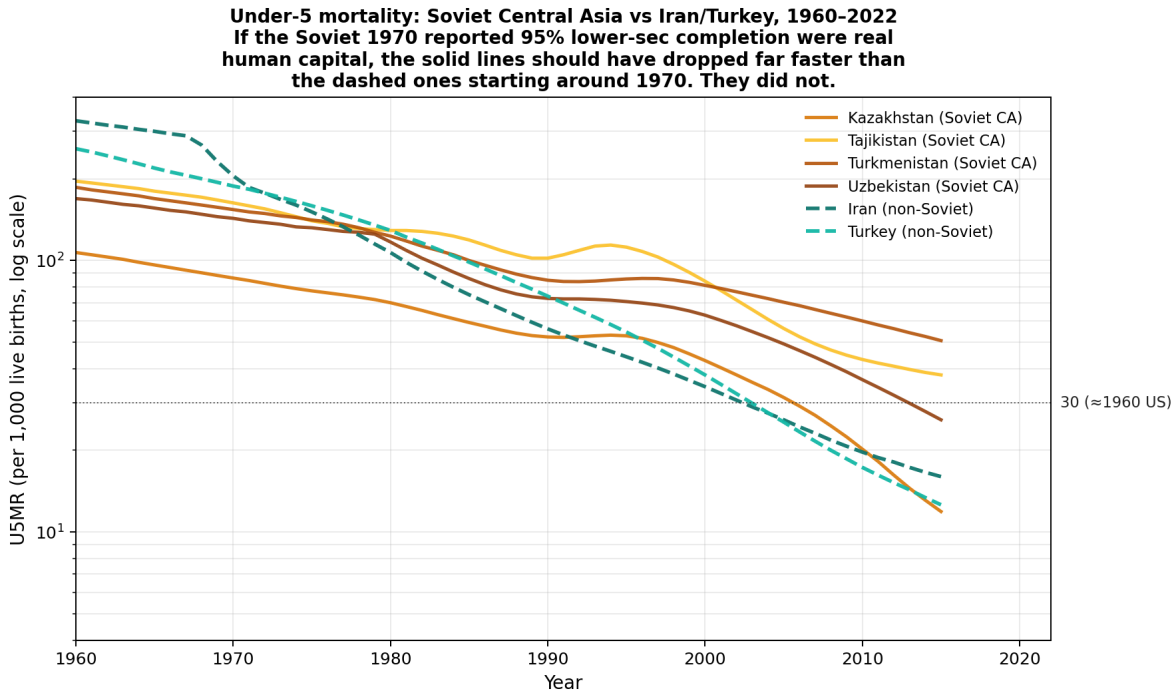

Figure 8: Under-5 mortality trajectories, 1960–2010. Central Asian and Slavic USSR republics (colored lines) start 1960 with large U5MR head starts over Iran (327 per 1,000) and Turkey (258). By 2010, Iran (20) has overtaken every Central Asian country except Kazakhstan. Under the mechanism, the country with four times the reported education in 1970 should dominate its neighbor permanently; no later catch-up should be able to reverse a head start built on five-times-larger secondary completion. The crossover says the head start was not real. *(Source: World Bank WDI. See* `scripts/post_soviet_u5mr_trajectories.py`*.)*

Iran 1960 → 2010: 327 → 20 per 1,000 (94% decline). Kazakhstan 1960 → 2010: 107 → 20 (81% decline). Turkey 2010 at 17 beats every Central Asian country except Kazakhstan. If the 1970 Soviet 95% had been real, Central Asia should have dominated Iran and Turkey permanently. Iran and Turkey are running on school systems that delivered actual cognitive skill to the mothers whose children survived.

TFR tells the same story more quietly. Central Asian fertility trajectories from 1960 onward are indistinguishable from Iran's and Turkey's. There is no 1991 kink — the trajectories run smoothly through Soviet dissolution as if the state had had no independent effect on the household decision. What moved was the same thing that moved in Iran: mothers becoming literate in their own dialect, one generation ahead of the crossing.

9.4 Barro-Lee's Partial Correction

Barro-Lee v3.0 is an independent reconstruction of the same series using different source surveys and interpolation. It is systematically more conservative than WCDE on Soviet-era reporting. The Goskomstat hypothesis predicts that switching measures should reduce the phenotype-fit bias, and it does — unevenly. Table 16 reports the mean phenotype residual (actual outcome minus model prediction given reported education) by subgroup, under a pooled OLS fit with year fixed effects trained on the complement of the contested cohort. A positive TFR residual means the subgroup had more children than its reported education predicted; a negative LE residual means it lived shorter; a positive log-U5MR residual means it lost more children. All three signs point the same direction: reported education was too high.

Table 16. Phenotype residual by subgroup under Barro-Lee reached-secondary (age 25–34), 1960–1990 snapshots. Bias reported in standard deviations of the non-contested fit. A subgroup "passes" when bias is indistinguishable from zero.

Subgroup	n	TFR (SDs)	LE (SDs)	log U5MR (SDs)
Central Asia + Caucasus (USSR)	16	+1.15	−1.79	+2.78
Slavic west (USSR)	8	+0.28	−1.93	+1.90
Baltics (USSR)	12	−0.01	−1.14	+0.71
Warsaw Pact (non-USSR)	20	−0.42	−0.49	+0.25
Yugoslavia + Albania	16	−0.59	+0.09	−0.15

Two things the table tells us. First, Goskomstat specifically: Warsaw Pact states had their own national statistical offices; Yugoslavia ran its own; both subgroups pass the phenotype test under Barro-Lee on all three outcomes. The 15 Soviet republics fail at LE and U5MR; the 8 Central Asian and Caucasus republics fail at TFR as well. The anomaly is not socialism. It is one statistical office.

Second, outcome specificity. Under the corrected measure, TFR fits for the Baltics and the Slavic west — their fertility decisions are in line with their real education. Where they fail — LE and U5MR — they fail in the direction of

shorter lives and more child deaths than the reported education should allow. The outcome partition is the first hint of the thesis this chapter converges on.

9.5 The Cohort Shadow

Figure 9 decomposes the USSR residuals by year. The LE panel carries the diagnostic: the residual plateaus in the 1970s near -8 years, widens through 2000 to a peak of -8.5 years, then begins to decay. This is not a 1990s transition-crisis spike, which would show as a narrow peak around 1995 with small residuals before and after. It is a cohort shadow.

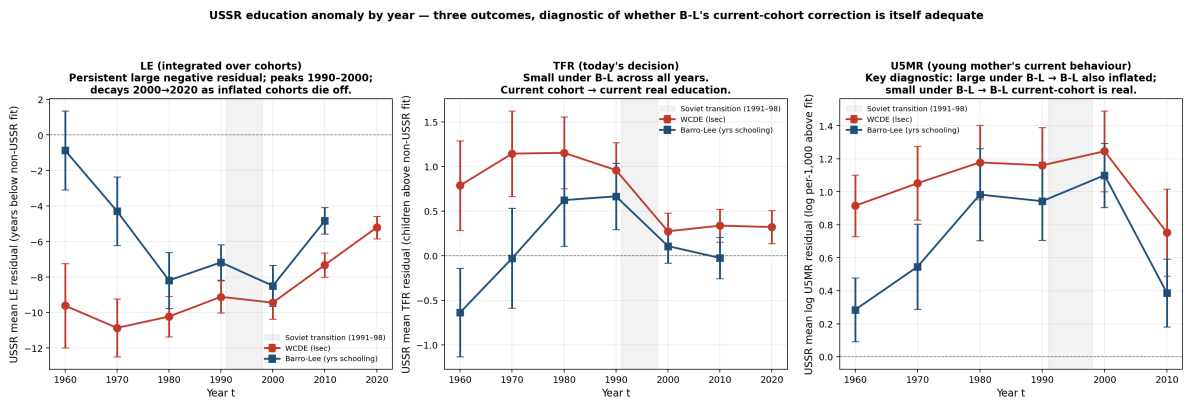

Figure 9: USSR mean phenotype residual by year, by measure. LE (left) shows a broad 1970–2000 plateau and decay after 2000 — the signature of inflated 1960–80 credential cohorts aging into peak-mortality years. TFR (middle) approaches zero under Barro-Lee at every year: fertility is a current decision, and Barro-Lee's current-cohort correction catches it. log U5MR (right) remains large under Barro-Lee for 1980–2000, because a young mother's health decisions depend on her real cognitive skill, not on her reported credential. *(Source: WCDE v3, Barro-Lee v3.0, World Bank WDI.* `scripts/ussr_residual_by_year.py`*.)*

The mechanics: LE in year t integrates mortality experience across all cohorts alive in year t. The 1960–1980 Soviet cohorts were reported as carrying ∼90% lower-secondary completion; under the phenotype test their real education looks more like 60–70%. Those cohorts, born 1935–55, reached peak-mortality ages (70–85) roughly 2005–2040. If their inflated-credential status did not carry real cognitive resources — information processing, health literacy, planning capacity — their mortality experience reflects real schooling, not reported schooling. The predicted signature is a large negative LE residual peaking around 2000–2015 and decaying through 2030. This is what we see.

TFR carries no such shadow under Barro-Lee. A 2000 fertility decision is made by a woman aged 20–30, educated in the 1980s–90s under progressively less-hollow Soviet and post-Soviet secondary systems, with Barro-Lee's current-cohort correction catching her actual schooling. Her fertility is in line with her real education by the time she makes the decision. This is why TFR corrects and LE does not.

U5MR tells the sharpest story. Under Barro-Lee reached-secondary the mean USSR log-U5MR residual is $+0.29$ in 1960, $+0.98$ in 1980, $+1.10$ in 2000 ($t = 5.7$), $+0.39$ in 2010. Peak U5MR bias is in 1980–2000 — the decades when the 1960–80 inflated-credential cohorts were themselves mothers of young children. The paper's claim on U5MR has always been that it loads on the mother's cognitive depth, not her completion certificate (Chapter 5). The Soviet case makes the claim visible: where the certificate said "secondary-completed" but the depth was not there, the child died anyway.

9.6 Hanushek's HLO Is Compounded Parental Transmission

The last forty years of development economics has increasingly treated test-score-based "learning outcomes" as the real human-capital variable, with completion treated as a noisy proxy. The Hanushek–Woessmann "knowledge capital" line makes this argument most explicitly: test scores predict growth, years do not, and where years and tests disagree the test is what matters.

The Soviet case is the first hard test of that line against the paper's generational mechanism. Over the contested cohort, HLO secondary scores and reported lower-secondary completion disagree catastrophically: Kyrgyzstan's 2009 HLO of 350 against 2010 reported completion of 99%; Albania's 412 against 96%. If HLO is the real human-capital signal and completion is noise, the paper's headline claims should survive stripping the contested cohort, which they do (Section 9.7). If completion is the real signal and HLO is downstream, HLO should be predicted by past completion at a lag consistent with parental transmission, not at lag zero.

Figure 10 resolves it. On a 77-country panel excluding USSR, HLO secondary today is regressed on lower-secondary completion at year $2010 - L$, for $L = 0$

to 60 in 5-year steps. The peak R^2 is not at lag 0 (current schooling, $R^2 = 0.504$). It is at lag 10 ($R^2 = 0.539$). Primary completion's peak sits at lag 25 ($R^2 = 0.549$) — exactly the parental window. At lag 60 (the great-grandparent generation), primary-completion R^2 is still 0.489.

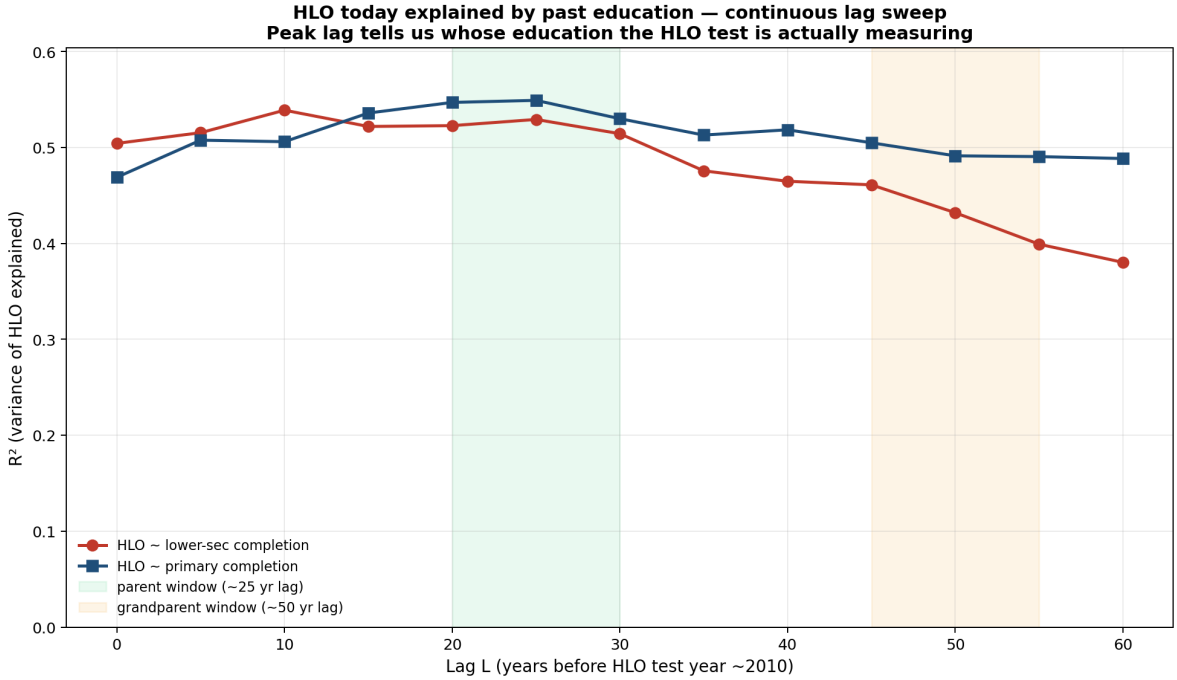

Figure 10: HLO secondary today regressed on WCDE completion at year $2010 - L$, for $L = 0, 5, \ldots, 60$. Peak lag for lower-secondary sits at $L = 10$; peak lag for primary sits at $L = 25$ (the parental window). R^2 at $L = 60$ remains near 0.49 for primary: great-grandparent primary completion explains nearly half the cross-country variance in today's teenagers' test scores. Current schooling (lag 0) is a worse predictor than either. *(Source: Angrist et al. 2021 for HLO, WCDE v3 for completion. `scripts/hlo_vs_education_lag_sweep.py`.)*

The direct test on the parental window: HLO secondary today regressed on lower-secondary completion in 1990 (the parent cohort's 20–24 education) gives $R^2 = 0.523$, $\beta = +1.94$ HLO points per percentage point, $t = 9.1$, correlation 0.72, on 77 countries with USSR excluded. **Fifty-two percent of cross-country variance in Hanushek's cognitive-skills measure is the paper's quantity measure, lagged one generation.** The residual variance is what school systems add *above* PT: China ($+111$ points), Viet Nam ($+111$), Singapore ($+93$), Portugal ($+76$), Japan, Finland, Korea — East Asian education reforms and select rich systems do deliver real cognitive depth above what the parental generation transmitted. The negative residuals — South Africa (-116), Ghana

(-108), Albania (-86), Montenegro (-72) — mark school systems that undershoot the parental baseline, a credential-inflation diagnostic independent of the Soviet case.

HLO is not an alternative to the paper's mechanism. It is the paper's mechanism measured three generations downstream.

The horse race Hanushek called for — completion versus test scores as predictors of TFR, U5MR, and LE — partitions exactly as the mechanism predicts. On the clean cross-country panel (USSR excluded, HLO averaged 2000–2017, completion at 2000): for TFR, quantity (primary completion) dominates at $\beta_z = -0.64$ ($t = -7.4$); adding quality contributes $\beta_z = -0.25$ ($t = -2.9$), taking R^2 from 0.70 to 0.72. Primary completion alone drives the fertility transition; test-score quality adds little. For log U5MR and LE, the picture reverses: quantity is insignificant ($t = -1.7$ for U5MR, $t = -1.1$ for LE); quality dominates ($t = -7.6$ for U5MR, $t = +7.7$ for LE). Fertility is a threshold flip that runs on primary completion even where schooling is shallow; child survival and adult longevity require cognitive depth and do not respond where credentials are hollow.

Hollow education is not education — but exposure still matters for TFR.

9.7 The Panel Survives Exclusion, Strengthened

The Table 3 headline specification reruns with the 15 USSR republics excluded. The parental-education coefficient rises from 0.483 ($n = 1665$, 185 countries) to 0.538 ($n = 1539$, 171 countries); within-R^2 rises from 0.457 to 0.507. The headline LE forward-prediction R^2 rises from 0.350 to 0.366. The headline TFR forward-prediction R^2 rises from 0.439 to 0.444. The headline log-U5MR forward-prediction R^2 rises from 0.635 to 0.659. Every coefficient of interest moves in the same direction: upward.

This is not what survivorship looks like for a contaminated claim. It is what survivorship looks like for a claim whose signal was being attenuated by the contaminated subgroup. USSR inclusion was not producing the paper's headline results. It was dampening them. Removing the dampening factor returns the cleaner estimate of the mechanism in market economies. Appendix A.2.4 reports the full table with all specifications; the 171-country row joins WCDE,

Barro-Lee, and post-1970 as one more robustness row confirming the direction and magnitude of the headline.

One sentence summarizes the chapter. The paper's mechanism identifies where the Soviet data was wrong; the Soviet data, once excluded, confirms the mechanism more sharply. It is the kind of agreement between theory and data that one does not get by accident.

10. The Natural Experiments

The panel establishes the patterns. The natural experiments isolate the mechanism under designs cleaner than cross-country regression can deliver — shocks that isolate one variable while holding the others constant, or within-country cases that eliminate confounders entirely. Each subsection below tests a prediction stated in Chapter 7.

10.1 The Famine Test (Prediction 7)

The shock test asks whether education survives crisis. The famine test asks the converse: does education *prevent* the most extreme form of crisis?

Every major famine since 1950 occurred in a low-education setting. Of 21 famines in the dataset, 19 occurred where lower secondary completion was below 50%. The median education level at the time of famine was 19.6%; the mean was 25.4%. The two exceptions — North Korea (1996, 100% completion) and Yemen (2018, 68%) — both required external force physically preventing food access: totalitarian state control of distribution in the first case, naval blockade in the second. Where the educated population retained agency, famine did not occur.

The near-miss cases make the pattern sharper. Seventeen countries faced shocks comparable to or worse than those that produced famines elsewhere — Cuba's 35% GDP collapse when Soviet food imports were cut by 80% (1993), post-WWII Japan and Germany, Armenia under simultaneous blockade and war (1993), Ukraine under Russian invasion and grain blockade (2022) — and none experienced famine. Their median education was 71.6%, against 19.6% for famine countries. The difference is significant at $p < 0.00001$ (Mann–

Whitney U). The near-miss cases span the same decades as the famine dataset (1945–2022), so the comparison is not an era effect.

Educated populations do not starve. They reorganise supply chains, ration collectively, substitute crops, demand state response, and migrate strategically. Uneducated populations, facing the same shock, lack the household-level capacity to adapt and the political capacity to compel response. The mechanism is the home niche operating through the household: the same cognitive and behavioural repertoire that drives the fertility transition also drives famine prevention. The test assumes the energy substrate (Section 2): every shock in the dataset was a distribution failure, not absolute absence of food in the region. Educated populations respond by solving the distribution problem; uneducated populations cannot.

Bihar and Kerala, 1966. The within-country case eliminates every confounder except education. The 1965–66 national drought crashed India's grain production by 19%. Bihar experienced famine: 70,000–130,000 excess deaths (Dyson & Maharatna 1992). Kerala — India's most food-deficit state, 40% import-dependent, hit so severely that food riots erupted and the UK Parliament identified it as one of the "most severely hit areas" — had no famine. Same Constitution, same free press, same central government, same democratic institutions. Bihar's literacy was 22% (female: 9%); Kerala's was 55% (female: 39%). (The 1966 figures are literacy rates — the only education statistic available at state level for that year — not the lower secondary completion measure used elsewhere in the paper; the gap runs in the same direction under any available measure.)

Sen's claim that "no famine has ever taken place in a functioning democracy" (Sen 1999) cannot explain the Bihar–Kerala divergence. Democracy was constant across both states. Education was not. Myhrvold-Hanssen (2003) challenges Sen directly on this point, arguing that Bihar 1966–67 meets any reasonable definition of famine and that the Indian media, "although free and independent, did not provide reliable information." Drèze himself (quoted in Brass 1986) described the evidence that Bihar was a success story as "precious little."

What the Bihar–Kerala case isolates is that education is a community-level shield, not a national one. The state apparatus was the same across both; the

Constitution was the same; the press was the same. What differed was how many households in each state collectively made educated decisions about food, migration, and political pressure. Where most households can and do, the population forms a shield that national institutions alone do not provide. Where most households cannot, no national apparatus substitutes for the missing mechanism. The same pattern extends into the deeper historical record. The Madras Presidency famines of 1876–78 killed an estimated five million people (Davis 2001) under the same imperial administration that was feeding its own educated population at home. Bengal 1943 and the earlier Irish famines of 1845–49 fit the same structure. The population that starves is the uneducated one regardless of which state nominally governs it; the educated population does not starve, even when its government is the one presiding over the starvation.

Kerala's own history confirms the mechanism. In 1943, Travancore (pre-independence Kerala) experienced famine under the same structural vulnerability — extreme food-import dependence — killing approximately 90,000 people. The difference between 1943 and 1966 was not democracy (Travancore had an elected legislature from 1932) but education: literacy had risen from approximately 30% to 55%. The educated population built the institutions — universal public distribution, food committees, political mobilisation — that prevented the same structural vulnerability from producing the same outcome.[17]

10.2 Cambodia: Destructive Disruption (Prediction 3)

Cambodia is the direct test of destructive disruption: what happens when the educated population itself is destroyed, as opposed to shocks that damage institutions, income, or infrastructure while leaving the carriers intact (Section 10.3). Together this subsection and the next test Prediction 3 as a pair — destruction of the carrier resets progress; destruction of everything else does not. Schools are the delivery mechanism; the educated adult is the loading channel. The Khmer Rouge destroyed the second and left the first repairable — buildings came back within a decade, but rebuilding the channel required a generation of its own operation. The country experienced severe disruption

[17] See `scripts/famine_education_test.py`.

under the Khmer Rouge (1975–1979), during which the education system collapsed. The data shows two distinct stalls. The first is the exogenous shock itself. The second — a generation later — is the mechanism's prediction: the home-niche baseline is the binding constraint (Section 5.1).

1975	Khmer Rouge takeover; completion 10.1%
1979	Khmer Rouge falls; educated population destroyed
1985	Completion 9.5% — **first stall** (system collapse)
1993	Paris Accords; UN transitional authority ends
1995	International reconstruction; completion jumps to 35.1%
2010	Plateau at 31–36% — **second stall** (parental baseline)
2011	Recovery as post-disruption cohort's children enrol

First stall (1975–1985). Completion was 10.1% in 1975, fell to 9.4% by 1980, and was still 9.5% in 1985 — a decade of zero progress.

Recovery and second stall (1990–2010). After the Paris Accords and UN transitional authority (1991–1993), schools were rebuilt with sustained international investment. Completion jumped to 35.1% by 1995 on the back of that reconstruction. Then it stalled again. From 1995 to 2010, completion plateaued at 31–36%, unmoved despite continued external funding. The buildings were there; the teachers were there; the money was there. Progress did not come. The 35% plateau is what competing-priority expansion looks like when it interacts with a parental education shadow: international reconstruction dispersed resources across health, infrastructure, governance, and education simultaneously, while the parental cohort remained frozen at ~10%. Korea proves singular priority can push far beyond any parental baseline — but Cambodia after 1991 was not Korea. The plateau reflects dispersed investment meeting a damaged generational base.

The first stall is the exogenous shock — the regime destroyed the education system. The mechanism predicts the second. It is the generational shadow: the children reaching secondary school in 1996–2010 were born approximately 1982–1996 — their parents were the cohort whose own education was frozen at ~10% during 1975–1985 — the pre-disruption level, preserved in households even as the school system collapsed. Countries that started at Cambodia's level in 1960 reached a median of 21% by 1985. The children of the frozen cohort inherited a 10% parental baseline instead of a ~21%

one; the plateau reflects that missing growth. Recovery from 2011 onward corresponds to the post-disruption cohort's children finally dominating the school-age population. The regime fell in 1979, but education remained frozen through 1985 (9.5%). Twenty-five years from the end of the educational disruption (1985) lands at 2010, matching the time-to-agency lag predicted by the mechanism (Prediction 1).

Countries that started at Cambodia's level in 1960 reached a median of 46% by 2015. Cambodia reached 36%. The buildings came back in 1991. Progress came back in 2011. That twenty-year gap is the home-niche shadow.

The shadow is not over. The grandparent channel (Section 7) predicts a second, deeper lag. Children reaching adulthood in 2020–2025 have parents from the post-reconstruction cohort (35–36% completion) but grandparents from the frozen cohort (9–10%). At low baselines, grandparent education predicts child outcomes independently of parent education, with a coefficient nearly twice the parent's ($\beta_{gp} = -0.059$ vs $\beta_p = -0.033$ for fertility). The Khmer Rouge fell in 1979; the parental shadow lifted in 2011; the grandparent shadow persists through at least 2035, when the first post-disruption cohort's children become grandparents themselves.

10.3 Non-Destructive Shocks (Prediction 3)

Where Cambodia tested what happens when the carrier is destroyed, the shock test asks the converse: what happens when institutions, income, or infrastructure are shattered while the educated population itself survives? Exogenous shocks that kill people depress life expectancy. They do not reverse the fertility transition. The pattern holds across every shock type in the dataset.

Table 18. Exogenous shocks and development outcomes.

Shock type	Country	LE effect	TFR effect	Edu effect
Civil war	Sri Lanka	Delayed 12 yr	On schedule	None
State collapse	Russia	−5.0 yr, recovered	Unaffected	None
Pandemic (HIV)	South Africa	−9.0 yr, recovering	Continued falling	None

Shock type	Country	LE effect	TFR effect	Edu effect
Income destruction	AFC (5 countries)	None	None	Continued rising
Population destruction	Cambodia	Stalled	Delayed	**Reset**

LE effect: deviation from pre-shock trajectory. TFR effect: deviation from education-predicted path. Sources: World Bank WDI, WCDE v3. See `scripts/verify_russia_south_africa.py` *for Russia and South Africa;* `scripts/robustness/asian_financial_crisis.py` *for the AFC; Sri Lanka and Cambodia in Sections 22 and 10.2.*

TFR is a household decision running on the home niche — internal to the household, unreachable by war, virus, or collapsing state. LE depends partly on whether external conditions allow survival. When the two diverge — TFR continuing its education-predicted decline while LE collapses — the divergence identifies the shock as external rather than educational.

Russia (99% lower secondary completion, 1990; Barro-Lee v3.0 and WCDE v3 agree at 1990 but WCDE over-reports secondary completion for 1960–1980 cohorts, with phenotype implications developed in Chapter 9). The Soviet dissolution crashed life expectancy from 69.5 to 64.5 (1988–1994) — 5 years lost in six. TFR, already at 1.89, continued below replacement throughout — 1.20 by 2000, never reversing. The state dissolved; the educated population did not. LE returned to its pre-collapse level by 2009 (68.7) and reached 73.1 by 2019 — surpassing the Soviet-era peak by 3.6 years. Education predicts the recovery trajectory, not immunity to state collapse.[18]

South Africa and the HIV epidemic. South Africa tests Prediction 3 under a health shock whose biology is exogenous to education. Between 1990 and 2005 LE fell from 62.9 to 53.9 (the largest peacetime LE reversal in modern history) while TFR fell from 3.72 to 2.41 and primary completion rose from 78% to 91% — the post-apartheid expansion reaching the Black majority at scale. The 1990 lower-secondary headline of 65% was an apartheid aggregate; the population bearing the shock sat below the behavioural- response

[18] See `scripts/verify_russia_south_africa.py`.

depth.

The composition matters because primary and lower-secondary depth carry different halves of the response. Primary drove the fertility decline: it was expanding through the shock, and TFR fell on schedule (R^2=0.65 for TFR-primary). Lower-secondary carries the behavioural response to novel pathogens; in the affected majority it remained too shallow. Acquisition-stage factors were genuinely biological — low traditional circumcision prevalence (60% biological reduction per three RCTs) and circular mining migration driving spread. Once information existed, the education–HIV gradient reversed: the Botswana 1996 school-reform natural experiment found each additional year of secondary schooling caused an 8.1-pp cumulative HIV reduction (De Neve et al. 2015), and Baker et al. (2017) formalise the pattern as the Population Education Transition curve — educated populations change behaviour first once novel-risk information exists. Post-ARV, LE recovered to 66.1 by 2019. Full mechanism detail — apartheid stratification, circumcision prevalence in the HIV belt, and the Population Education Transition curve — is in Appendix A.2.6.[19]

Cambodia (Section 10.2) is the exception that proves the rule: the only shock that breaks the household mechanism is the one that destroys the household.

10.4 The Colonial Test (Prediction 5)

The famous settler mortality instrument does not measure institutions. It measures whether Protestant colonisers — who built schools — survived.

Acemoglu, Johnson & Robinson (2001) use settler mortality as an instrument for institutional quality: where European settlers survived, they built inclusive institutions; where they did not, they built extractive ones. The instrument cannot distinguish this from an alternative channel: where *Protestant* settlers survived, they built *schools*. The Reformation made mass literacy a theological obligation (Section 5.1). The Counter-Reformation removed it. Protestant colonisers (Britain, the Netherlands) transplanted mass education systems. Catholic colonisers (Spain, Portugal, France, Belgium) did not. AJR's instrument captures this difference and attributes it to institutions.

[19] See `scripts/verify_russia_south_africa.py`.

The data separates the two channels. Among 99 former colonies, education at independence (1950 lower secondary completion) explains 47% of the variance in current GDP per capita. Polity5 institutional quality explains 11%. Colonial-era education (1900 birth cohort, people schooled under colonial rule) explains 35%. Coloniser religion alone explains 7% — but adding religion to a model that already contains education raises R^2 from 0.462 to 0.462. Religion predicts GDP only because it predicts education. Once education is in the model, religion adds nothing. The channel is religion → schools → education → development, not religion → institutions → development (Table A7).

The 2SLS first-stage diagnostics confirm this ordering: coloniser religion is a strong instrument for education at independence (first-stage $F = 10.8$, above the Stock & Yogo threshold) but a weak one for Polity2 institutional quality ($F = 1.4$). Full IV machinery — first-stage F-statistics, Wu-Hausman, second-stage coefficient ($\hat{\beta} = +0.059$ for education) — is in Appendix A.2.7.

Latin America is the critical case. Spanish and Portuguese colonies were settler colonial — Europeans came and stayed in large numbers, built cities, established legal systems. AJR's framework predicts that settlement should produce inclusive institutions and therefore development. It did not. AJR require an ad hoc distinction between "inclusive" and "extractive" settler colonialism to explain why Latin America underperforms. The education account needs no such distinction: Catholic settlers brought no mass education tradition. Spain had 0.6% primary completion for its own 1875 birth cohort; Portugal had 0.1%. They could not transplant what they did not possess. The educational base at independence was correspondingly low (mean 11% lower secondary, 1950), and development followed the education, not the institutions.

The horse race also extends to within-country variation. In a country-FE panel of 103 former colonies (1985–2015, outcomes every five years, education lagged one generation, polity2 contemporaneous, 721 observations), education beats polity2 on every outcome the paper tracks: log under-5 mortality (edu R^2=0.632, polity R^2=0.245, Δ=+0.387), child education (Δ=+0.327), total fertility rate (Δ=+0.163), and life expectancy (Δ=+0.125). The margin is largest precisely where the mechanism predicts — child survival mediated by maternal education, not by colonial legal tradition. See `scripts/robustness/colonial_all_outcomes.py`.

The settler mortality instrument is a Protestant education instrument. See `scripts/robustness/colonial_vs_institutions.py`.

The Protestant channel connects to Easterlin's (1981) original insight: the divergence in economic growth between early and late industrializers tracked the divergence in mass schooling — driven originally by the Reformation's insistence on personal scripture-reading, which produced mass literacy across Northern Europe. The distribution of development in 1960 was substantially the distribution of schooling in 1860. Goldin & Katz (2008) document the same transmission within the United States: the "human capital century" of 1910–1940, driven by the high school movement, explains the majority of twentieth-century income growth — a within-country longitudinal result consistent with the cross-country generational mechanism estimated here.

11. The Cases

Every country that developed did so one generation after sustained education investment — under every political system, at every income level, on every continent.

Table 20. Year of development (TFR < 3.65 AND life expectancy > 69.8) and relationship to prior education investment. Thresholds are 1960 United States values (World Bank WDI).

Country	Dev.	TFR crossed	LE crossed	Onset	Rate	Lag	Gen.
Taiwan	**~1970**	~1970	~1970	1950s	2.15	~20 yr	**1**
S. Korea	**1987**	1975	1987	1953–65	2.13	~25 yr	**1**
Cuba	**1974**	1972	1974	1961 (40%)	2.27	~13 yr	**1**
Bangladesh	**2014**	1995	2014	1990s	1.30	~24 yr	**1**
Sri Lanka	**1993**	1981	1993	1940s–50s	1.20	~42 yr	**2**
China	**1994**	1975	1994	1950s+CR	1.50	~42 yr	**2**
Kerala†	**~1982**	~1973	~1981	Early 20thC	—	~65 yr	**3**
India	**2017**	1996	2017	1950s	0.87	~67 yr	**3**
Uganda	**—**	TFR 4.39	LE 67.7	None	—	—	—

Developed = year both thresholds crossed. Expansion onset = year or level at which sustained educational expansion began. Rate = average annual change in lower secondary completion from expansion

onset to development crossing, in percentage points per year (see `scripts/expansion_rate_predicts_crossing.py`*). Lag = years from onset to development crossing. Generations = number of ~25-year cycles. Cuba's high starting base (40%) and the 1961 reconstruction of losses from the post-revolution exodus shortened the lag below the typical ~25-year generation. † Kerala figures estimated from India Sample Registration System and census records; all other figures from World Bank WDI direct measurement.*

The earliest modern example is Japan (1872): the Meiji compulsory education ordinance, implemented through repurposed temples with community teachers, no substantial budget. The sequence was mandate before resources — the state declared compulsory education and built the infrastructure afterward. Japan industrialised within a generation of these reforms despite negligible domestic fossil fuels, demonstrating that the mechanism does not require natural resource endowments.

Korea is the calibration case: 2.13 pp/yr, the fastest sustained expansion in the WCDE dataset. All other rows in Table 20 are independent tests. The expansion rate determines generations to crossing (Section 5.1): benchmark pace compresses it to one generation; India's 0.87 pp/yr spreads it across three.

Three parameters predict crossing time: starting base, expansion rate relative to the Korea benchmark, and structural disruption. Disruption delays LE crossing — Sri Lanka's 12-year TFR-to-LE gap (1981→1993) maps onto the civil war; China's 19-year gap (1975→1994) reflects Deng's dismantling of the barefoot doctor network. Korea, Taiwan, Cuba, and Bangladesh had no comparable disruption. The predicted depths match: Korea-pace cases in 20–34 years; half-Korea-pace in 42–45 years; third-Korea-pace in 60–70 years. The crossing dates are the test, not the inputs.

Each case below confirms one or more of the predictions stated in Chapter 7. Read together, the cases demonstrate that the mechanism holds across regime type (authoritarian, democratic, socialist, post-colonial, military), ideology (Confucian, Islamic, Catholic, Buddhist, Marxist-Leninist, Hindu-majority), and income level — $1,038 per capita at Korea's expansion, $1,159 at Bangladesh's crossing. This is Prediction 5 in its most direct form.

11.1 Taiwan and Korea (Predictions 1 and 5)

Taiwan and Korea are the timing and regime-independence test. Both were authoritarian during expansion; both crossed one generation after sustained expansion began. Taiwan and Korea crossed earliest — both at benchmark pace, both state-driven from colonial bases built by Japan (25% and 18% completion by 1950). Korea expanded at 2.13 pp/yr, the fastest sustained rate in the WCDE dataset; Taiwan followed a nearly identical trajectory. Both had functioning markets — what Sen calls "growth-mediated security" — but the growth was mediated by education, not the reverse. State-driven singular expansion compressed development to 20–34 years against the 40–70 years seen in competing-priority cases.

American colonial education in the Philippines was similarly deep — 22% lower secondary completion by 1950, comparable to Korea. In 1960, the Philippines was one of the most prosperous countries in developing Asia: GDP per capita of $1,124 against Korea's $1,038 (constant 2017 USD), ahead of Thailand ($592), Indonesia ($598), and far ahead of India ($313) and China ($241). The Philippines had both the educational base and the income advantage. But no post-independence government sustained the educational investment. Korea made education the singular priority; the Philippines drifted at roughly half that pace. The result: comparable colonial base, comparable income, divergent post-colonial trajectory. Education policy is a choice, not an inheritance. Income does not make the choice for you.

11.2 Kerala (Prediction 5 at sub-national scale)

Kerala tests the mechanism at sub-national scale: the same transition, the same lag, inside a larger country that has not crossed. Kerala crossed at ~1982 — the paradigm case of competing-priority expansion. Education had been building since the early twentieth century through social reform movements — gradual accumulation through social reform and state investment — competing-priority, not singular-priority at Korea pace. TFR crossed by ~1974; LE followed ~8 years later. Kerala had extensive state provision — public health, food distribution, land reform — but fertility declined because educated women made the decision, not because provision was available (Section 6.7). The Emergency sterilisations (1975–77) confirm the polarity: forced

on the least-educated northern states, they brought down the government (Gwatkin 1979; Vicziany 1982); Kerala needed no coercion. State coercion either arrives after education has done the work, or fails where education has not.

The sub-national test has been run systematically. Drèze & Murthi (2001), using district-level panel data for 1981 and 1991, found that women's education was the single most important predictor of fertility differences across Indian districts, while urbanisation, poverty reduction, and male literacy showed no significant association. The cross-country result replicates within a single country at district level.

11.3 Four Further Cases

Each row below identifies the specific prediction the case most directly confirms.

Case	Crossed	Prediction tested and result
Sri Lanka	1993	**Prediction 3** (non-destructive shock). TFR crossed 1981; LE climbed to 69.0 by 1988, fell to 67.3 during the civil war, recovered to cross 69.8 in 1993. War disrupted the economy and life expectancy without breaking the home niche.
Myanmar	—	**Prediction 5** (regime independence). 73 years of military rule, GDP $1,025 (2015), education at 0.6 pp/yr. TFR fell from 5.9 to 2.3; LE rose from 44.1 to 65.3 — consistent with 43.5% completion. The regime suppressed speed; it could not reverse what the home niche had already carried forward.

Case	Crossed	Prediction tested and result
Cuba	1974	**Prediction 5** (socialist regime) and 6 (high starting base, short lag). 40.3% completion in 1960; post-revolution exodus removed the professional class. The 1961 campaign deployed 268,000 volunteers (Prieto 1981), replacing an elite that left with a mass baseline that stayed. Cuba and Taiwan — Soviet-aligned and US-allied — crossed within four years.
Bangladesh	2014	**Prediction 4** (income independence). 11.4% completion in 1960, GDP $1,159 at crossing — the lowest income at which the paper's dataset records a development crossing. Sustained commitment to girls' education from the 1990s. TFR decline tracks girls' secondary expansion, not contraceptive distribution (Mamun & Bongaarts 2022).

11.4 China (Prediction 4 — the support-led residual is education)

China is the direct test of income-independence against the support-led-security thesis. Drèze and Sen attributed China's outcome to direct health provision; the panel shows the residual was education. China crossed in 1994. Three corrections to the standard narrative.

First, what is called an educational catastrophe was the largest educational expansion in Chinese history. The standard framing conflates the disruption of university education — a small percentage of the population — with the overall trajectory. For rural China — 80% of the population (National Bureau of Statistics 1982) — the Cultural Revolution era produced the largest lower secondary gains in the WCDE 1870–2015 record: +10.6pp for the 1975

cohort, +15.0pp for the 1980 cohort. Community schools (民办学校) brought secondary education to villages that had none (Pepper 1996; Unger 1982; Gao 2008).

Second: China is the most direct test of Drèze and Sen's support-led-security thesis, and the thesis fails. Drèze & Sen (1989, Table 10.6) show China's under-5 mortality at 32% of what GNP predicted — the largest deviation in their sample — and attribute it to "support-led security": barefoot doctors (赤脚医生), direct provision. But Sen regressed on income and found a residual. That residual is education.

When life expectancy and under-5 mortality are matched on mean years of schooling (age 20–24, ± 0.5 years of schooling, any calendar year) rather than GNP, China's advantage reverses. China's LE was *below* education-matched peers from 1965 to 1991 — 6.6 years below at 1965, still 2.7 below at 1980, converging only in 1991 — and its U5MR was *above* education-matched peers until 2000.[20] The barefoot doctors were not producing exceptional outcomes; China was underperforming what its education level predicted.

The barefoot doctors and the community school teachers came from the same CR-era rural mobilisation (Gao 1999). The doctors were local, culturally embedded, drawn from the villages they served — optimally delivered provision. When Deng dismantled the system after 1980, they left, because rational people freed from revolutionary ideology redirected their labour toward self-interest. The LE convergence rate was statistically unchanged: $+0.31$ years/year before 1981, $+0.30$ after ($\beta_3 = -0.007$, $p = 0.82$). LE continued to rise — from 64 to 69.8 by 1994 — because the educational baseline built into rural households did not leave with the doctors. Drèze & Sen (1989, pp. 215–221) document the resulting health crises and rising rural medical costs (Gao 2008; Liu et al. 2003), but these crises left no detectable mark on the trajectory.

China crossed to *above* education-predicted LE in 1992 and to *below* education-predicted U5MR around 2000 — two decades after the barefoot

[20] See `scripts/cases/china_mean_years.py`. Mean years of schooling computed from WCDE v3 proportions (age 20–24, both sexes); LE and U5MR from World Bank WDI. Peer pool is all country-years within ± 0.5 mean years of schooling of China's value at each year. The result is robust to peer-pool bandwidth: LE and U5MR gaps retain the same signs and comparable magnitudes at ± 0.25 and ± 1.0 mean years (`scripts/cases/china_band_sensitivity.py`).

doctors were gone. Mean years of schooling rose from 5.9 (1965) to 8.0 (1980) to 9.6 (2000), tracking both outcomes across both eras.

Third, the TFR threshold was crossed in 1975 — five years before the one-child policy. China's fertility trajectory mirrors South Korea and Thailand's, both of which achieved equivalent TFRs without compulsory policy (Cai 2010; Miller et al. 2018).

12. The Dark Parallel

The positive case has been stated: education is the substrate on which development runs. What the positive case does not say, on its own, is what the same substrate does when it is pointed at something other than a population's own flourishing. This chapter says it.

12.1 Capacity, Not Virtue

The mechanism produces capacity for coordinated collective action. That capacity forms the community shield of Section 10.1 — food reorganised under drought, fertility decisions made at the household level, child mortality reduced without waiting for the state. The same capacity, pointed outward at a rival population, organises armies, plans blockades, and sustains a war economy long enough to break the other side. Pointed inward at a subjugated group within a population, it organises domination at whatever scale the population's educational base permits.

The shield and the amplification are the same mechanism, rotated. An educated population is not a peaceful population. It is a capable population. What the capability is directed at is a political matter, not a biological one. The preceding chapters described what the mechanism does when it is pointed at a population's own survival. This chapter describes what it does when it is not.

12.2 Demographic Pressure

Malthus (1798) identified the structural pressure: human populations, absent a check, outgrow the resources that sustain them. When growth outruns carrying capacity, the options narrow. Populations expand into new territory,

or contract through starvation, or are destroyed in conflict with neighbouring populations doing the same. Peaceful coexistence is a function of slack — resource abundance relative to population. When slack disappears, the kill-or-be-killed structure forms. There is no middle ground when the resource base will not support the bodies.

The educational substrate supplies the organised response to that pressure at scale. Uneducated populations under Malthusian pressure can riot, migrate in disordered waves, starve. Educated populations under the same pressure can organise armies, project navies, build and sustain settler economies on other continents, administer conquests across distance. The pressure is the same; the instruments are different. This is why the great engines of inter-group violence correlate with demographic peaks more reliably than they correlate with particular ideologies. The ideological frames participants use to describe what they are doing — religious, national, civilisational — are the organising symbolism. The pressure underneath is demographic and material. When the pressure releases, the frames lose their conscript force.

The ideological frames are not epiphenomenal, but they are arbitrary. Out-group identification is the species' baseline orientation at the group boundary. Hare & Woods (2020) argue that human self-domestication selected for in-group cooperation — the positive trait that makes large-scale human coordination possible at all — and heightened out-group hostility is what that trait produces when a boundary is drawn. The two are consequences of the same evolutionary history but operate differently: in-group bonding holds the population together, out-group orientation governs how it meets another. Human populations will find an out-group; which category gets elevated to that role — religion, ethnicity, language, nation, caste, class — is contingent on which markers are locally available and which a directing class finds useful. The content of the othering is arbitrary. What is not arbitrary is that demographic pressure combined with out-group identification produces violence at whatever scale the educational substrate permits.

12.3 The European Window

European expansion from approximately 1500 to approximately 1960 has been read as educated Europeans dominating uneducated others. The reading

is too clean. Europe in that window was not a uniformly educated civilisation. It was a small literate elite — clergy, merchants, administrators, military officers, later industrialists and engineers — sitting atop a largely uneducated mass population whose fertility had not yet begun to decline. Mass education in Europe arrived late. Scotland after Knox was exceptional; most of the continent reached mass schooling only in the last century and a half. England reached an Education Act in 1870; France reached the Ferry Laws in 1881–82; Prussia earlier but unevenly. The European fertility transition arrived in the same window, and for the same reason (Section 6.7).

What that configuration produced was a machinery, not a virtue. The educated elite had enough organisational capacity to build states, navies, finance, navigation, and industrial infrastructure (Section 3 on the tool sequence). The uneducated masses supplied the demographic pressure — pre-transition fertility produced a population boom that outran European agricultural yields — and the bodies: the soldiers, sailors, settlers, and conscripts who carried the projection outward. The populations the projection reached had neither the elite capacity nor the mass-level community shield; they absorbed the costs.

Ordinary European migrants, soldiers, sailors, and settlers were human beings responding to demographic and material pressure with the instruments the mechanism supplied. They are kin to the Mughal peasant conscripted into imperial service, the Qing subject sent to settle frontier provinces, the Roman citizen shipped to a colonia, the Phoenician family boarding for Carthage, the Bantu lineage absorbing smaller neighbours on the way south, the Arab trader extending the reach of caliphates across the Indian Ocean. Ordinary people in every imperial age did what ordinary people do: they took the opportunities available, they accepted the costs they did not see any way to refuse, and they died at rates they did not choose. There is nothing in the European window that is not also in the long human record of organised expansion under pressure. The scale is larger because the capacity is larger; the humans are what they have always been.

The Chola maritime expansion across the Bay of Bengal in 1025, reaching Srivijaya and Kedah, instances the same structure from an Indian base centuries before the European maritime projection — the substrate-and-pressure combination produces outward projection in whatever geography it obtains,

and the form it takes (continental, maritime, tributary, settler) reflects local conditions rather than population-specific tendencies.

The directing class at the top of the imperial machinery made decisions whose costs fell on populations unable to resist. They are not named here because the paper is about a mechanism, not about individuals. They are also not defended.

The postcolonial account that locates the current state of the Global South primarily in colonisation cannot carry the weight it is asked to carry. But the account that locates those conditions in post-independence failure is equally uncalibrated. Post-independence states have operated, almost without exception, under the competing-priority regime (Section 5.3): education in the national plan, the budget, the five-year strategy, the international compact. Europe, for most of its four-century imperial ascent, operated under no priority at all. On the regime dimension I use, the post-independence Global South has operated at a level Europe itself did not reach until late in its own imperial age. What varies across the post-independence countries is speed, not direction. A handful — Korea, Taiwan, Cuba, Bangladesh, Sri Lanka (Section 11) — made education the singular priority and compressed the transition to one generation. Most operate under competing priority and are proceeding slower. The paper neither celebrates the first group as exceptional nor indicts the second as negligent. Competing priority is itself a regime above anything Europe ran under for most of the period during which Europe dominated the world.

Everyone — coloniser, colonised, postcolonial — has been human in the same way, operating under different regimes at different moments.

12.4 The Evolutionary Baseline

Hierarchies, empires, out-group violence, and organised domination are not unique to the educational window, and not unique to Europe. They are the baseline condition of settled human societies going back to the first states in Mesopotamia and the Yellow River, and of pre-literate human societies before that. The biological chapters establish the substrate: de Waal (2013) on primate hierarchy; Wrangham & Peterson (1996) on coalitionary male violence as a deep ancestral trait shared with chimpanzees; Hare & Woods (2020)

on the in-group orientation that accompanies self-domestication. The human species is not a naturally peaceful population interrupted by occasional lapses. Out-group violence and hierarchical domination are the baseline; inter-group peace at large scale is the historical anomaly and is the phenomenon that requires explanation.

The educational substrate does not create hierarchical domination. It amplifies the reach of whatever hierarchical arrangement a population's capacity permits. Small educated elites built the first states, the first organised religions, the first slave systems, the first conquest empires. Larger educated populations built larger and more durable versions of each. When mass education finally arrived in one region, the reach of organised violence scaled proportionally. This is not a charge against education; it is a description of what capacity does.

The honest reading is not that education corrupts an otherwise peaceful species. It is that the species is what it has always been, and education raises the ceiling on what the species can accomplish — in both directions.

13. Convergence and Peace

The preceding chapter described the mechanism pointed outward under demographic pressure. This chapter looks forward from convergence. The argument lands in two parts — empirical and structural — and closes with the signature already visible and the kind of peace the mechanism supports.

13.1 The Empirical Sequence

The demographic transition has, in every case the dataset covers, followed the extension of mass education. Pre-transition European fertility was high because the population was largely uneducated; educated elites alone were not numerous enough to reduce aggregate fertility. Once mass schooling reached most of each country's population, aggregate fertility fell. The same sequence is visible in every post-war case in the paper's sample: fertility falls one generation after secondary completion rises (Section 6.7). The mechanism is the same in each case and the timing is the same in each case, because the mechanism is the same species-level trait operating in all of them.

13.2 The Malthusian Engine Runs Out of Fuel

The Malthusian engine that powers organised inter-group violence has one off-switch the species has ever found, and the preceding chapters have described it. When most of a population reaches the educational threshold, the community shield forms, household-level fertility decisions arrive, population growth slows or reverses, and the demographic pressure that makes populations expandable-by-force or forced-to-expand releases. The window that produced the past four centuries of asymmetric organised violence was the window during which one region had arrived at the substrate and the rest had not. Convergence closes that window.

13.3 The Four Radii Reverse

The reversal is already partly visible. The four radii of educational effect (Section 5.2) describe how an educated population's capacity extends outward — household, kin and identity, beyond identity, global — with decreasing durability as the arc widens. Under demographic pressure, the outer radii previously produced the populations that got killed: conquered, displaced, starved, shipped across oceans in chains, left to die in famines the imperial administration preferred not to relieve. The same outer radii now produce GAVI vaccine delivery, SDG commitments, global health funding, and educational aid flowing from already-converged populations toward populations that have not yet crossed. The mechanism is the same four-radii structure; the direction has reversed. The post-2000 drop in global under-five mortality, driven substantially by vaccine delivery from the already-developed world, is the most visible signature: the educated populations that previously let foreign children die now keep them alive. This is not new altruism. It is the outer radii of populations that have crossed the threshold extending toward populations that have not — under the same four-radii structure that previously pointed the other way, when the populations at the centre were under pressure the convergence has now released.

13.4 The Peace That Remains

The paper predicts peace. Not the peace of virtue — the species will remain what it has always been, with its hierarchies and its arbitrary out-group ori-

entations intact. The peace that follows when the Malthusian engine runs out of fuel and the arbitrary othering humans always do loses the demographic pressure that has scaled it into mass violence for most of recorded history. Motive-removal is primary; defensive capacity is the backstop. Peace follows not from the brittle equilibrium of mutual deterrence but from the dissolution of the demographic configuration that made outward projection structurally rational — the nuclear era, capacity without motive-removal on both sides, produced suspended near-conflict across four decades, not peace. What remains is the ordinary evolutionary baseline at a scale the species has not previously experienced: a planet on which most populations have the capacity to feed themselves, to defend themselves, and to pressure their own governments.

The peace claim and the development claim are therefore the same claim, through the same mechanism: completing the convergence is what dissolves the pressure that produced the four-century window, and the decision that extends education is also the decision that extends peace.

This is the claim the mechanism supports.

14. Difficulties on the Theory

The preceding chapters made the positive case. This chapter engages the major alternative accounts in development on biological grounds, not only on data. Each identifies something real; each stops one level short of the substrate that makes the thing it identifies possible. Sen and Drèze are the exception: the capabilities frame is this paper's frame, read one layer deeper, and Section 14.2 completes rather than displaces them.

14.1 Smith and the Loaded Labour

Smith (1776) founded the modern account of wealth on specialisation. The pin factory scaled because each worker performed a narrow task, and the tasks composed into a sequence that produced more pins per hour than any single generalist could. The causal claim in *The Wealth of Nations* is that the division of labour is the engine; political economy follows from it.

What the pin factory required, and what Smith took as given, was labour that could be loaded with specialised tasks at all. The workers were Scottish, and Scotland in Smith's lifetime already carried the highest parish-level literacy in Europe — a legacy of Knox's 1560 instruction that every parish maintain a school so every child could read scripture. By the time Smith observed the factory, categorical literacy (Chapter 4) was widespread in Scottish labour markets. A generalist labourer could receive a written instruction and act on it. The foreman did not need to demonstrate each step; he could describe it.

Smith's account starts after the loading. The substrate that made loadable labour possible — the cultural-transmission channel (Chapter 3) running through written language into categorical cognition — was inside the frame of the factory by the time he observed it, not outside. He attributed to specialisation what belonged to the substrate that specialisation presupposed. The founding error propagated: Smith to Becker (Section 14.7) to Sen (Section 14.2) all work in a space that assumes away the cognitive-substrate step whose presence their frames depend on.

The cost is not conceptual alone. Policy built on Smith tells countries to specialise into export markets to get rich. Korea did the opposite; Taiwan did the opposite; Bangladesh did the opposite. Each built the substrate first and the specialisation followed. The bio claim predicts this ordering; Smith's frame, taken as primary, reverses it.

14.2 Sen and Drèze: Two Routings, One Mechanism

Sen (1999) and Drèze and Sen (1989) are the closest frameworks to the one I make. The distance is a single step: what generates the capabilities.

Sen argues that development is what people are able to do and to be, not what they earn. The list of capabilities — to live a long life, to be literate, to participate politically — is the right list, and measuring development in those terms is a correction of the GDP-primary frame. Drèze and Sen's claim that democracy with a free press prevents famine is a genuinely novel observation largely right on its own evidence.

What the frame underdetermines is the mechanism that produces the capabilities in the first place. Sen treats them as parallel expressions of human

freedom — coequal axes of wellbeing. The bio claim is that they are serial consequences of a single substrate. Educated cognition makes categorical choice possible (Chapter 4); categorical choice, applied at the household scale, produces the fertility transition, the mortality transition, and the political capacity that the capabilities list comprises. The list is the output of one mechanism, not a set of independent dimensions.

The distinction matters most for the growth-mediated / support-led dichotomy. Drèze and Sen describe two routings of fiscal resources — through market growth or through state service provision — and identify democratic accountability as the feature that determines which populations receive which routing. Both routings, in their own data, produce development only where they reach an educated population. The China case (Section 11.4), matched on mean years of schooling to the fastest movers in the support-led list, shows the residual attributed to "support-led" routing is education itself. Bihar and Kerala (Section 10.1), under one constitution and one national press, isolate the mechanism: where most households are educated, the community-level shield forms and famine does not arrive; where most households are not, the shield does not form and press coverage alone does not substitute.

The frame is the same frame, read one layer deeper. The fiscal routing is how resources arrive; the educated household is where they become the capability.

14.3 The Deaton Objection

Deaton (2013) attributes the post-1950 mortality decline to global health technology diffusion largely independent of education. The argument is: vaccines, antibiotics, oral rehydration therapy, and safe-water knowledge are cheap, globally available, and easily applied; once the technology is invented, populations benefit without needing to understand the science behind it; the curve rises because the technology spreads.

The distribution of outcomes does not agree. Vaccines have been globally available since the 1960s; under-five mortality fell sharply in some populations and barely moved in others over the same decades. The technology diffused; the outcomes did not, uniformly. Between the existence of a vaccine and a vaccinated child sits a household decision — to locate the clinic, to trust the schedule, to complete the series, to act on subsequent infection warn-

ing signs. Each of those is a small act of categorical reasoning on a medical claim. Educated mothers make those decisions reliably; uneducated mothers make them variably. Gakidou et al. (2010), decomposing the global mortality decline, attributed roughly half of the post-1970 reduction in under-five mortality to gains in women's education.

The panel confirms the channel. Residualized GDP — income stripped of education's contribution — has never exceeded $R^2 = 0.019$ as a predictor of under-five mortality or life expectancy in the paper's sample (Section 8.9). If technology were operating independently of education, residualized income should retain predictive power; it does not. What remains, when education is partialled out, is noise.

The technology is necessary. It is not sufficient. The mechanism that converts the technology into outcomes is the educated household, one child at a time, and the community-level threshold at which that conversion becomes routine.

14.4 The Institutional Challenge

Acemoglu and Robinson (2012) argue that institutions — inclusive versus extractive — are the primary determinant of long-run development. The question is which way the causal arrow runs.

Institutions are built by educated populations. The civil servants, judges, journalists, health workers, and administrators who constitute institutional capacity are the educated cohort. Without the underlying educated population, institutional reform produces forms without function. Controlling for institutions is controlling for a product of education.

Country fixed effects absorb time-invariant institutional characteristics. Time-varying institutional change is itself driven by education. Qatar ($69,000 per capita, stable institutions by standard measures) delivered 4.8 percentage points below its generational education baseline in 2015. The institutions were present; the educational commitment was not; the development did not compound. The Philippines (Section 11.1) confirms the same point from the opposite direction: equivalent initial institutions, slower education, slower convergence.

India has had democratic institutions, an independent judiciary, and a free

press since 1947. China had none. China expanded lower secondary completion from 10% to 75% in 40 years (1950–1990) at 1.6 pp/yr — under Mao, the Cultural Revolution, and the Great Leap Forward. India expanded from 10% to 37% in the same 40 years at 0.7 pp/yr — with every institutional advantage the literature claims to matter. India's institutions did not accelerate education; China's absence of them did not prevent it. Among countries still expanding, mean expansion rates were 1.06 pp/yr in 1950–1975, 0.86 in 1975–2000, and 0.94 in 2000–2015. If institutions drove educational expansion, rates should have risen as institutions matured. They did not.

Autocracy does not predict education speed. Merging Polity5 time-varying regime scores with the WCDE completion data (160 countries, 1950–2020, lagged 20 years to match the regime that made the schooling decision), polity2 explains less than 0.7% of the variance in education gain rates (Table A5). Democracies are slightly faster on average: 10.3 pp/decade against 8.1 for autocracies at the 20-year lag.

What autocracy produces is not speed but *variance*: 76% of autocratic countries fall below the democratic median gain rate, but the fat right tail includes Korea and Cuba. Autocracies populate both the fastest and the slowest education trajectories in the dataset. Among 50 countries that transitioned, 30 were faster under democracy (paired mean $= +0.6$ pp/decade, $p = 0.57$).

Regime type is a variance amplifier, not a cause. The decision variable is education policy (Section 15), and it is orthogonal to regime type. See `scripts/robustness/regime_education.py`.

The colonial instrumental-variable contest (Section 10.4) makes the same point from the institutional literature's own preferred instrument. Settler mortality — the instrument that grounds the institutionalist identification strategy — turns out to predict education at first-stage $F = 10.8$ and institutions at $F = 1.4$. The effect the instrument identifies is the educational one. Tables A7 and A9 give the full results.

Table A5. Regime type and education gain rate. Polity5 time-varying regime scores (polity2, -10 to $+10$) merged with WCDE lower secondary completion (both sexes, age 20–24). Education at age 20–24 reflects schooling 15–20 years earlier; lagged specifications attribute the education out-

come to the regime that made the decision. 160 countries, 1950–2020. See `scripts/robustness/regime_education.py`.

Regime lag	R^2(polity2)	Polity coef	Auto mean	Demo mean	n intervals
Concurrent (0 yr)	0.0015	+0.056	8.5	9.5	1,782
15-year lag	0.0067	+0.122	8.1	10.0	1,574
20-year lag	0.0050	+0.106	8.1	10.3	1,472

Gain rate in percentage points per decade. Polity coef: the slope of the democracy score on gain rate (positive = more democratic regimes gain faster). R^2: share of gain-rate variation explained by democracy score alone — less than 1% at every lag. Auto = strong autocracies (polity2 ≤ -6); Demo = strong democracies (polity2 $\geq +6$). Both invest at roughly the same rate.

The colonial test puts education and institutional quality into a bivariate horse race on log GDP per capita across 99 former colonies. Education dominates at every measurement horizon; adding coloniser religion on top of education does not raise the fit.

Table A7. Colonial education vs. institutions as predictors of current GDP. 99 former colonies classified by coloniser identity and religion (Protestant: Britain, Netherlands; Catholic: Spain, Portugal, France, Belgium, Italy). Education data from WCDE v3; GDP from World Bank WDI (constant 2017 USD). See `scripts/robustness/colonial_vs_institutions.py`.

Predictor	R^2 (log GDP 2020)
Current education (2020 lower sec)	0.696
Education at independence (1950 lower sec)	0.462
Colonial-era education (1900 cohort primary)	0.348
Polity2 2015 (institutional quality)	0.115
Coloniser religion (Protestant = 1)	0.055
Education 1950 + coloniser religion	0.462

Each row is a separate bivariate OLS regression on log GDP per capita 2020, except the final row which includes both regressors. Adding coloniser religion

to education raises R^2 from 0.462 to 0.462: religion predicts GDP only through its effect on education.

The institutionalist literature identifies the effect of institutions on income using settler mortality or coloniser religion as instruments. Running the same instrumental-variable design with education as the alternative endogenous regressor reverses the usual reading: the instrument predicts education at first-stage $F = 10.8$ and institutions at $F = 1.4$. The effect the instrument identifies is the educational one.

Table A9. 2SLS instrumental-variable contest: education vs. institutions. Coloniser religion (Protestant = 1) as the excluded instrument. First-stage F > 10: strong instrument (Stock & Yogo 2005). 84 former colonies with complete data. See `scripts/robustness/iv_2sls_colonial.py`.

	First stage		Second stage (log GDP)	
Endogenous variable	F-stat	Strength	$\hat{\beta}$	t
Education 1950 (lower sec)	10.8	Strong	+0.059	3.21
Polity2 (institutions)	1.4	Weak	+0.349	1.56†

†*Institutional coefficient is inflated by weak-instrument bias ($F = 1.4 < 10$); the 2SLS estimate is unreliable. Wu-Hausman test for the education specification: $p = 0.85$ (OLS and 2SLS agree). The instrument predicts education, not institutions.*

14.5 Geography as Substrate, Not Substitute

Diamond (1997) argues that geography — the distribution of domesticable plants and animals, the east–west axis of Eurasia, the biogeographic inheritance of different continents — explains the differential trajectory of human populations. The argument is historical and long-run; it is not primarily about the post-1950 development window the panel here covers. On its own horizon it is largely right. Populations that inherited rich substrates of domesticable species developed settled agriculture earlier, accumulated food surpluses, and built the first stratified societies on those surpluses.

What the geographic frame cannot do is substitute for the mechanism that

translates the geographic inheritance into developmental outcomes at any given point in time. The substrate for which environments could be rebuilt — the long juvenile dependency (Chapter 2) and the cultural-transmission channel (Chapter 3) — has geographic prerequisites in its deep evolutionary history; that is not in dispute. What geography does not do is fix the moment at which a particular country decides whether to put children in classrooms, and at what pace.

Korea and the Philippines share a climatic latitude, a maritime geography, a set of tropical crop constraints, and equivalent colonial histories at mid-century. Korea compressed the transition to one generation; the Philippines did not. The geographic inheritance was the same; the educational regime was not. The panel is full of such pairings.

Geography is the substrate on which the mechanism runs. It is not the mechanism. The confusion of the two levels is the same shape as Smith's confusion of specialisation with the loaded labour specialisation operates on.

14.6 Easterlin's Question

Easterlin (1981) asked the question I answer: *Why isn't the whole world developed?* His own answer — that the cognitive and organisational substrate for modern development had to be built country by country, and that mass schooling was the mechanism — is the direct ancestor of the account presented here. Lutz and Kebede (2018) extended the empirical side of it; the panel here extends both.

The question's persistence in the literature four decades after Easterlin asked it reveals what the discipline of development economics has done with the answer he gave. The frame that took hold instead was growth-first, with the claim that education would follow, adjusted to whichever framework was ascendant: human capital in the 1980s, institutions in the 2000s, schooling-versus-learning in the 2010s. None of those frames permits Easterlin's answer, because each subordinates education to something prior. The substrate-is-primary claim is what those frames cannot absorb.

The whole world is not yet developed because the substrate-level investment is generational, and because most countries, in most decades since 1950, did

not make the singular-priority decision (Section 15). The countries that did — Korea, Taiwan, Cuba, Bangladesh, Sri Lanka — compressed the transition to one generation. The rest proceed slower under competing-priority regimes. Easterlin asked the right question and waited for the answer the panel now supplies.

14.7 Human Capital and the Wage Frame

Schultz (1961) and Becker (1964) founded the human capital school on the observation that education raises individual wage productivity and can be treated, economically, as an investment yielding private returns. The framework is the operating frame of labour economics and much of education policy. Measurement of the private return to individual schooling is well developed within it.

The bio claim is that the private labour-market return is the secondary channel. Education's primary developmental effect is not on the educated person's wages but on their children's lives — through the cognitive and decision-making traits transmitted parent-to-child (Chapter 5). A mother's education shows up in her children's mortality, nutrition, and school completion at effect sizes that dwarf its effect on her own wages. Gakidou et al. (2010) attributed roughly half of the post-1970 reduction in under-five mortality globally to gains in women's education — a population-level intergenerational effect, not a labour-market effect.

Human capital theory treats this downstream effect as an externality and sometimes adds it to private returns. The bio claim reverses the ordering. The intergenerational population-level effect is primary; the private wage return is the smaller, more variable side channel. Korea, Taiwan, Singapore, and every other country that made education the singular priority received the developmental compounding whether or not the policy frame named the mechanism correctly.

The framework's narrower consequence is a misdirection of measurement. Decades of Mincerian wage-return estimation now anchor the policy question of whether schooling is "worth it" in developing countries. The answer from wage data is contingent and often uncertain. The answer from intergenerational outcome data is uniform and large. The disagreement is a frame choice,

not a measurement dispute.

14.8 Schooling and Learning

Pritchett (2013) argues that schooling in many developing countries fails to produce learning: children sit in classrooms for years and finish unable to read, do arithmetic, or reason at grade level. The measurement is accurate; cross-national assessments confirm wide gaps between years of schooling and measured proficiency. The policy consequence Pritchett draws is that schooling quality, not schooling coverage, is the development variable.

The bio claim disagrees on the mechanism. Duration of exposure to the categorical, symbolically organised environment of formal schooling is what restructures cognition (Chapter 4). Test scores measure one narrow output of that restructuring at a particular moment in a child's life; they do not capture the restructuring that has already taken place. The restructuring itself is observable at the neural level, not inferred from behaviour: Dehaene et al. (2010) document the physical reorganisation of the brain's visual processing pathways in newly literate adults, and numeracy recruits parietal circuits unavailable to innumerate populations (Gordon 2004; Frank et al. 2008; Chapter 4). These structural changes persist whether or not the curricular content on which they were installed is retained for a particular test. The panel is the population-scale evidence. Years in school predict life expectancy, under-five mortality, and fertility at effect sizes an order of magnitude larger than test-score variation across countries at the same mean years of schooling. A child with nine years of low-quality schooling in rural Bangladesh reaches adulthood with a different cognitive architecture than a child with three years of high-quality schooling, and the household-scale behaviour that follows reflects the first child's architecture, not the second's score.

The schooling-without-learning frame is a level-of-measurement mistake. Learning is measured in content absorbed; the bio claim is that schooling's primary effect is categorical — the brain's reorganisation toward symbolic and propositional thought — and that content is the disposable vehicle on which duration rides (Section 4.3). This is a strong claim, and it implies that the curriculum wars that dominate education policy are second-order. The historical record is the test.

Four successful educational expansions ran through content regimes with nothing in common at the level of curriculum and produced the same cognitive outcome and the same subsequent development trajectory.

Scotland after 1560 taught scripture. Knox's parish schools were designed to make every child capable of reading the Bible, because Protestant salvation required direct textual access and Knox did not trust intermediaries. The content was religious. The effect was the highest parish-level literacy in Europe by Smith's lifetime — the loaded labour Smith observed in the pin factory came out of scripture-based schooling (Section 14.1).

France under the Ferry Laws (1881–1882) taught secular classical literature, republican history, and a standardised national curriculum designed specifically to displace religious instruction. The content was the opposite of Knox's. The effect was mass literacy across the French population, independent of whether the pupil retained the Latin declensions or the list of French kings.

Cuba in 1961 taught revolutionary doctrine. The literacy brigades used primers built around political slogans, Castro's speeches, and revolutionary narrative. The content was propaganda in the technical sense: controlled, ideological, deliberately shaped. The effect was mass literacy in a single year and a subsequent development trajectory — fertility transition, mortality transition, life expectancy tracking high-income countries despite low GDP — that Section 22 documents.

Bangladesh expanded schooling through the 1980s and 1990s with a curriculum that, by Pritchett's own measurement, produced substantial learning deficits. The content was weak in exactly the sense that motivates the schooling-without-learning critique. The effect was a fertility transition and a mortality transition that place Bangladesh among the paper's policy overperformers (Section 8.10).

Scripture, classics, revolutionary doctrine, weak modern curriculum. Four content regimes, one cognitive outcome. Content did not select which way the brain reorganised. Duration selected whether it reorganised at all. The curriculum wars mistake the vehicle for the payload.

Coverage is the primary policy variable. Quality is a second-order adjustment on top of coverage, not a substitute for it.

15. The Decision

Every country that developed made the same decision: sustained investment in education. The variable is speed.

Every government on earth now claims to prioritise education (UN 2015, SDG 4; UNESCO GEM 2024). Competing priority is the norm. The decision I call for is singular priority — for education through the full dependency window, not for stopping at nine years. Singapore is the existing demonstration: when the option is given, roughly three-quarters of the cohort continues into tertiary education.

The question for the remaining 20% is whether evidence can now substitute for the contingent motivations that drove every previous breakthrough. Knox, Meiji, and Korea bypassed the cost-benefit frame by never entering it; the commitment preceded any calculation. Evidence-based commitment would bypass the frame from the other direction: by showing that the calculation has already been done across 185 countries and the answer is always the same — education first, everything else follows.

A worked example, looking forward. Uganda's TFR was 4.39 in 2022, down from 5.56 in 2013 — a decline of roughly 0.13 per year sustained over a decade. Lower secondary completion reached 48.8% in 2025, expanding at approximately 6 percentage points per five-year period. The women whose fertility will determine the 2030 figure were educated by 2025; their exposure is already locked in. Run the mechanism forward on these inputs and Uganda crosses the 3.65 threshold by 2030.

This is illustration, not a load-bearing test. The framework's empirical case rests on the 185-country panel: 154 countries crossed, every one through education, income residualizing to zero across four outcomes, the four-generational lag-decay signature visible at every horizon. A single forecast a decade out, on a country still on the curve, can be perturbed by any number of local shocks — drought, regional conflict, a measurement revision, a survey delay. What Uganda demonstrates is what prediction looks like when the mechanism is run forward in a case it did not train on. Whether the 2030 release lands at 3.6 or 3.8 is interesting; it is not what the framework rests on.

One scope note, since the question is often raised: the material question that haunted every earlier call for universal development — whether the earth's energy base could sustain decent lives for everyone — is being resolved in parallel with, but independently of, the development decision. The solar–wind–battery transition is changing the cost and scale of energy capture on a trajectory that relieves the forward-looking resource constraint (Smil 2017). I take no position on its speed. The point is structural: the substrate is becoming tractable without further political choice in each country where it is needed. The variable that still requires a choice, in every country where development has not arrived, is education (Section 2).

No leader has yet made the decision on the basis of evidence alone. I provide the evidence.

There is one species, through one biological channel of cultural transmission, converging across every political system and every continent onto a single developmental form; where it has not arrived, the variable that still awaits is the decision.

Acknowledgments

Wolfgang Lutz's datasets made this analysis possible and his work on education and human capital set the terms on which it stands. He will recognise the question I am answering as his teacher's.

Richard Wrangham established what the human species was before education reached it — what cooking made, what violence shaped, what self-domestication permitted.

Sarah Blaffer Hrdy established that the household which transmits is never just two parents. The cooperative breeding that made the eighteen-year dependency period viable is the precondition I inherit.

Melvin Konner's *The Evolution of Childhood* is the biological foundation I build on. The eighteen years that separate the human juvenile period from every other species is the fact from which everything else follows. *The Tangled Wing* has been a longer, quieter presence.

The CARTA symposia at UCSD provided a year's immersion in the scientific

conversations underlying the biological arguments in this paper — evolutionary biology heard as a living community rather than a finished literature.

Frans de Waal spent his life showing what primates reveal about what is possible — the bonobo's imperfect, conflicted cooperation alongside the chimpanzee's lethal hierarchy. Without the first the household does not hold; without understanding the second we cannot see what education overcomes. He is missed.

Hunger and Public Action was one of the most important books I have read. The ten years I spent working to secure its Creative Commons release were a privilege; Jean Drèze was consistently supportive throughout. The views in this paper are my own. Drèze and Sen saw the right anomalies and asked the right questions. This paper is one answer.

Code and statistical analysis were written by Claude (Anthropic) under my specification and architectural direction. Prose drafting was assisted by Claude. Every empirical claim in this paper was verified against the data by reproducible scripts, not by trusting the language model's output; all scripts and data are published with the paper. The theoretical framework, analytical design, test specifications, arguments, conclusions, and all errors are my own.

A. Robustness

This appendix contains the econometric checks referenced in Sections 8 and 10. The body reports the headline specifications; the checks below confirm that staggered adoption, alternative cutoffs, alternative lag lengths, alternative estimators, alternative data sources, and the Colonial Test first-stage diagnostics leave the ordering and the coefficient signs unchanged. None of these checks is required by the biological claim; they are placed here for readers who work in traditions that treat identification as the load-bearing element of an empirical argument.

A.1 Two-Way Fixed Effects and Staggered-Adoption Diagnostics

Adding year fixed effects to the headline specification collapses the full-panel parental coefficient from 0.483 to 0.083. Table A1 documents the collapse across four parental-completion subsamples. The paragraphs below explain why the collapse is the mechanical signature of two-way FE under staggered treatment, not evidence against the mechanism, and present the Goodman-Bacon weight decomposition and the Callaway-Sant'Anna recovery.

Table A1. Two-way fixed effects (country + year): child lower secondary completion on parental education. R^2 is the partial within-R^2 (incremental variance explained beyond country + year FEs). See `scripts/robustness/twfe_child_edu.py`.

Model	Parental edu β	R^2 (within)	n	Countries
(1) FE + year: parent (< 30%)	**0.739*** **	**0.154**	783	137
(2) FE + year: parent (< 20%)	1.032***	0.145	600	118
(3) FE + year: parent (< 10%)	1.019***	0.067	358	85
(4) FE + year: parent (all)	0.083	0.009	1,665	185

Country-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. *Model (4) does not reach conventional significance ($p = 0.07$) — see explanation below.*

Why Model 4 shows β=0.083 and why it does not matter. When each country is compared only to its own past (one-way fixed effects), a 10-point increase in parental education predicts a 4.8-point increase in child education one generation later (β=0.483). Adding year fixed effects — which also control

for which *decade* an observation falls in — collapses this to β=0.083. The question is whether this collapse reflects a real weakness in the evidence or a flaw in the statistical method. On the active-expansion subsamples (Models 1–3 above), the collapse is far milder: β=0.739 to β=1.019, all $p < 0.01$.

The decline is the mechanical signature of a now well-documented two-way-FE pathology. When treatment is staggered — different countries begin expanding education in different decades — the 2WFE estimator is a weighted average of *all* pairwise comparisons, including "forbidden" ones that compare newly-expanding countries to countries already at high completion (Goodman-Bacon 2021). A decomposition on my panel (`scripts/robustness/goodman_bacon_decomposition.py`) gives the weights directly: 7.2 percent from clean not-yet-treated comparisons (weighted β = 11.3), 21.3 percent from treatment-timing pairs (weighted β = 1.1), and 71.5 percent from already-treated countries used as controls for newly-treated ones (weighted β = 9.6 but attenuated toward zero by heterogeneous treatment effects). The 0.083 figure is a weighted average in which three-quarters of the weight sits on comparisons that make no causal sense; it is not an unbiased estimate of anything.

The Callaway–Sant'Anna (2021) estimator — designed explicitly for staggered adoption and limited to the clean comparisons only — recovers the effect. Treating the 10 percent lower secondary completion threshold as the event (the crossing from "no priority" to "active expansion"), crossing the threshold raises child completion by 7.9 percentage points one generation later (clustered SE 1.7, 95 percent CI 4.4–11.0); the effect compounds to +21.4 pp by year 35. Pre-treatment event-study coefficients are indistinguishable from zero (Figure 6).

A caveat on method. CS and Goodman-Bacon were built for binary-staggered settings; the treatment variable here is continuous. I discretise at the 10 percent threshold because that is where the mechanism turns on: below 10 percent the state's education system is not yet reaching most households, and the home niche has no cohort of educated adults to propagate from. Robustness to thresholds at 5% and 15% is a natural next test; I do not claim the particular cutoff is the only defensible one.

What I do claim is that the 83 percent 2WFE collapse is not evidence against

education's effect. The pathology is visible in the weights, the clean comparisons give a positive estimate, and the staggered- adoption-appropriate estimator confirms it. I keep the one-way country-FE specification as the headline because it corresponds to the "how much do educated countries transmit to their children" test the paper sets up — not because it happens to produce a larger number.

A.2 Other Robustness Checks

A.2.1 Alternative Cutoffs

Table A3. Robustness to alternative cutoffs. Fixed effects regressions of child lower secondary completion on parental education (or log GDP per capita) across all cutoffs from 10% to 90%. At every cutoff education leads GDP in within-country predictive power; the 30% cutoff used in Table 3 is not an optimum but one point along a smooth curve. Produced by `scripts/residualization/by_gdp_cutoff.py`.

Cutoff	n	countries	Edu β	Edu R^2	GDP β	GDP R^2	Edu/GDP R^2
<10%	275	58	2.022	0.590	15.535	0.296	2.0×
<20%	469	83	1.704	0.665	15.502	0.258	2.6×
<30%	629	105	1.376	0.699	13.659	0.214	3.3×
<40%	740	112	1.185	0.685	15.469	0.223	3.1×
<50%	829	116	1.053	0.697	17.311	0.247	2.8×
<60%	906	120	0.902	0.663	18.059	0.259	2.6×
<70%	969	125	0.819	0.655	18.232	0.270	2.4×
<80%	1018	128	0.739	0.626	17.839	0.280	2.2×
<90%	1086	137	0.663	0.587	17.584	0.284	2.1×
Full	1665	185	0.483	0.533	15.787	0.245	2.2×

Country fixed effects; clustered standard errors. "Cutoff" restricts the sample to country-years in which parental lower secondary completion is below the stated threshold (active-expansion subsample). "Full" is the unrestricted panel. The within-country education R^2 peaks at the 30% cutoff (0.699) and declines only mildly outside that band; the education/GDP R^2 ratio stays between 2.0× and 3.3× across the full 10%–90% range. No cutoff reverses the

ordering. The 30% choice in Table 3 is where active expansion is most informative — not an optimum for the coefficient.

A.2.2 Alternative Lag Lengths

Table A5. Robustness to alternative lag lengths. Education R^2 vs. residualized GDP R^2 across lags 15–30 years. Entry-cohort design (entry $\geq$ 10%, ceiling $\leq$ 90%), country fixed effects, lower secondary completion. The T+25 specification is the theoretically motivated anchor, not a cherry-picked optimum; the result holds at every lag tested. See `scripts/robustness/lag_sensitivity.py`.

	Life exp.		TFR		U-5 mort.		Child edu	
Lag	Edu	Resid	Edu	Resid	Edu	Resid	Edu	Resid
15 yr	0.455	0.001	0.569	0.000	0.409	0.005	0.725	0.009
20 yr	0.451	0.002	0.509	0.000	0.328	0.013	0.617	0.006
25 yr	0.474	0.002	0.482	0.000	0.280	0.019	0.524	0.005
30 yr	0.457	0.000	0.473	0.001	0.232	0.005	0.461	0.004

"Edu" = how much of each country's outcome change is predicted by education. "Resid" = how much is predicted by income after stripping out education's contribution. At every lag tested and every outcome, education explains 23% or more of the variation; leftover income never exceeds 1.9%. n=822 (152 countries) for LE and TFR; n=787 (147 countries) for U-5 mortality; n=856 (157 countries) for child education.

A.2.3 Long-Run Panel and Threshold Robustness

Table A7. Long-run panel countries (1900–2015, 28 countries). Selected on the criterion of self-determined education policy and data quality; colonial-era data for colonised countries reflects colonial investment rather than domestic policy decisions and is excluded.[21]

[21] The 28 countries: *East Asia* — Japan, Taiwan, South Korea. *Western Europe* — United Kingdom, France, Germany, Sweden, Norway, Denmark, Finland, Netherlands, Belgium, Switzerland, Austria, Italy, Spain, Portugal. *New World* — United States, Canada, Australia, New Zealand. *Latin America* — Argentina, Chile, Uruguay, Cuba, Costa Rica. *Other* — Sri Lanka (British Ceylon, active colonial education investment, known anomaly), Hong Kong.

Table A8. Development threshold robustness: crossing dates under three specifications (World Bank WDI data; Taiwan excluded — not in WDI). The loose spec uses generous thresholds; the main spec uses the 1960 USA values; the strict spec tightens to TFR < 2.5 and LE > 72.6 (USA 1975).

Case	Loose	Main	Strict	Shift
Cuba	1964	1974	1980	16 yrs
South Korea	1978	1987	1993	15 yrs
Sri Lanka	1975	1993	2005	30 yrs
China	1982	1994	2001	19 yrs
Bangladesh	2005	2014	2019	14 yrs

Note: Every country crosses under every specification, and the ordering never changes. The threshold choice shifts dates by 14–19 years for fast expanders (Korea 2.13 pp/yr, Cuba 2.27 pp/yr, China 1.50 pp/yr) and 14–30 years overall, but no country that crosses the loose spec fails to cross the strict one. Sri Lanka's 30-year shift reflects the civil war holding life expectancy near the threshold while the educational base accumulated beneath it; the binding constraint was disruption, not the threshold definition. Taiwan (1970 loose, 1980 strict; WCDE data) is consistent with Cuba under all specifications but cannot be computed from WDI data. See `scripts/robustness/threshold_robustness.py`.

A.2.4 Data-Source Robustness

The paper uses WCDE v3, which reconstructs historical education distributions from census microdata, DHS surveys, and demographic back-projection. Three tests verify that the main forward-prediction and residualization results are not artifacts of WCDE's reconstruction methodology: replication on the independent Barro-Lee v3.0 dataset (Lee & Lee 2016) with different source surveys and interpolation; residualization on Barro-Lee; and restriction of WCDE to post-1970 observations, where data quality is highest and reconstruction plays the least role.

Specification	LE R^2	TFR R^2
WCDE v3 (main)	0.335	0.388

Specification	LE R^2	TFR R^2
Barro-Lee v3.0 ("at least some secondary," age 15–24, 146 countries)	0.320	0.287
Barro-Lee v3.0 (mean years of schooling)	0.400	0.428
WCDE v3, post-1970 only	0.313	0.281
WCDE v3, 171-country panel (15 USSR republics excluded; 25-yr forward lag, country FE)	0.366	0.444
Barro-Lee residualization (GDP with education's contribution removed)	≤ 0.003	≤ 0.003

All specifications significant at $p < 0.0001$, same sign and direction. The finding that income has no independent predictive power is not specific to one dataset; the pre-1970 decades add statistical power but are not driving the result. The natural experiments (Cambodia, Bihar–Kerala, Asian financial crisis, Russia, South Africa) rest on WDI outcome data and documented historical events, not on WCDE reconstruction, and are unaffected by any data-source concern.[22]

Chapter 9 develops the credibility case and the four-signature phenotype test that motivates the 171-country row, including the subgroup split under Barro-Lee that identifies the bias as specifically Goskomstat rather than socialist in general. Every headline coefficient in Table 3–Table 10 moves in the same direction under the 171-country exclusion: upward. USSR inclusion was attenuating the paper's signal, not producing it.[23]

A.2.5 Specification Robustness

Six additional checks address standard concerns about the Table 3 headline. They are summarised in Table A11 and described in turn below.

Table A11. Specification robustness. Each row reports the parental-education coefficient (or its semi-elasticity in the PPML/log rows) under the noted check, with country FE and country-clustered SEs. "Active expansion" restricts parent $< 30\%$ to match the Table 3 sample definition.

[22] See `scripts/robustness/barro_lee_replication.py`. Barro-Lee v3.0 source: `github.com/barrolee/BarroLeeDataSet`.

[23] See `scripts/robustness/ussr_exclusion_panel.py` for the exclusion panel and `scripts/which_edu_measure_is_correct.py` for the subgroup split.

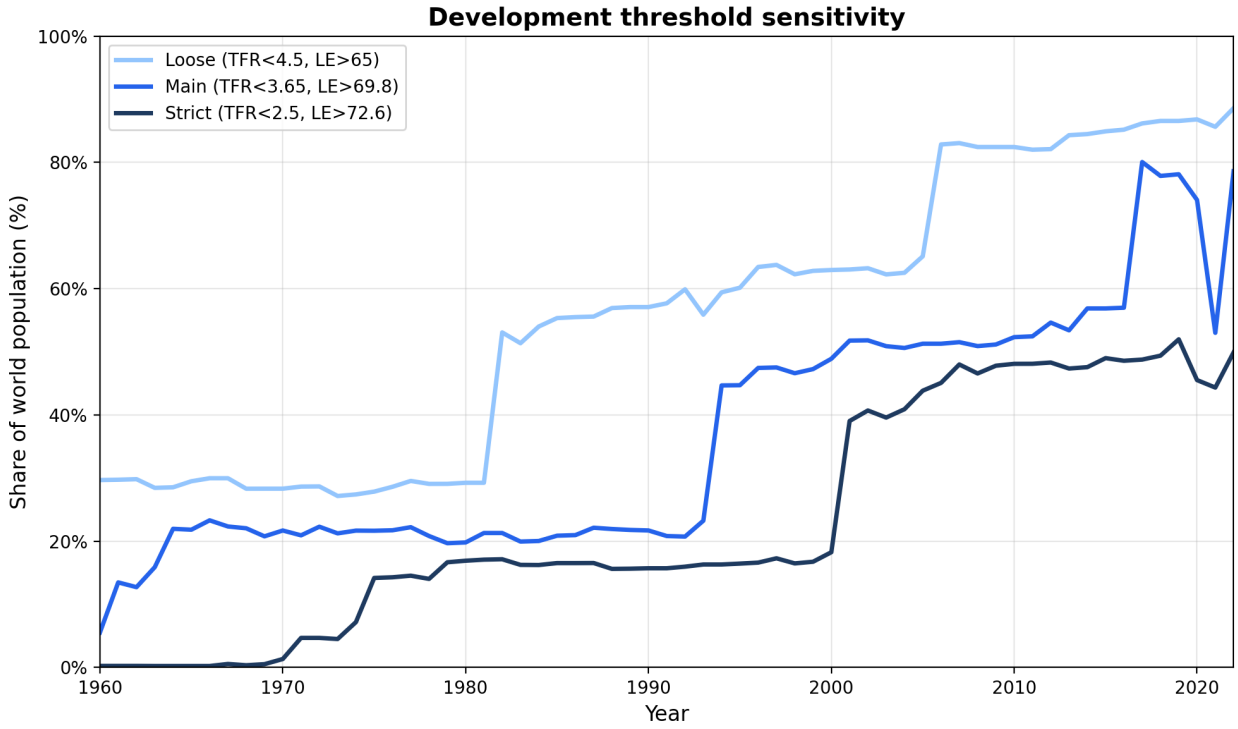

Figure 11: Cumulative share of world population crossing development thresholds under three specifications: loose (TFR < 4.5, LE > 65), main (TFR < 3.65, LE > 69.8), and strict (TFR < 2.5, LE > 72.6). The shape is identical across specifications — the same China jump, the same steady rise. The threshold choice shifts the level; it does not change the story. *(Source: World Bank WDI, WCDE v3. See* `scripts/figures/cumulative_developed_thresholds.py`*)*

Check	β (or elast.)	SE	N	Count
5-year baseline (Table 3 col 1)	1.376	0.083	629	
10-year aggregation, active expansion	1.437	0.078	416	
Annual (interpolated), active expansion	1.326	0.078	3,548	
Balanced subpanel (9/9 obs), active expansion	1.490	0.124	423	
Within-year cross-cohort (20–24 on 45–49), active expansion	1.965	0.051	1,747	
PPML, TFR_{T+25} (semi-elast. per pp)	-0.91%	0.14%	590	
PPML, U5MR_{T+25} (semi-elast. per pp)	-3.30%	0.36%	579	
Log LE_{T+25} (semi-elast. per pp)	$+0.36\%$	0.04%	590	
Log TFR_{T+25} (semi-elast. per pp)	-0.82%	0.11%	590	
Log U5MR_{T+25} (semi-elast. per pp)	-2.59%	0.28%	579	

All rows significant at $p < 0.01$. Produced by `scripts/robustness/period_length.py`, `balanced_panel.py`, `cross_cohort_within_year.py`, `ppml_outcomes.py`, *and* `log_outcomes.py`.

Period length. Aggregating child cohorts at 5-year (baseline), 10-year, and annual (interpolated) resolution leaves the active-expansion coefficient in the 1.33–1.44 range. The annual grid mechanically multiplies the observation count roughly fivefold; the point estimate barely moves. Time-aggregation is not driving the headline.

Balanced panel. The WCDE cohort grid is balanced by construction at the full-panel level (every country has 9 cohort years 1975–2015), so the headline is not a composition artefact. Inside the active-expansion subset, forty-seven countries satisfy the $< 30\%$ cutoff in all nine cohort years; re-estimating on this deepest-observed subset gives $\beta = 1.49$, slightly above the unbalanced headline of 1.36. Balance does not weaken the effect.

Within-year cross-cohort. The headline regression uses a 25-year forward lag: child education at t on parental education at $t - 25$. A standard concern is that shocks occurring between $t - 25$ and t (global MDGs, the Asian financial crisis, HIV) are absorbed by the forward-lag coefficient. The WCDE v3 microdata (`prop_both.csv`) gives the lower-secondary-completion share for every 5-year age group in every country-year. In any given year, the age-20–24 share and the age-45–49 share are one generation apart but measured simultaneously: any shock in the intervening 25 years affects both cohorts, is

absorbed by country FE if country-specific, and cannot drive the coefficient. The within-year cross-cohort coefficient is $\beta = 1.97$ (SE 0.05) in the active-expansion sample, directionally stronger than the forward-lag headline. The transmission mechanism does not depend on the forward-lag design choice.

PPML. TFR and under-5 mortality are non-negative and right-skewed; the standard log-linear regression drops zeros and is sensitive to heteroscedasticity in $E[\log y \mid x]$. Poisson PML (Santos Silva and Tenreyro 2006) is consistent under arbitrary heteroscedasticity provided the conditional mean is correctly specified. Under PPML with country FE, a one-percentage-point rise in parental lower secondary completion lowers TFR at $T + 25$ by -0.91% (SE 0.14%, $p < 0.001$) and under-5 mortality by -3.30% (SE 0.36%, $p < 0.001$). GDP's coefficient is insignificant in both equations ($p = 0.24$ and $p = 0.69$), consistent with the residualization result.

Log outcomes. Putting LE, TFR, and U5MR in logs gives elasticities directly comparable to the PPML semi-elasticities. A one-percentage-point rise in parental completion predicts, one generation later: life expectancy $+0.36\%$, total fertility -0.82%, under-5 mortality -2.59%. All three are significant at $p < 0.001$ with country FE and log GDP included. The levels specification reported in Table 7 gives the same sign and ordering; the choice of functional form does not materially change the substance.

Wooldridge strict-exogeneity test. The regression-based test from Wooldridge (2010, §10.5) adds the one-period lead of parental education (at $t - 20$) to the FE regression and tests whether its coefficient is zero. On the full panel the lead enters with $\hat{\gamma} = 1.275$ (SE 0.114, $t = 11.1$); on the active-expansion sample $\hat{\gamma} = 0.655$ (SE 0.281, $t = 2.3$). In both samples strict exogeneity is formally rejected. This is expected rather than surprising: parental lower secondary completion is a highly persistent stock variable, so parent education at $t - 20$ is near-perfectly correlated with parent education at $t - 25$, and a test based on the lead has nearly zero power to distinguish mechanical persistence from a true exogeneity violation. Identification in this paper does not rely on strict exogeneity: the Callaway–Sant'Anna event study in Section 8.6 uses not-yet-treated countries as the comparison set, the forward-lag design rules out reverse causality by construction, and the within-year cross-cohort result above removes the lag altogether. The

Wooldridge statistic is reported here for completeness and should be read as a diagnostic on the persistence of the regressor, not as evidence against the design.

A.2.6 South Africa: Apartheid Stratification and the PET Curve

South Africa's 65% lower-secondary completion in 1990 is an aggregate masking apartheid stratification: near-universal white completion and much lower Black completion produced a bimodal distribution invisible in national data. The population bearing the brunt of HIV had far lower education than the headline figure suggests. The fall in TFR from 3.72 (1990) to 2.41 (2005) is the composition result of Section 8.9 operating as predicted. Primary education drives fertility decline hardest ($R^2 = 0.65$ for TFR), and Black primary was expanding through the shock. The fall is not a frozen home-niche baseline surviving external disruption — 1990 would be too early to carry that claim, with TFR still at 3.72, above the development threshold. It is the ordinary primary-to-fertility mechanism continuing to run while lower-secondary depth remained too shallow in the affected majority to deliver the other half of the composition: the behavioural response to a novel pathogen that deeper schooling carries.

The LE collapse runs through that secondary deficit. The Southern African epidemic was driven at the acquisition stage by factors education could not have prevented: low traditional circumcision prevalence (8–11% in the HIV belt vs 85–99% in Muslim West Africa), which three randomised trials showed provides a 60% biological reduction in male acquisition risk (Auvert et al. 2005); and circular mining migration that created connected sexual networks across the subcontinent. These are genuinely exogenous to education.

Education was in the story on both sides of the behavioural response. Early in the epidemic, more educated Africans were *more* likely to be HIV-positive: greater mobility, more partners, more agency exercised without information about a novel pathogen (Fortson 2008). Once information existed, the gradient reversed: educated populations changed behaviour first. In Botswana, a natural experiment exploiting a 1996 school reform showed that each additional year of secondary schooling *caused* an 8.1-percentage-point reduction in cumulative HIV risk (De Neve et al. 2015). In rural Uganda, educated indi-

viduals responded more to prevention campaigns, and the education–HIV gradient reversed from positive to negative between 1990 and 2000 (De Walque 2007). Baker et al. (2017) formalise this as the Population Education Transition curve: when a novel health risk enters a population, the educated are early adopters (positive gradient); once information exists, the educated change behaviour first (gradient reverses). The pattern replicates across cigarettes, processed food, and HIV — education is the mechanism in both directions, at the secondary-completion depth the behavioural response requires.

South Africa's 1990 lower-secondary figure, once stratified, sat below that depth for the majority who received the shock.

A.2.7 Colonial Test: IV First-Stage Diagnostics

A formal instrumental-variable test sharpens the Colonial-Test result (Section 10.4). The strategy uses AJR's own settler-mortality channel and asks what it actually identifies — not whether the instrument works, but whether its exogenous variation enters development through institutions or through schools. Two-stage least squares using coloniser religion as the excluded instrument produces a first-stage F-statistic of 10.8 for education at independence — above the Stock & Yogo (2005) threshold for instrument relevance — but only 1.4 for Polity2 institutional quality, well below the threshold. The second stage for education is significant ($\hat{\beta} = +0.059$, $t = 3.21$, $p = 0.002$); the institutional 2SLS is unreliable because the instrument is weak. The Wu-Hausman test is insignificant ($p = 0.85$), confirming that OLS and 2SLS agree when the endogenous variable is education — consistent with coloniser religion being exogenous to the education channel. AJR's instrument works, but it works *because* it predicts education, not because it predicts institutions. See `scripts/robustness/iv_2sls_colonial.py`.

A.2.8 Panel Summary Statistics and Country Coverage

Table A13. Summary statistics for analysis variables. Analysis panel: 185 countries, 1975–2015 at 5-year intervals, with parental education lagged 25 years (observed at $T-25$). Produced by `scripts/tables/summary_stats.py`.

Variable	n	mean	sd	min	max
Parental lower-sec completion (%, $T-25$)	1,665	42.0	33.3	0.0	100.0
Child lower-sec completion (%, T)	1,665	61.9	31.6	0.4	100.0
Log GDP per capita (const. 2017 USD)	1,466	8.28	1.49	4.98	11.67
Life expectancy at birth (years)	1,608	65.7	10.6	12.8	84.3
Total fertility rate (births/woman)	1,608	3.70	1.95	0.91	8.86
Under-5 mortality (per 1,000 live births)	1,566	68.3	69.9	2.2	338.0

The education+GDP panel covers 178 of 185 countries; seven are dropped because the World Bank WDI has no GDP series for them: North Korea, Taiwan, Micronesia (FSM), and four French overseas departments (French Guiana, Guadeloupe, Martinique, Reunion). Country coverage for every main specification is listed in `checkin/summary_stats.json` *under* `country_lists` *and is cross-checked by* `scripts/verify_country_names.py`.[24]

References

Acemoglu, D. & Robinson, J.A. (2012). *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*. Crown Publishers.

[24]Full 185-country panel (alphabetical, WCDE v3 naming): Afghanistan, Albania, Algeria, Angola, Argentina, Armenia, Aruba, Australia, Austria, Azerbaijan, Bahamas, Bahrain, Bangladesh, Belarus, Belgium, Belize, Benin, Bhutan, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cambodia, Cameroon, Canada, Cape Verde, Central African Republic, Chad, Chile, China, Colombia, Comoros, Congo, Costa Rica, Côte d'Ivoire, Croatia, Cuba, Curaçao, Cyprus, Czech Republic, North Korea, Democratic Republic of the Congo, Denmark, Dominican Republic, Ecuador, Egypt, El Salvador, Equatorial Guinea, Estonia, Ethiopia, Fiji, Finland, France, French Guiana, French Polynesia, Gabon, Gambia, Georgia, Germany, Ghana, Greece, Guadeloupe, Guatemala, Guinea, Guinea-Bissau, Guyana, Haiti, Honduras, Hong Kong (China SAR), Hungary, Iceland, India, Indonesia, Iran, Iraq, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Kiribati, Kuwait, Kyrgyzstan, Laos, Latvia, Lebanon, Lesotho, Liberia, Lithuania, Luxembourg, Macao (China SAR), Madagascar, Malawi, Malaysia, Maldives, Mali, Malta, Martinique, Mauritius, Mexico, Micronesia (FSM), Mongolia, Montenegro, Morocco, Mozambique, Myanmar, Namibia, Nepal, Netherlands, New Caledonia, New Zealand, Nicaragua, Niger, Nigeria, Norway, Palestine, Oman, Pakistan, Panama, Paraguay, Peru, Philippines, Poland, Portugal, Puerto Rico, Qatar, South Korea, Moldova, Reunion, Romania, Russia, Rwanda, Saint Lucia, Saint Vincent and the Grenadines, Samoa, Sao Tome and Principe, Saudi Arabia, Senegal, Serbia, Sierra Leone, Singapore, Slovakia, Slovenia, Solomon Islands, Somalia, South Africa, South Sudan, Spain, Sri Lanka, Sudan, Suriname, Eswatini (Swaziland), Sweden, Switzerland, Syria, Taiwan, Tajikistan, Thailand, North Macedonia, Timor-Leste, Togo, Tonga, Trinidad and Tobago, Tunisia, Türkiye, Turkmenistan, Uganda, Ukraine, United Arab Emirates, United Kingdom, Tanzania, United States, Uruguay, Vanuatu, Venezuela, Vietnam, Yemen, Zambia, Zimbabwe.

Angrist, N., Djankov, S., Goldberg, P.K. & Patrinos, H.A. (2021). Measuring Human Capital Using Global Learning Data. *Nature*, 592, 403–408.

Angrist, J.D. & Pischke, J.S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

Auvert, B., Taljaard, D., Lagarde, E., Sobngwi-Tambekou, J., Sitta, R. & Puren, A. (2005). Randomized, Controlled Intervention Trial of Male Circumcision for Reduction of HIV Infection Risk: The ANRS 1265 Trial. *PLOS Medicine*, 2(11), e298.

Becker, G.S. (1964). *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. National Bureau of Economic Research; Columbia University Press.

Baker, D.P., Collins, J.M. & Leon, J. (2017). Risk Factor or Social Vaccine? The Historical Progression of the Role of Education in HIV and AIDS Infection in Sub-Saharan Africa. *Prospects*, 38(4), 467–486.

Brass, P.R. (1986). The Political Uses of Crisis: The Bihar Famine of 1966–1967. *Journal of Asian Studies*, 45(2), 245–267.

Boyd, R. & Richerson, P.J. (1985). *Culture and the Evolutionary Process*. University of Chicago Press.

Cai, Y. (2010). China's Below-Replacement Fertility: Government Policy or Socioeconomic Development? *Population and Development Review*, 36(3), 419–440.

Caldwell, J.C. (1976). Toward a Restatement of Demographic Transition Theory. *Population and Development Review*, 2(3/4), 321–366.

Callaway, B. & Sant'Anna, P. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

Deaton, A. (2013). *The Great Escape: Health, Wealth, and the Origins of Inequality*. Princeton University Press.

Dehaene, S. (1997). *The Number Sense: How the Mind Creates Mathematics*. Oxford University Press.

Dehaene, S. (2009). *Reading in the Brain: The New Science of How We Read*.

Viking.

Dehaene, S., Pegado, F., Braga, L.W., Ventura, P., Nunes Filho, G., Jobert, A., Dehaene-Lambertz, G., Kolinsky, R., Morais, J. & Cohen, L. (2010). How Learning to Read Changes the Cortical Networks for Vision and Language. *Science*, 330(6009), 1359–1364.

Diamond, J. (1997). *Guns, Germs, and Steel: The Fates of Human Societies*. W.W. Norton.

De Neve, J.-W., Fink, G., Subramanian, S.V., Moyo, S. & Bor, J. (2015). Length of Secondary Schooling and Risk of HIV Infection in Botswana: Evidence from a Natural Experiment. *Lancet Global Health*, 3(8), e470–e477.

de Waal, F.B.M. (2013). *The Bonobo and the Atheist: In Search of Humanism Among the Primates*. W.W. Norton.

De Walque, D. (2007). How Does the Impact of an HIV/AIDS Information Campaign Vary with Educational Attainment? Evidence from Rural Uganda. *Journal of Development Economics*, 84(2), 686–714.

Drèze, J. & Murthi, M. (2001). Fertility, Education, and Development: Evidence from India. *Population and Development Review*, 27(1), 33–63.

Davis, M. (2001). *Late Victorian Holocausts: El Niño Famines and the Making of the Third World*. Verso.

Duflo, E. (2001). Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment. *American Economic Review*, 91(4), 795–813.

Drèze, J. & Sen, A. (1989). *Hunger and Public Action*. Clarendon Press.

Dyson, T. & Maharatna, A. (1992). Bihar Famine, 1966–67, and Maharashtra Drought, 1970–73: The Demographic Consequences. *Economic and Political Weekly*, 27(26), 1325–1332.

Easterlin, R.A. (1981). Why Isn't the Whole World Developed? *Journal of Economic History*, 41(1), 1–19.

Fortson, J.G. (2008). The Gradient in Sub-Saharan Africa: Socioeconomic Status and HIV/AIDS. *Demography*, 45(2), 303–322.

Frank, M.C., Everett, D.L., Fedorenko, E. & Gibson, E. (2008). Number as a cognitive technology: Evidence from Pirahã language and cognition. *Cognition*, 108(3), 819–824.

Gakidou, E., Cowling, K., Lozano, R. & Murray, C.J.L. (2010). Increased Educational Attainment and Its Effect on Child Mortality in 175 Countries between 1970 and 2009: A Systematic Analysis. *The Lancet*, 376(9745), 959–974.

Gao, M. (1999). *Gao Village: Rural Life in Modern China*. University of Hawai'i Press.

Gao, M. (2008). *The Battle for China's Past: Mao and the Cultural Revolution*. Pluto Press.

Goldin, C. & Katz, L.F. (2008). *The Race Between Education and Technology*. Harvard University Press.

Goodall, J. (1986). *The Chimpanzees of Gombe: Patterns of Behavior*. Belknap Press of Harvard University Press.

Goodman-Bacon, A. (2021). Difference-in-Differences with Variation in Treatment Timing. *Econometrica*, 89(5), 2261–2290.

Gwatkin, D.R. (1979). Political Will and Family Planning: The Implications of India's Emergency Experience. *Population and Development Review*, 5(1), 29–59.

Gordon, P. (2004). Numerical Cognition Without Words: Evidence from Amazonia. *Science*, 306(5695), 496–499.

Grantham-McGregor, S., Cheung, Y.B., Cueto, S., Glewwe, P., Richter, L., Strupp, B. & the International Child Development Steering Group. (2007). Developmental Potential in the First 5 Years for Children in Developing Countries. *The Lancet*, 369(9555), 60–70.

Hamilton, W.D. (1964). The Genetical Evolution of Social Behaviour, I & II. *Journal of Theoretical Biology*, 7(1), 1–52.

Hanushek, E.A. & Woessmann, L. (2008). The Role of Cognitive Skills in Economic Development. *Journal of Economic Literature*, 46(3), 607–668.

Hanushek, E.A. & Woessmann, L. (2012). Do Better Schools Lead to More

Growth? Cognitive Skills, Economic Outcomes, and Causation. *Journal of Economic Growth*, 17(4), 267–321.

Hare, B. & Woods, V. (2020). *Survival of the Friendliest: Understanding Our Origins and Rediscovering Our Common Humanity*. Random House.

Hawkes, K. (2003). Grandmothers and the evolution of human longevity. *American Journal of Human Biology*, 15(3), 380–400.

Herculano-Houzel, S. (2016). *The Human Advantage: A New Understanding of How Our Brain Became Remarkable*. MIT Press.

Heyes, C. (2018). *Cognitive Gadgets: The Cultural Evolution of Thinking*. Harvard University Press.

Hrdy, S.B. (2009). *Mothers and Others: The Evolutionary Origins of Mutual Understanding*. Harvard University Press.

Konner, M. (2010). *The Evolution of Childhood: Relationships, Emotion, Mind*. Harvard University Press.

Liu, Y., Rao, K. & Hsiao, W.C. (2003). Medical Expenditure and Rural Impoverishment in China. *Journal of Health, Population and Nutrition*, 21(3), 216–222.

Lutz, W. & Kebede, E. (2018). Education and Health: Redrawing the Preston Curve. *Population and Development Review*, 44(2), 343–361.

Lutz, W., Reiter, C., Özdemir, C., Yildiz, D., Guimaraes, R. & Goujon, A. (2021). Skills-adjusted Human Capital Shows Increasing Global Progress and Persistent Disparities. *Nature Human Behaviour*, 5, 701–711.

Myhrvold-Hanssen, T. (2003). Democracy, News Media, and Famine Prevention: Amartya Sen and the Bihar Famine of 1966–67. Masters thesis, University of Oslo.

Malthus, T.R. (1798). *An Essay on the Principle of Population*. London: J. Johnson.

Mamun, M.A. & Bongaarts, J. (2022). Revisiting the causes of fertility decline in Bangladesh: the relative importance of female education and family planning programs. *Asian Population Studies*, 18(2), 125–148.

Miller, G., Babiarz, K.S., Chin, Y. & Song, S. (2018). The Limits and Consequences of Population Policy: Evidence from China's Wan Xi Shao Campaign. NBER Working Paper No. 25130.

Mullainathan, S. & Shafir, E. (2013). *Scarcity: Why Having Too Little Means So Much*. Times Books.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.

National Bureau of Statistics of China. (1982). *Third National Population Census of the People's Republic of China*. China Statistics Press. [Rural population share: 79.7% in 1980.]

Pepper, S. (1996). *Radicalism and Education Reform in 20th-Century China*. Cambridge University Press.

Prieto, A. (1981). Cuba's National Literacy Campaign. *Journal of Reading*, 25(3), 215–221. [268,000 brigadistas mobilised for the 1961 campaign.]

Pritchett, L. (2013). *The Rebirth of Education: Schooling Ain't Learning*. Center for Global Development.

Santos Silva, J.M.C. & Tenreyro, S. (2006). The Log of Gravity. *Review of Economics and Statistics*, 88(4), 641–658.

Sapolsky, R.M. (2017). *Behave: The Biology of Humans at Our Best and Worst*. Penguin Press.

Schultz, T.W. (1961). Investment in Human Capital. *American Economic Review*, 51(1), 1–17.

Sen, A. (1999). *Development as Freedom*. Oxford University Press.

Smil, V. (2017). *Energy and Civilization: A History*. MIT Press.

Stearns, S.C. (1992). *The Evolution of Life Histories*. Oxford University Press.

Tomasello, M. (2014). *A Natural History of Human Thinking*. Harvard University Press.

Tooby, J. & Cosmides, L. (2010). Groups in Mind: The Coalitional Roots of War

and Morality. In H. Högh-Olesen (Ed.), *Human Morality and Sociality: Evolutionary and Comparative Perspectives* (pp. 191–234). Palgrave Macmillan.

Smith, A. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*.

UNDP. (1990). *Human Development Report 1990*. Oxford University Press.

UNESCO. (2023). *Global Education Monitoring Report 2023: Technology in Education — A Tool on Whose Terms?* Paris: UNESCO.

UNESCO. (2024). *Global Education Monitoring Report 2024/5: Leadership in Education — Lead for Learning*. Paris: UNESCO.

Unger, J. (1982). *Education Under Mao: Class and Competition in Canton Schools, 1960–1980*. Columbia University Press.

United Nations. (2015). *Transforming Our World: The 2030 Agenda for Sustainable Development* (A/RES/70/1). New York: United Nations.

Vicziany, M. (1982). Coercion in a Soft State: The Family-Planning Program of India. Part I: The Myth of Voluntarism. *Pacific Affairs*, 55(3), 373–402.

Wallace, A.R. (1870). *Contributions to the Theory of Natural Selection: A Series of Essays*. Macmillan.

Wrangham, R. & Peterson, D. (1996). *Demonic Males: Apes and the Origins of Human Violence*. Houghton Mifflin.

Wrangham, R. (2009). *Catching Fire: How Cooking Made Us Human*. Basic Books.

Wrigley, E.A. (2010). *Energy and the English Industrial Revolution*. Cambridge University Press.

World Bank. (2018). *World Development Report 2018: Learning to Realize Education's Promise*. Washington, DC: World Bank.

World Bank. World Development Indicators. Various years.

Development threshold data computed from World Bank WDI. USA 1960 benchmarks: TFR = 3.65, life expectancy = 69.8. Full analysis code, data, and verification outputs available at github.com/rkpagadala/education-of-nations.